SQANTI3 report

Unique Genes: 9389

Unique Isoforms: 15213

Gene Classification

Category	Genes, count
Annotated Genes	9383
Novel Genes	6

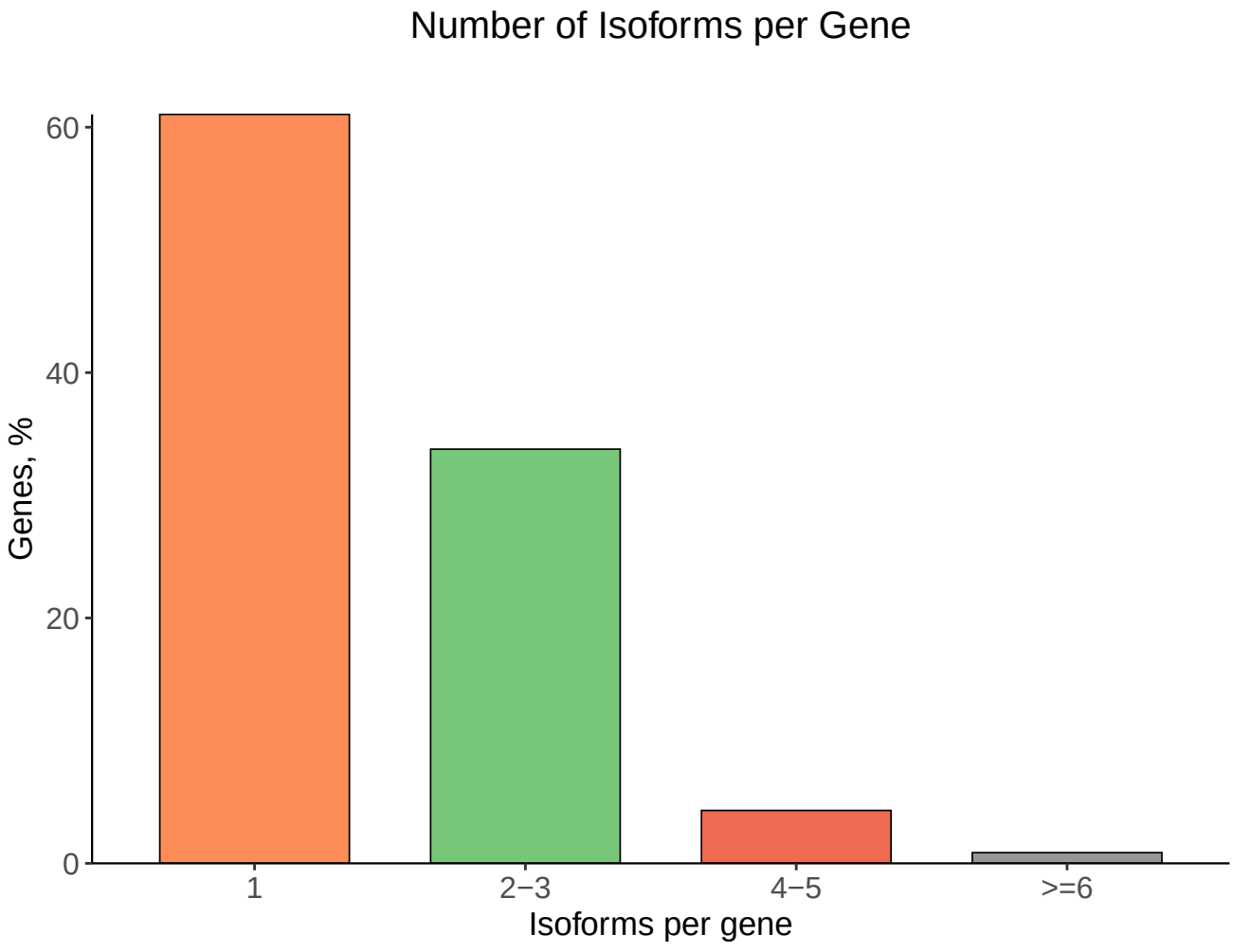
Transcript Classification

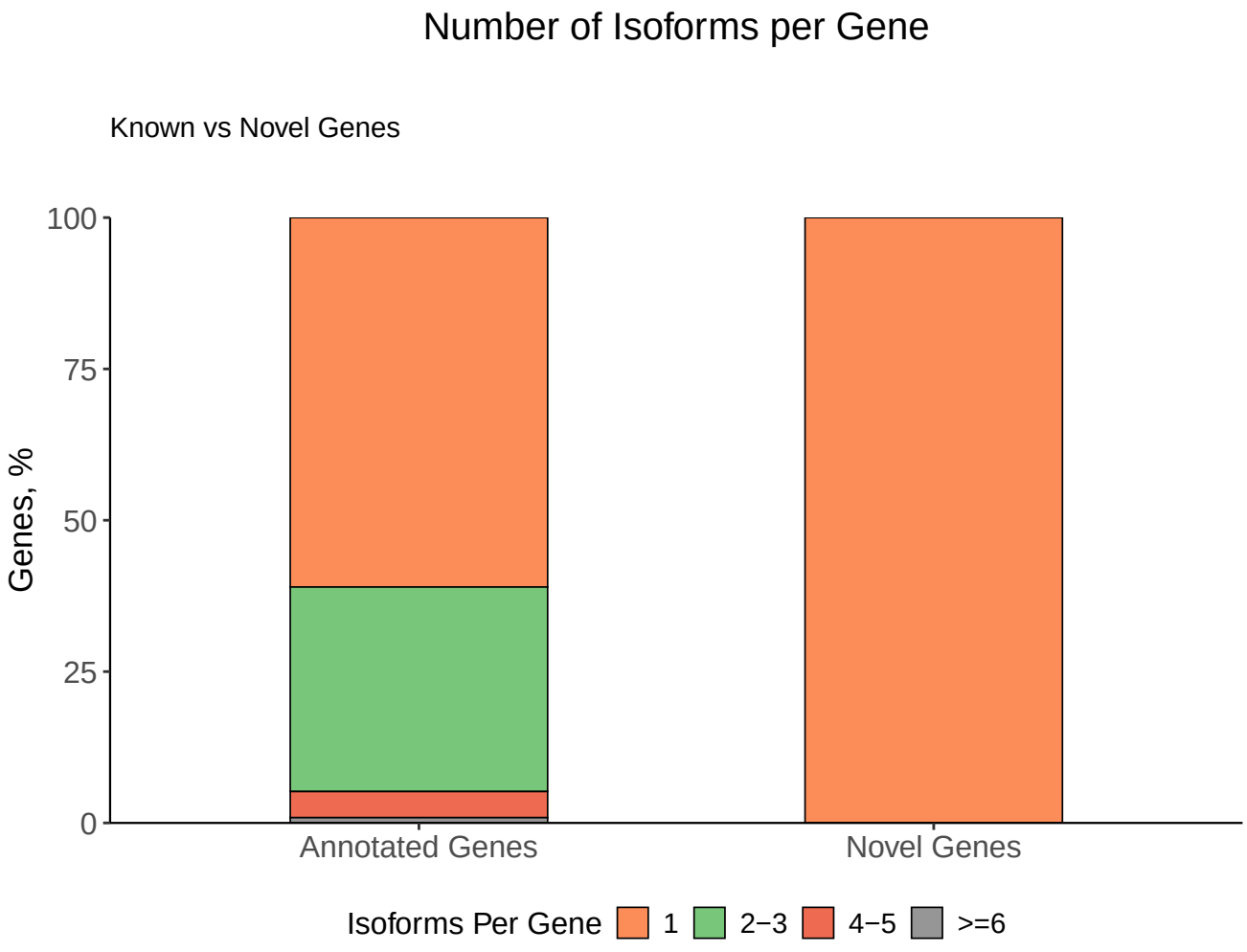
Category	Isoforms, count
FSM	14950
ISM	69
NIC	136
NNC	35
Genic Genomic	11
Antisense	6
Fusion	6
Intergenic	0
Genic Intron	0

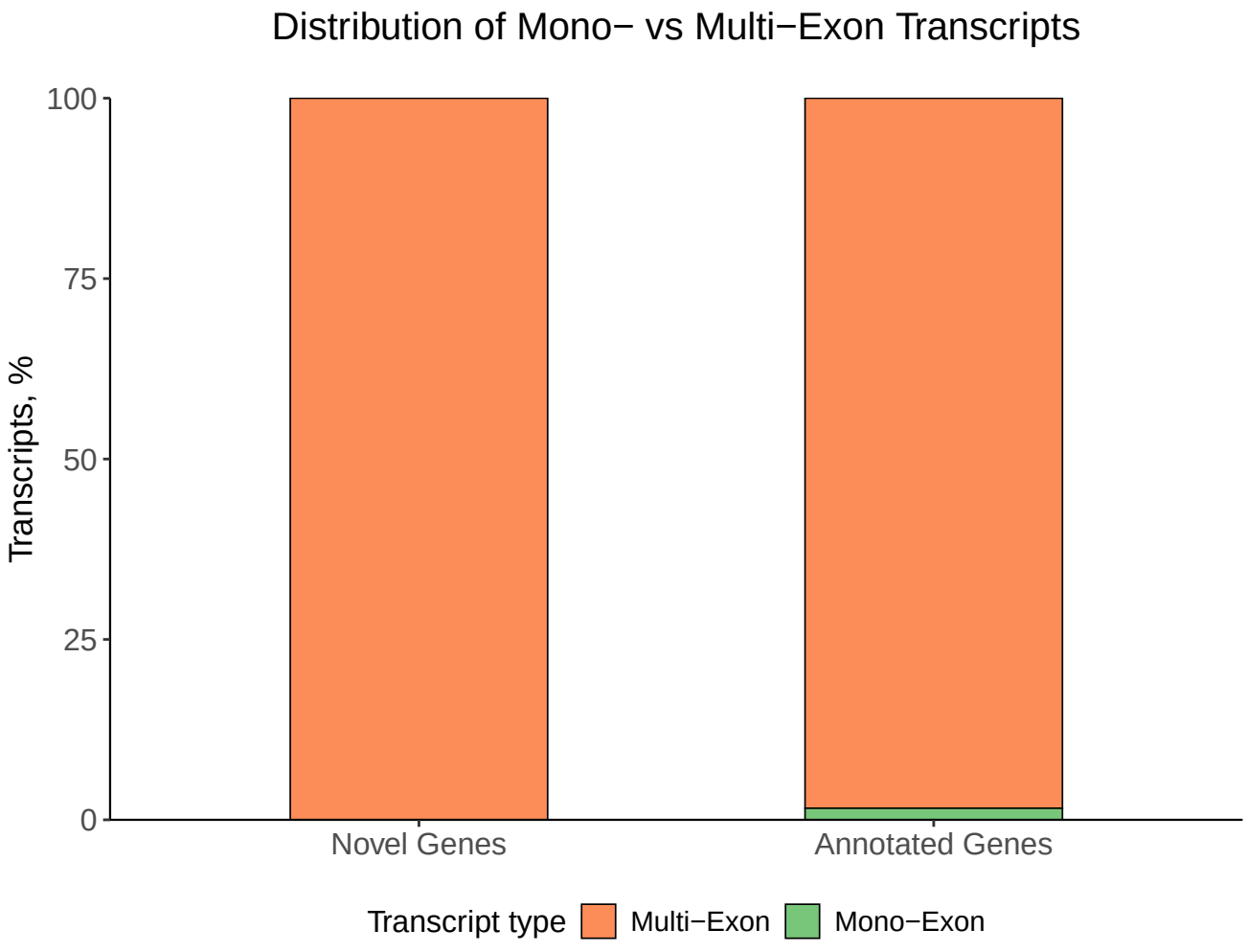
Splice Junction Classification

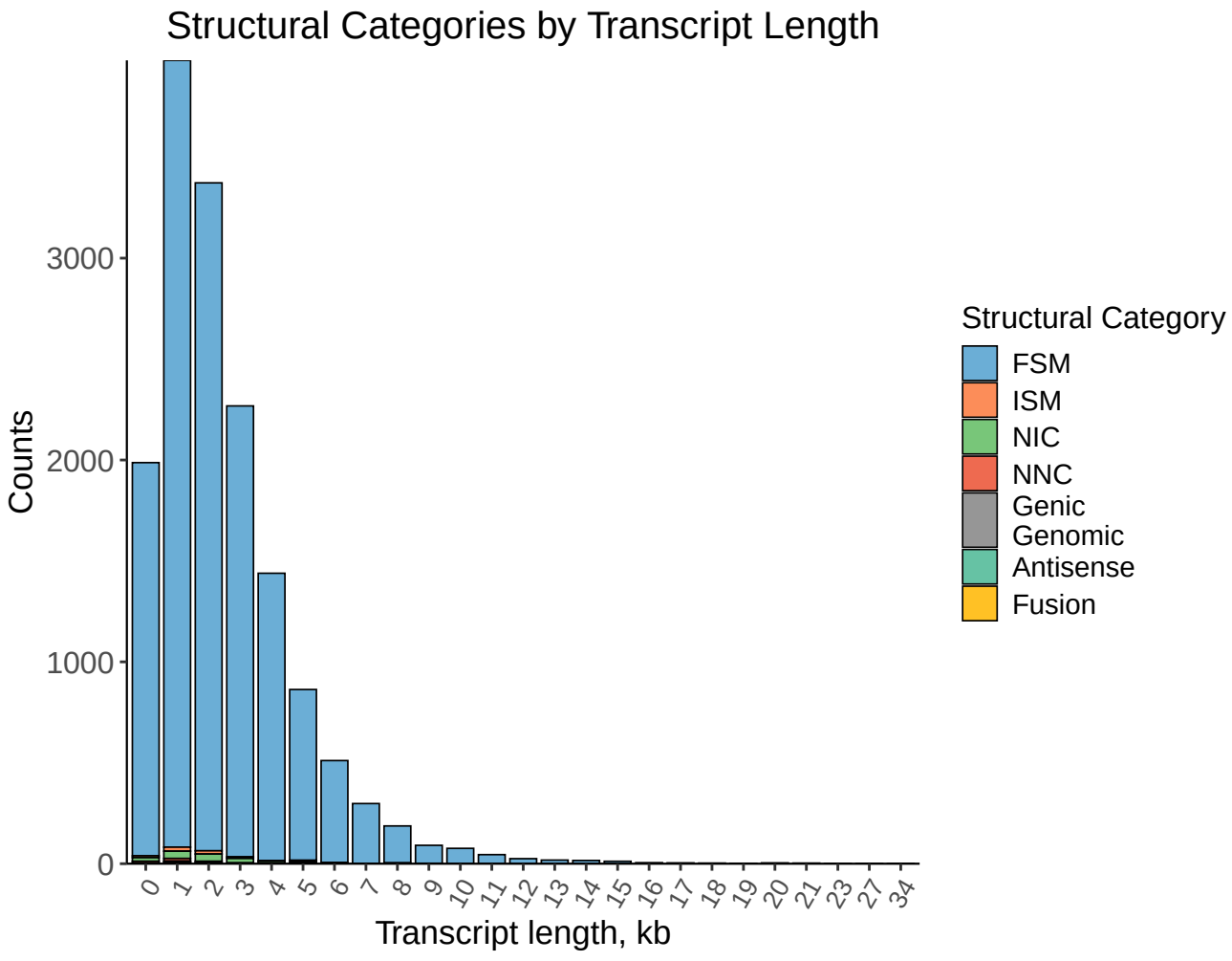
Category	SJs, count	Percent
Known canonical	93169	99.86
Known Non−canonical	46	0.05
Novel canonical	83	0.09
Novel Non−canonical	5	0.01

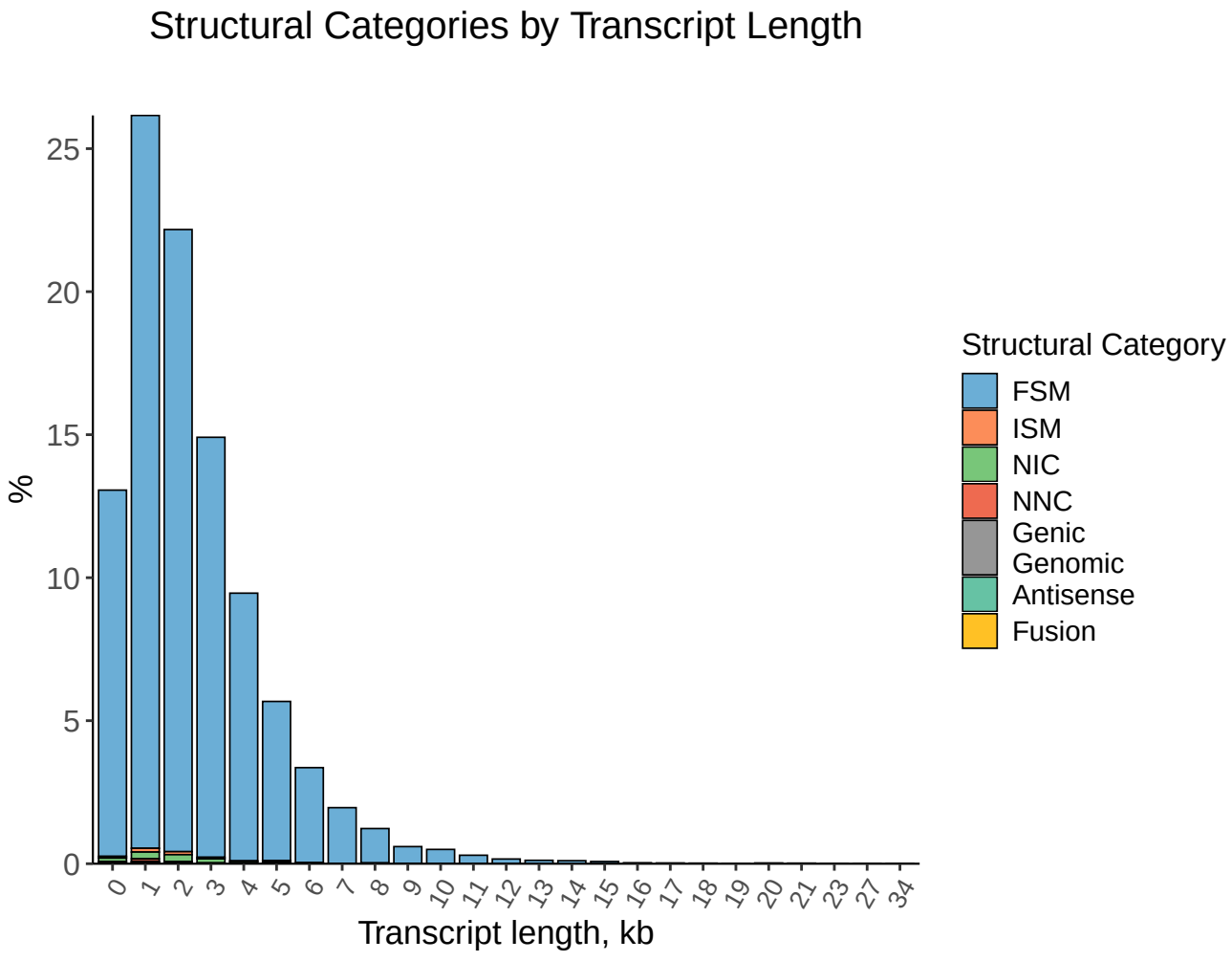
Gene Characterization

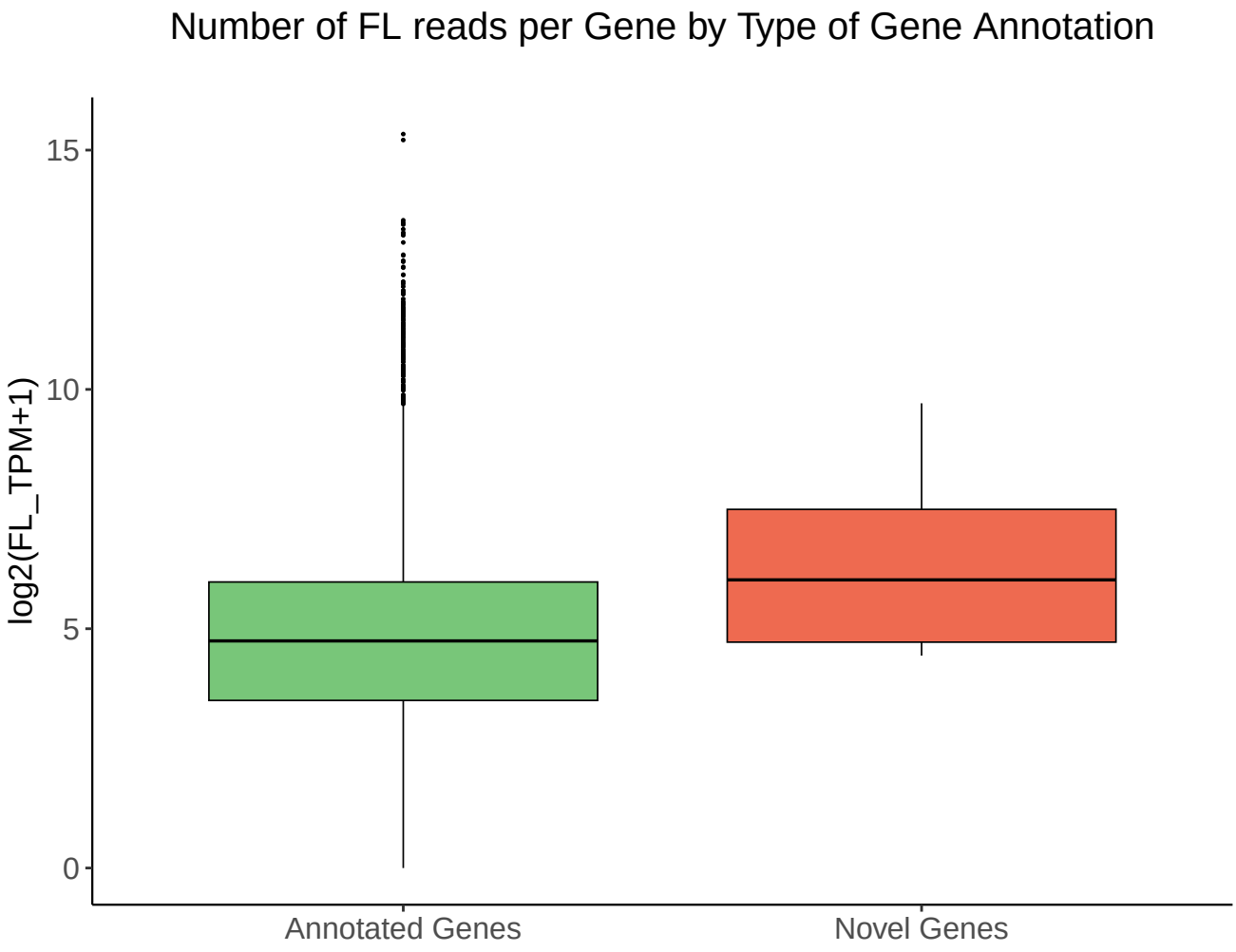

Number of Isoforms per Gene
Genes, %
60
40
20
0
1
2−3
4−5
>=6
Isoforms per gene

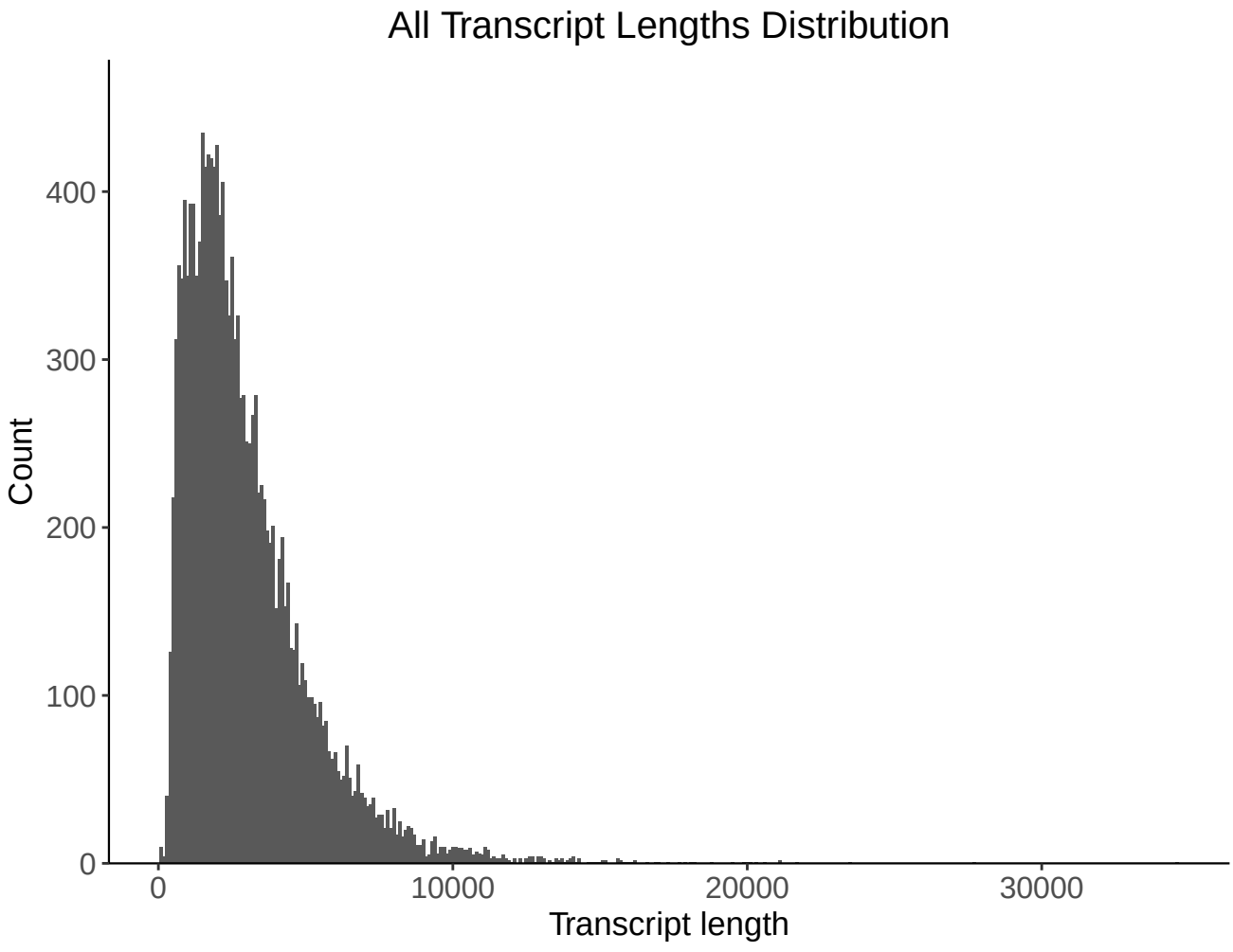

Number of Isoforms per Gene
Known vs Novel Genes
Genes, %
100
75
50
25
0
Annotated Genes
Novel Genes
Isoforms Per Gene
1
2−3
4−5
>=6

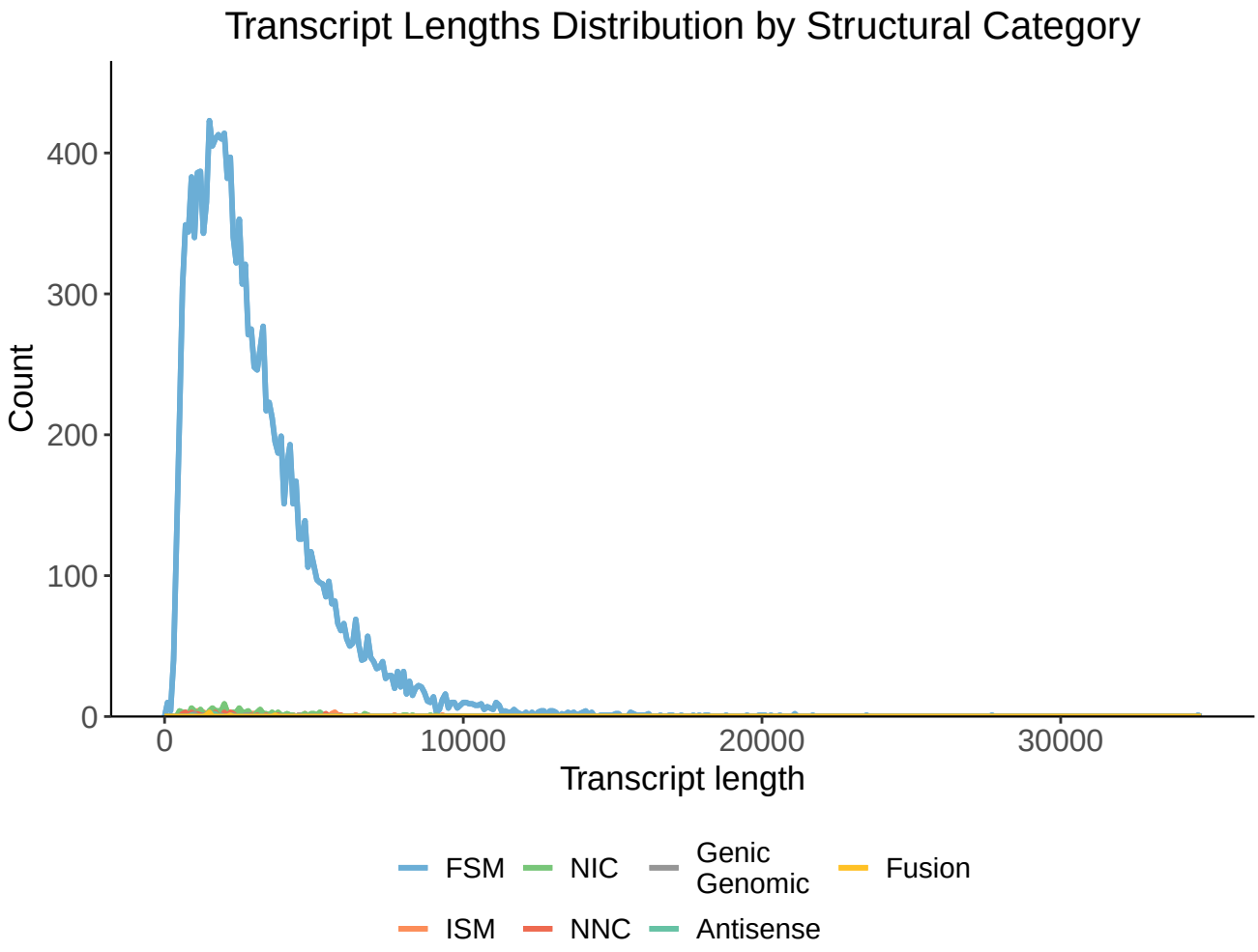

Distribution of Mono− vs Multi−Exon Transcripts
Transcripts, %
100
75
50
25
0
Novel Genes
Annotated Genes
Transcript type
Multi−Exon
Mono−Exon

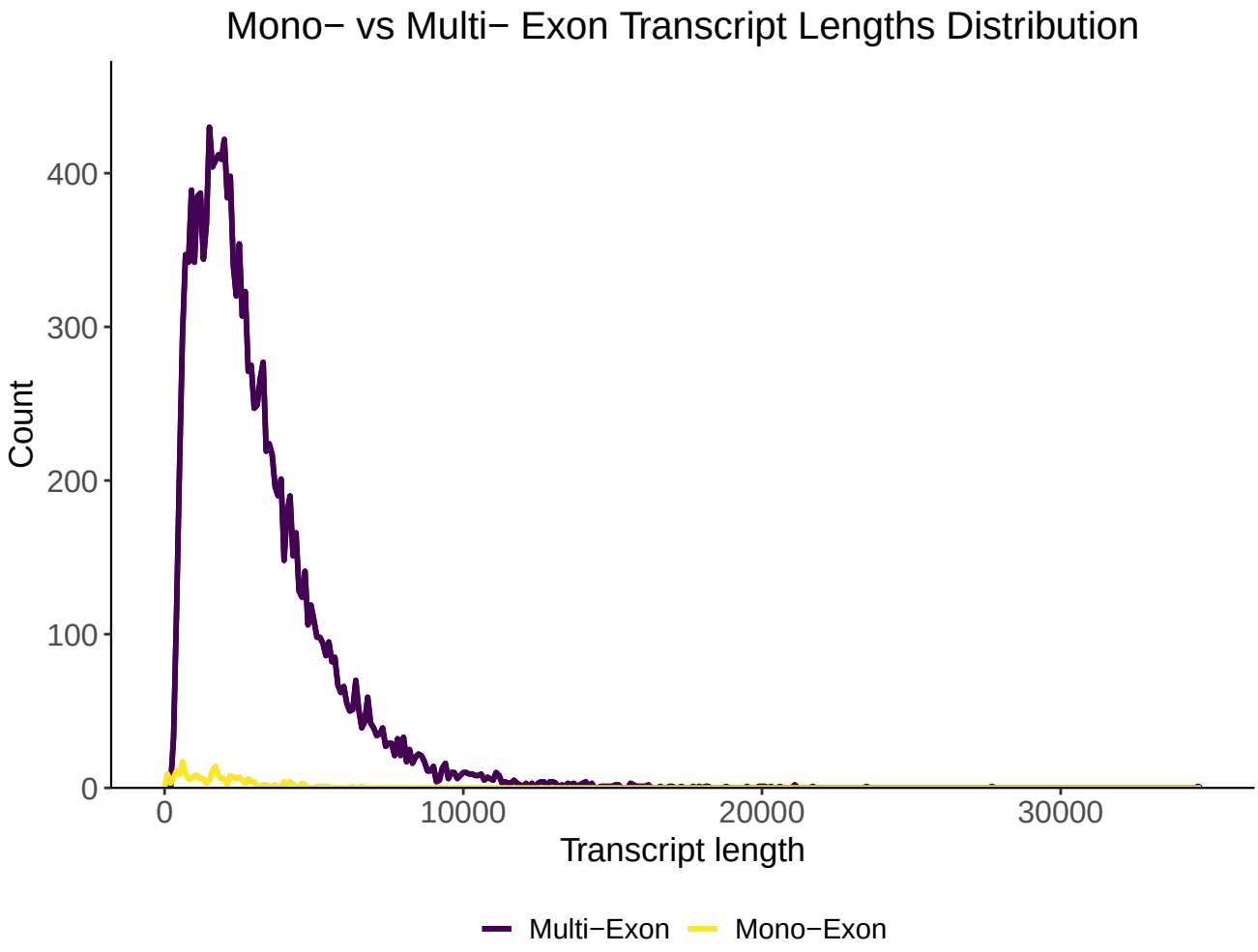

Structural Categories by Transcript Length
Counts
3000
2000
1000
0
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
23
27
34
Transcript length, kb
Structural Category
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion


Structural Categories by Transcript Length
%
25
20
15
10
5
0
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
23
27
34
Transcript length, kb
Structural Category
FSM
ISM
NIC
NNC
Genic
Genomic
Antisense
Fusion

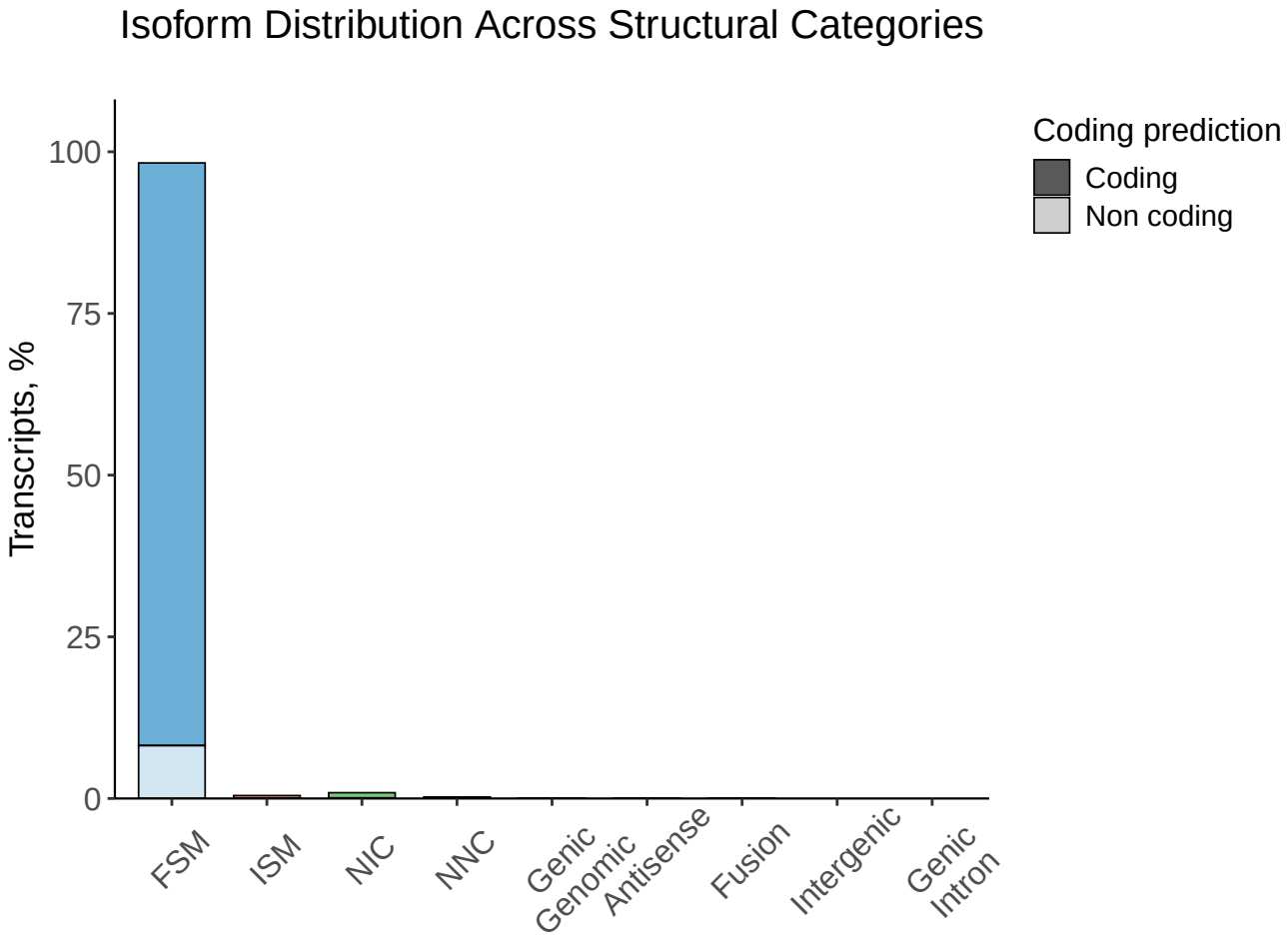

Number of FL reads per Gene by Type of Gene Annotation
log2(FL_TPM+1)
15
10
5
0
Annotated Genes
Novel Genes

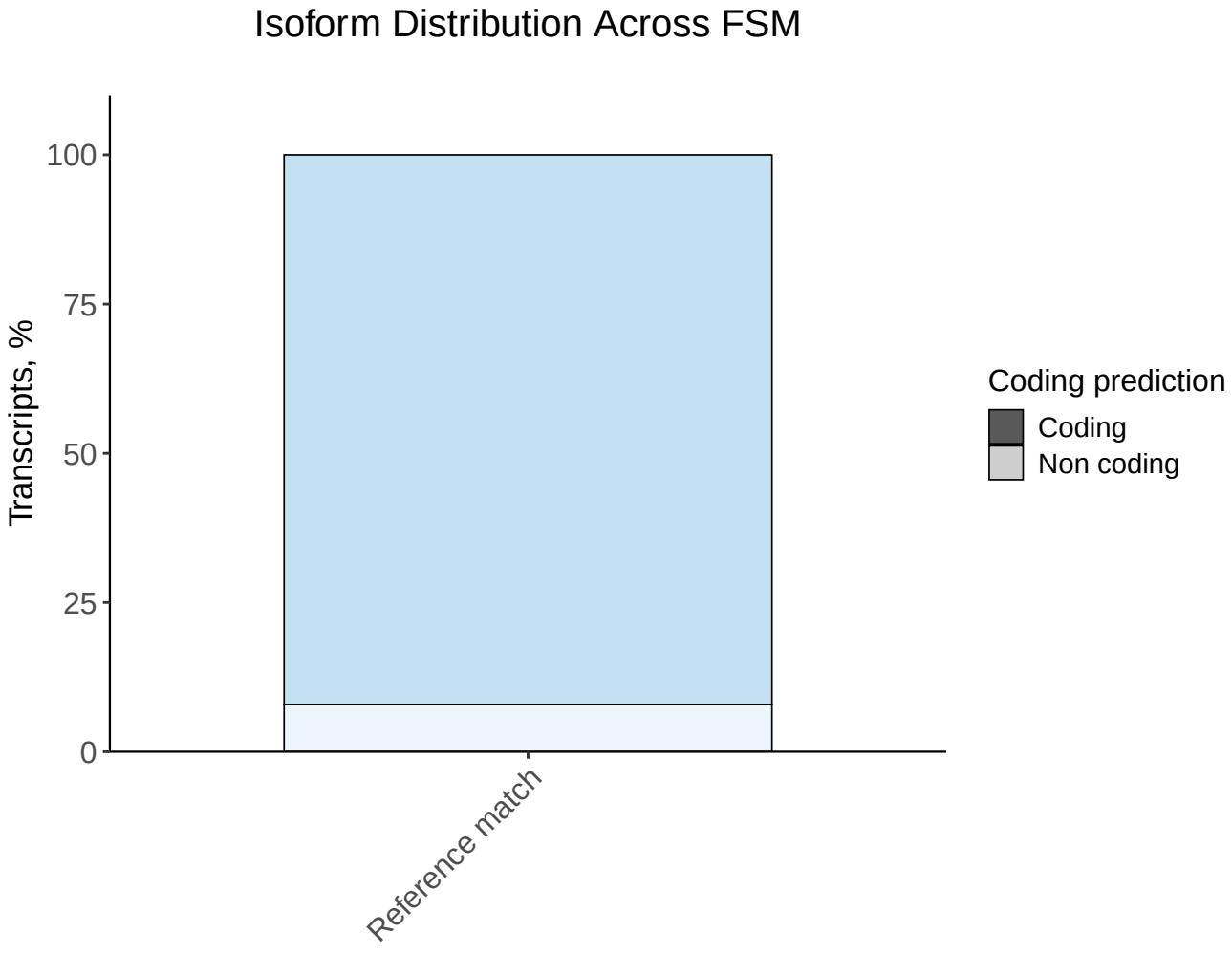

All Transcript Lengths Distribution
Count
400
300
200
100
0
0
10000
20000
30000
Transcript length

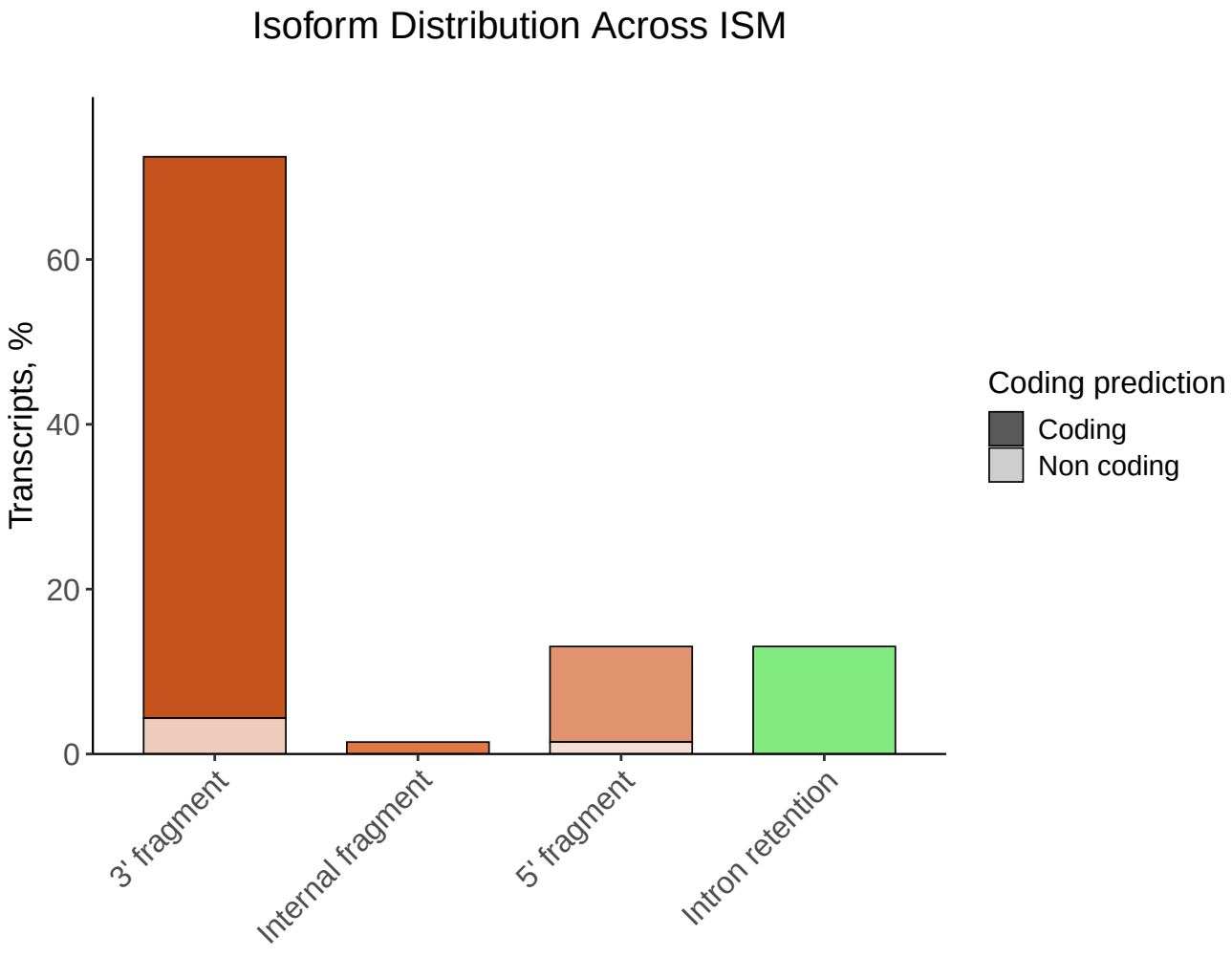

Transcript Lengths Distribution by Structural Category
Count
400
300
200
100
0
0
10000
20000
30000
Transcript length
FSM
NIC
Genic Genomic
Fusion
ISM
NNC
Antisense

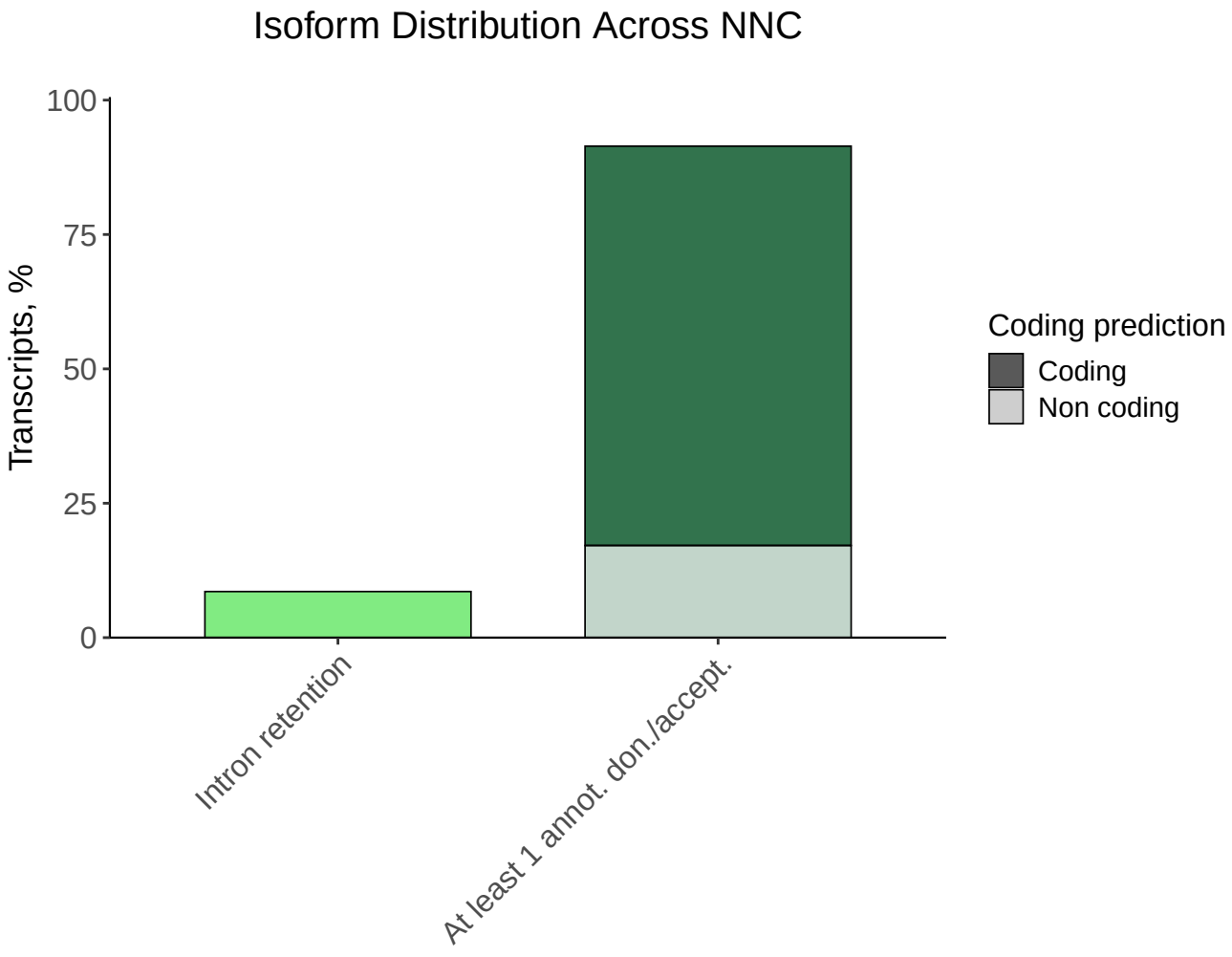

Mono− vs Multi− Exon Transcript Lengths Distribution
Count
400
300
200
100
0
0
10000
20000
30000
Transcript length
Multi−Exon
Mono−Exon

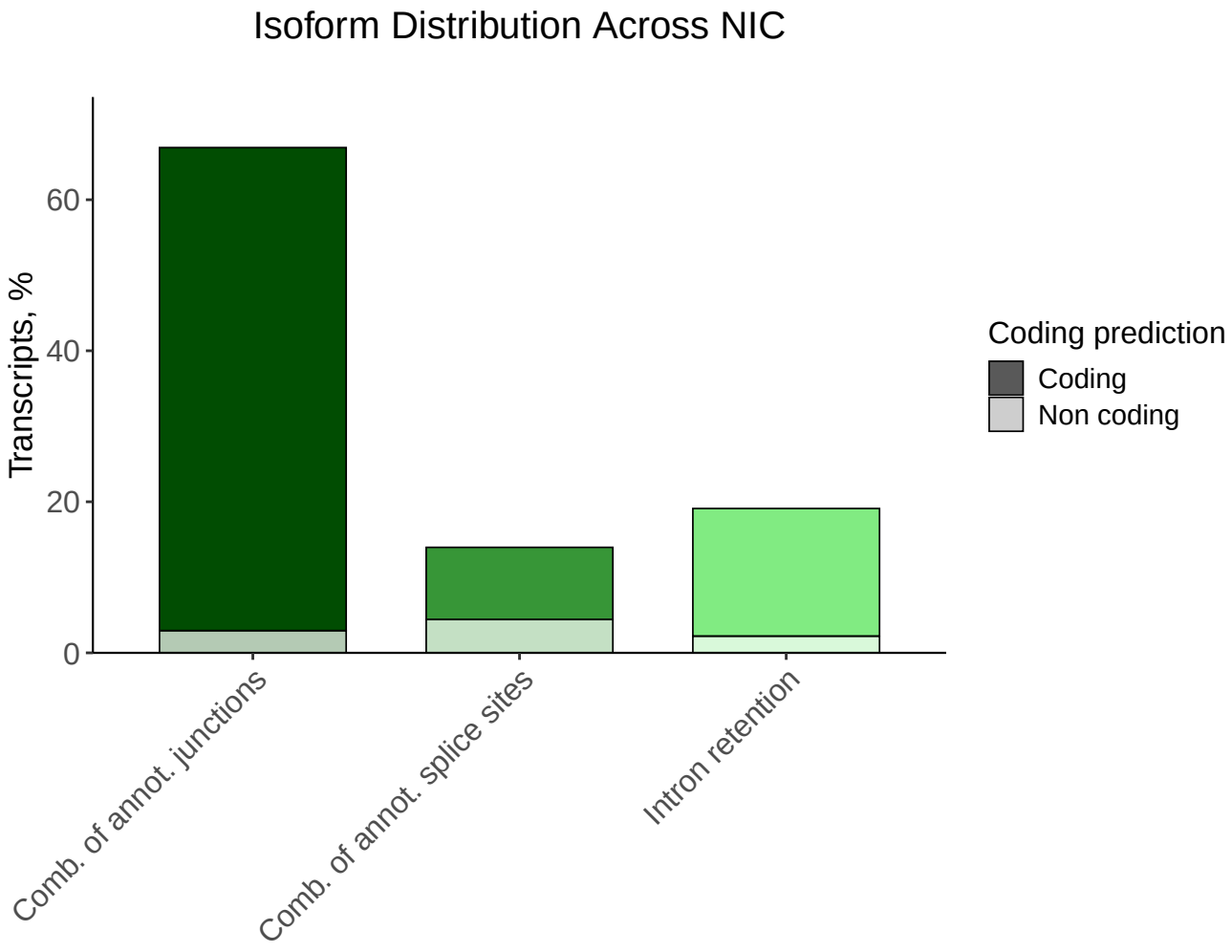
Structural Isoform Characterization

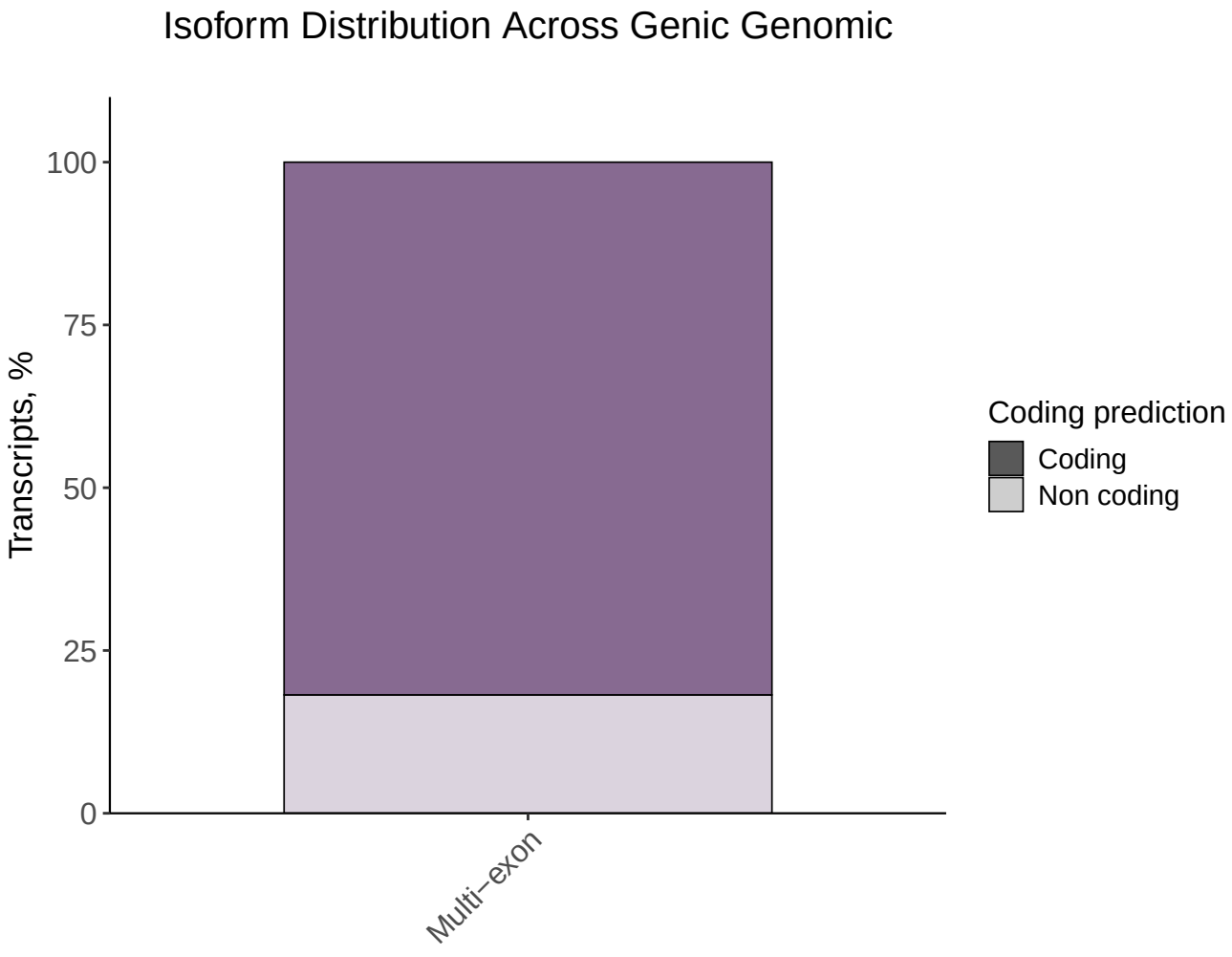

Isoform Distribution Across Structural Categories
Transcripts, %
100
75
50
25
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Intergenic
Genic Intron
Coding prediction
Coding
Non coding

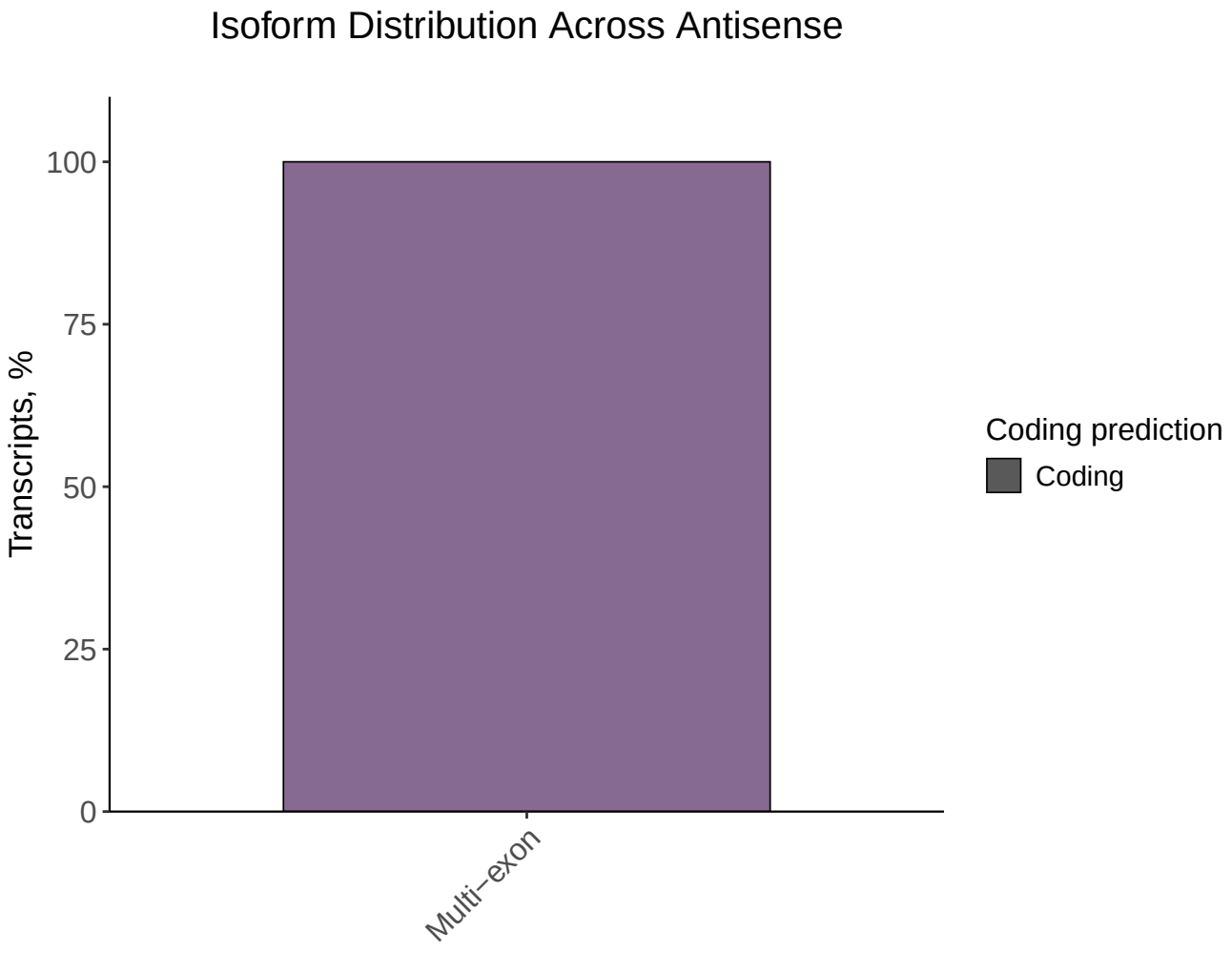

Isoform Distribution Across FSM
Transcripts, %
100
75
50
25
0
Reference match
Coding prediction
Coding
Non coding

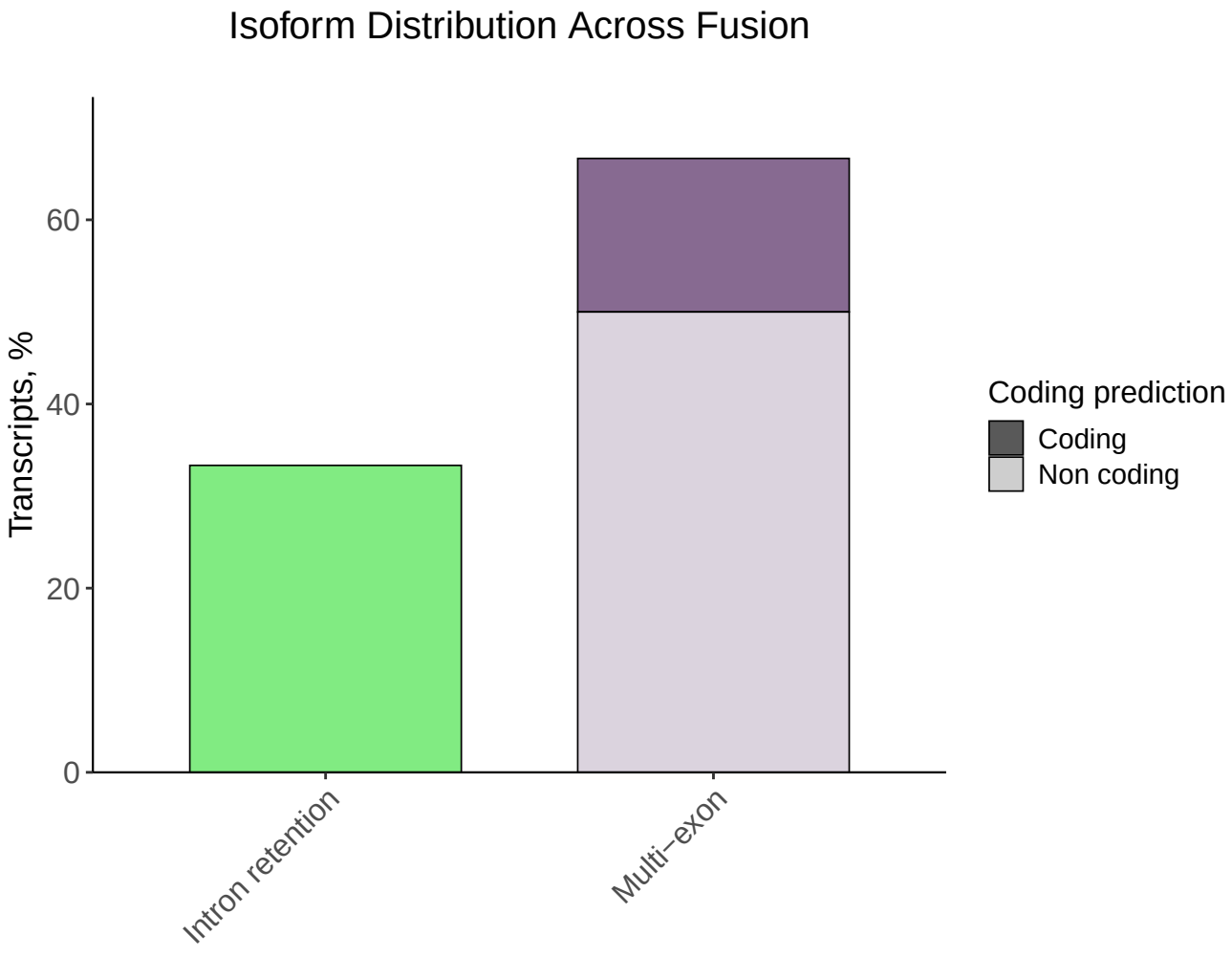

Isoform Distribution Across ISM
Transcripts, %
0
20
40
60
3' fragment
Internal fragment
5' fragment
Intron retention
Coding prediction
Coding
Non coding

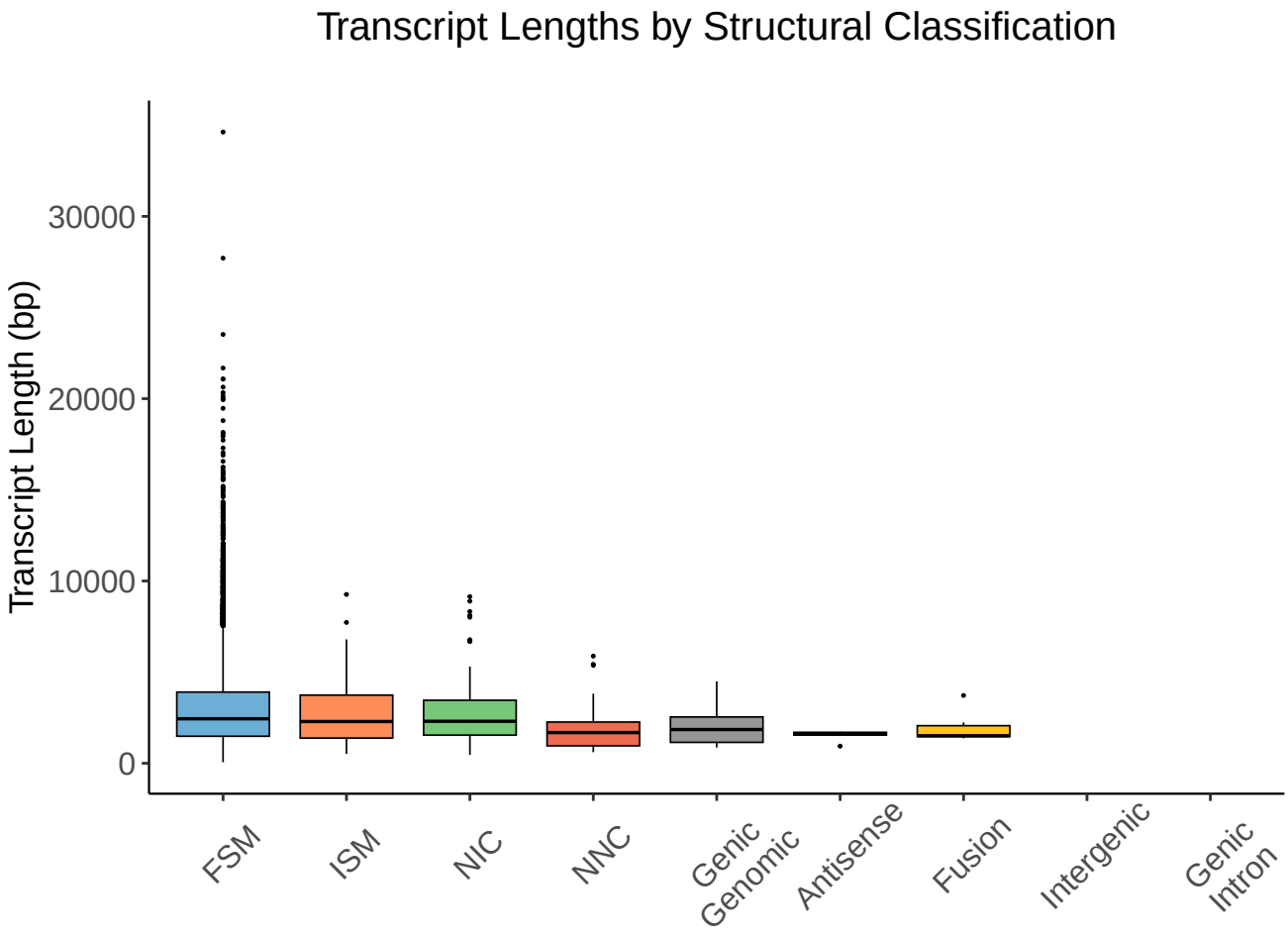

Isoform Distribution Across NNC
Transcripts, %
100
75
50
25
0
Intron retention
At least 1 annot. don./accept.
Coding prediction
Coding
Non coding

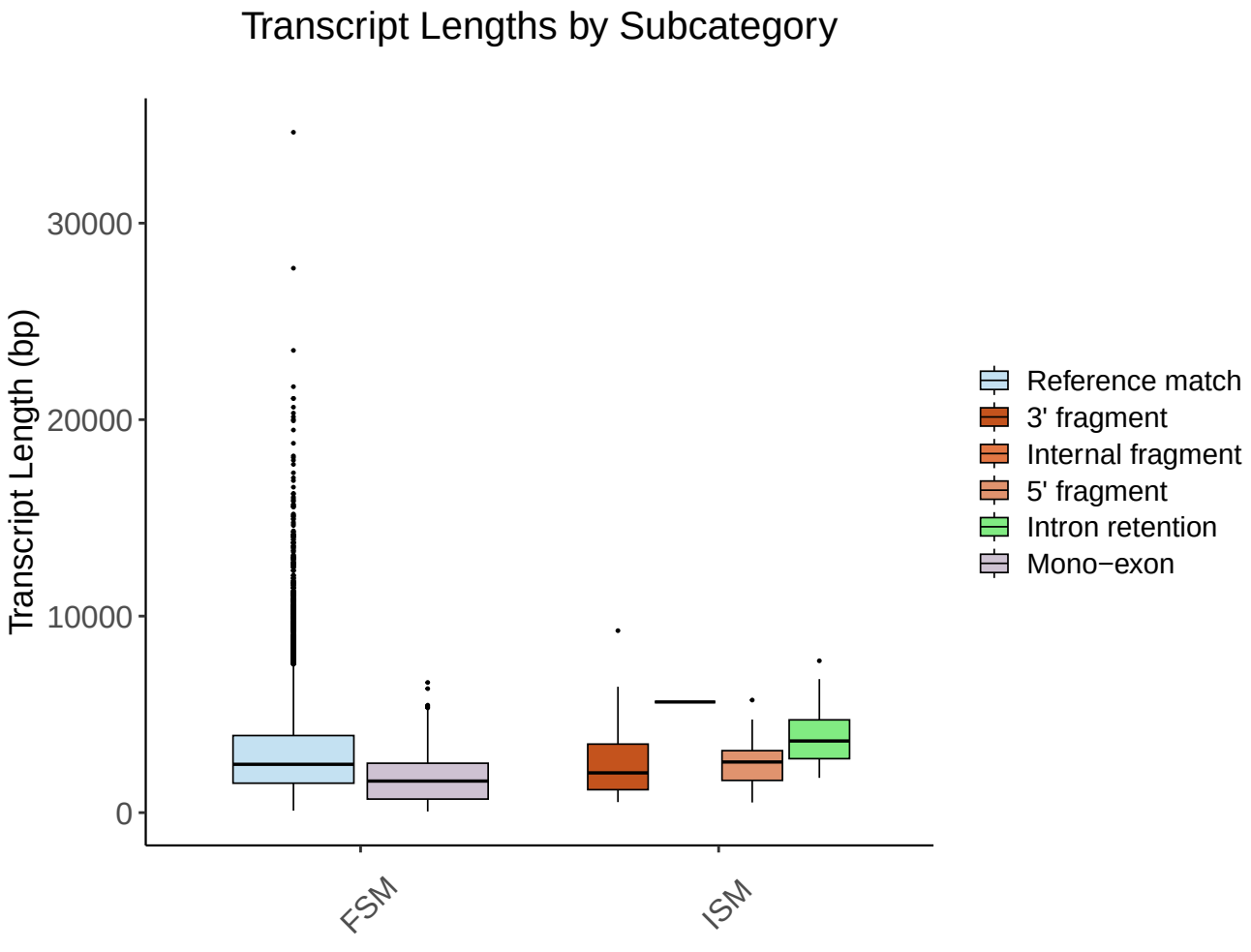

Isoform Distribution Across NIC
Transcripts, %
0
20
40
60
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
Coding prediction
Coding
Non coding

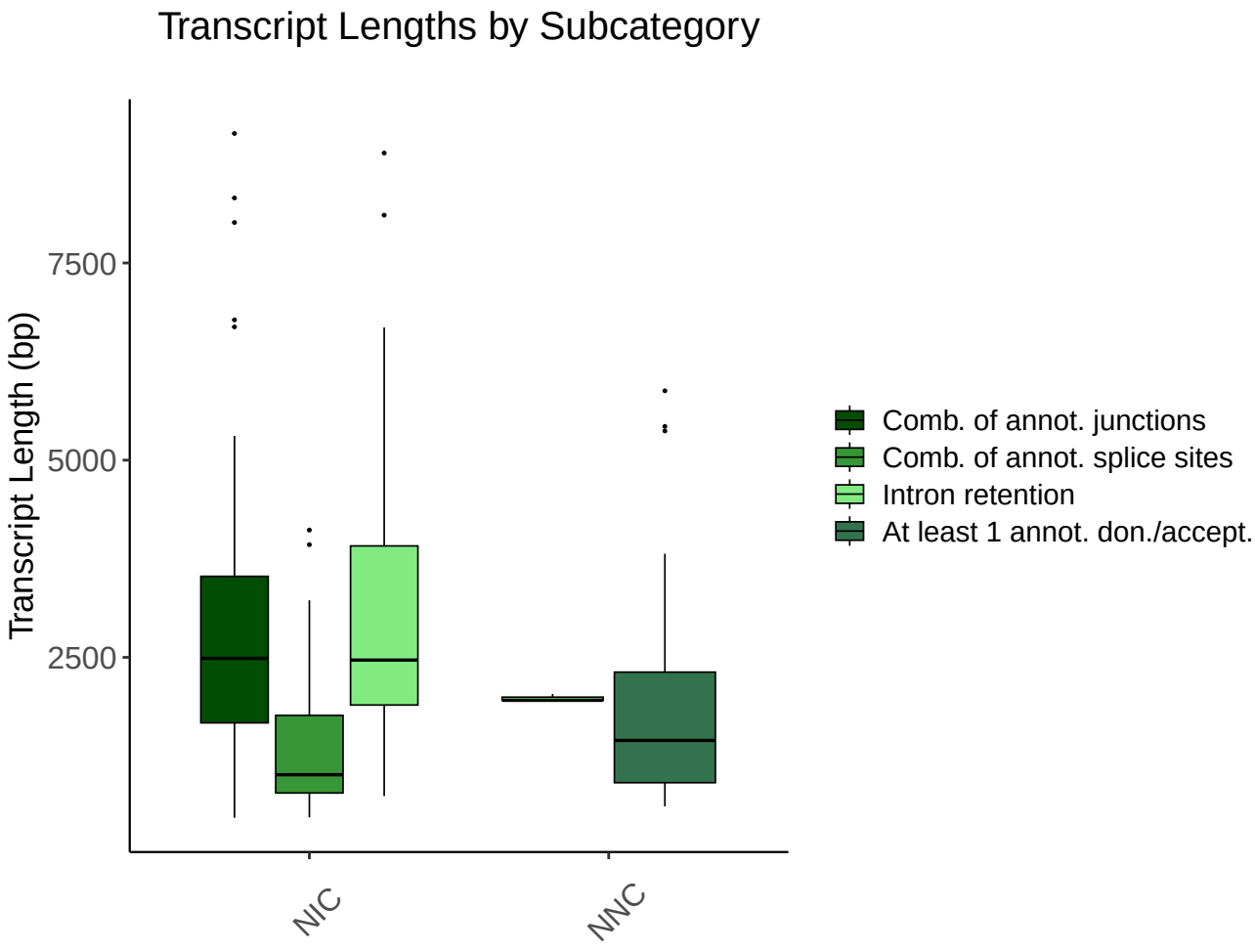

Isoform Distribution Across Genic Genomic
Transcripts, %
100
75
50
25
0
Multi−exon
Coding prediction
Coding
Non coding

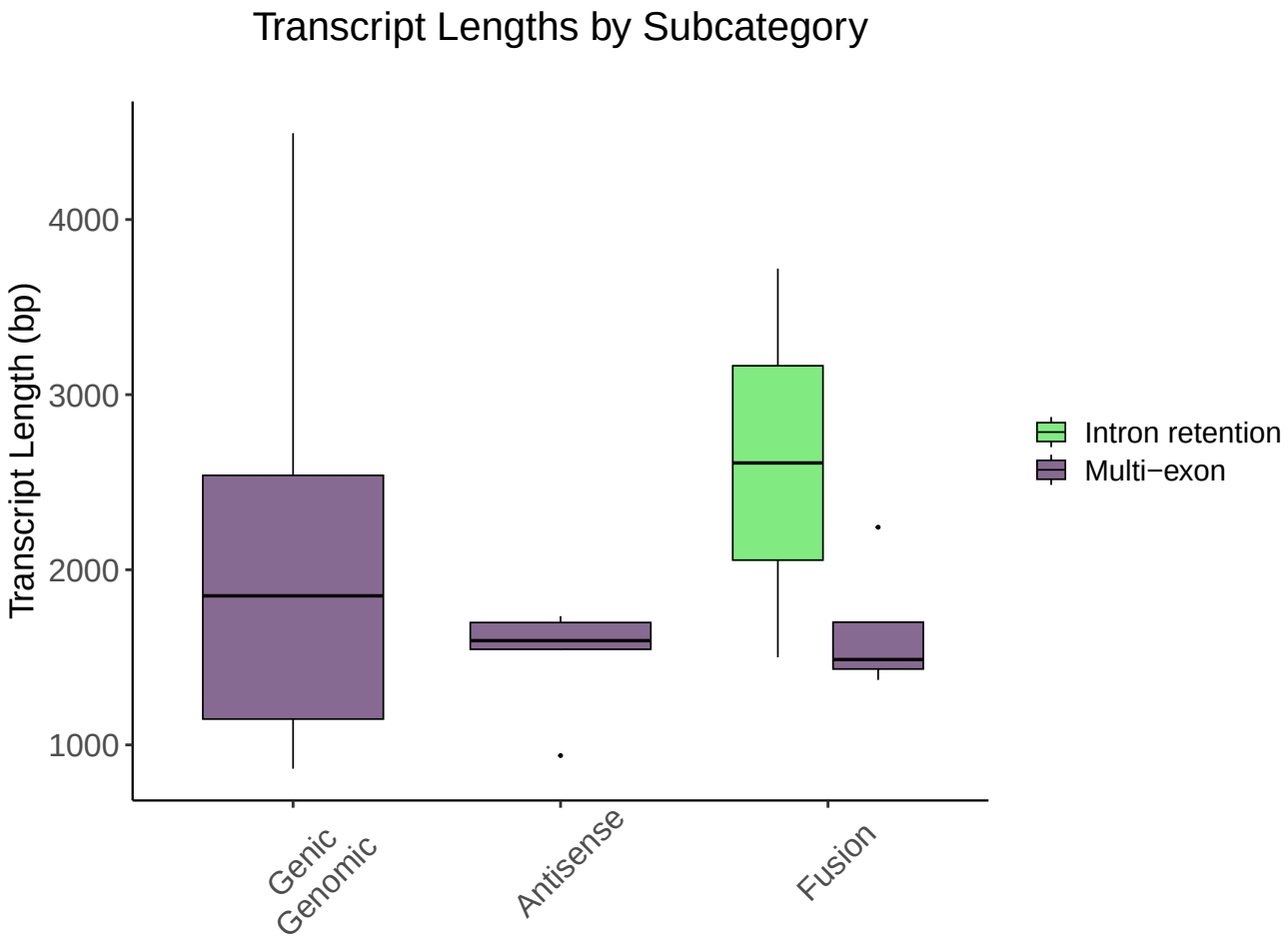

Isoform Distribution Across Antisense
Transcripts, %
100
75
50
25
0
Multi−exon
Coding prediction
Coding

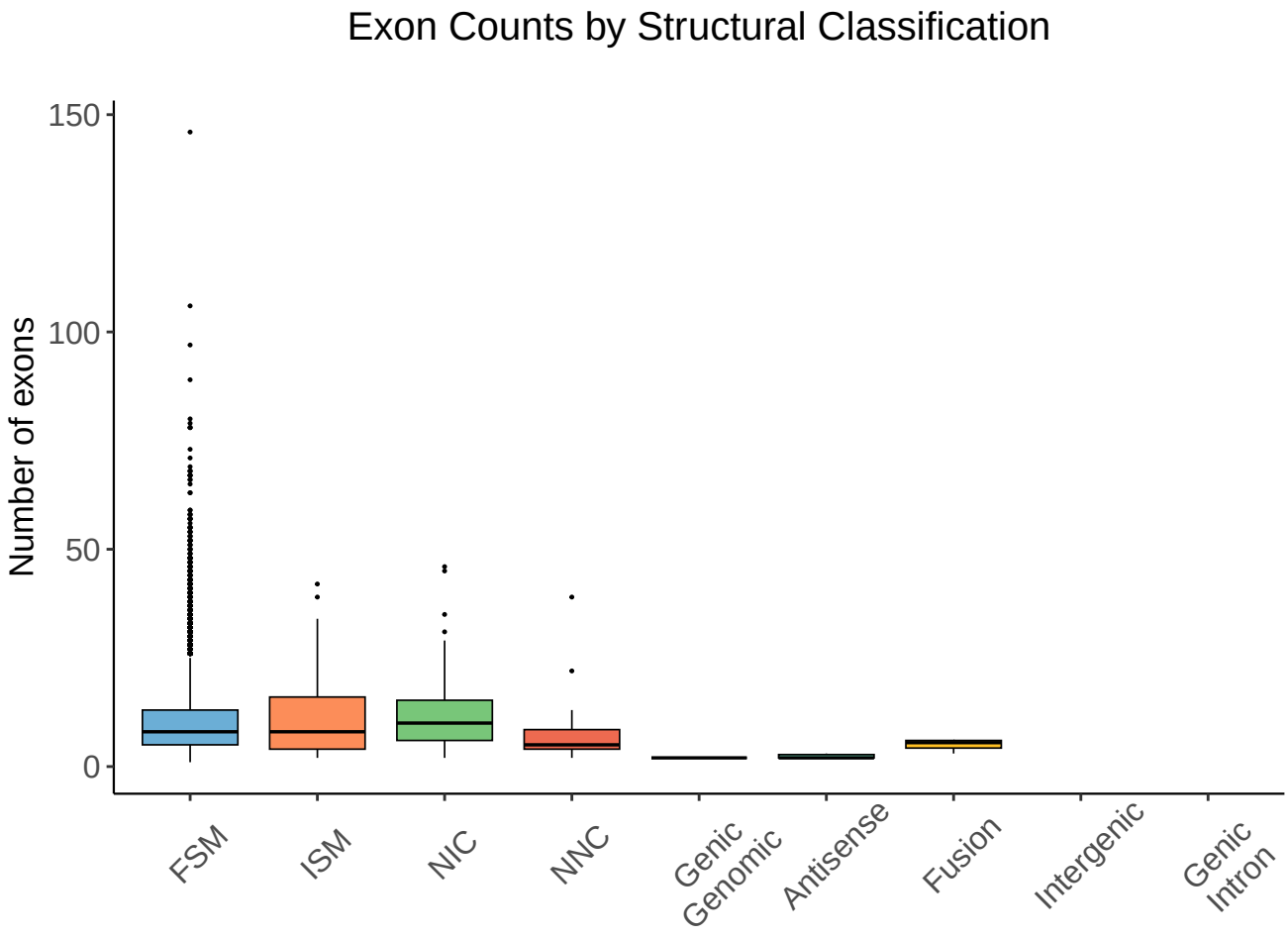

Isoform Distribution Across Fusion
Transcripts, %
0
20
40
60
Intron retention
Multi−exon
Coding prediction
Coding
Non coding

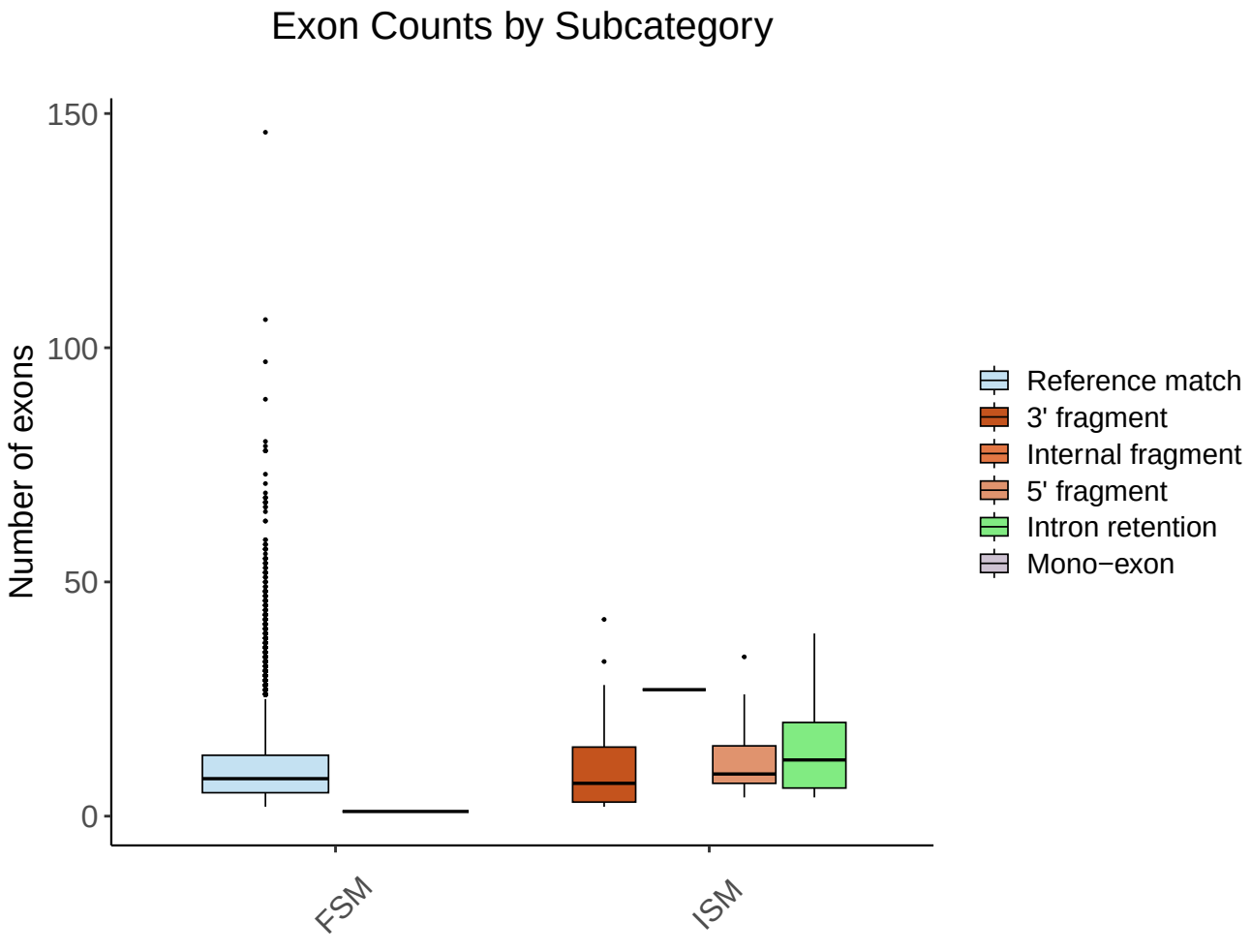

Transcript Lengths by Structural Classification
Transcript Length (bp)
30000
20000
10000
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Intergenic
Genic Intron

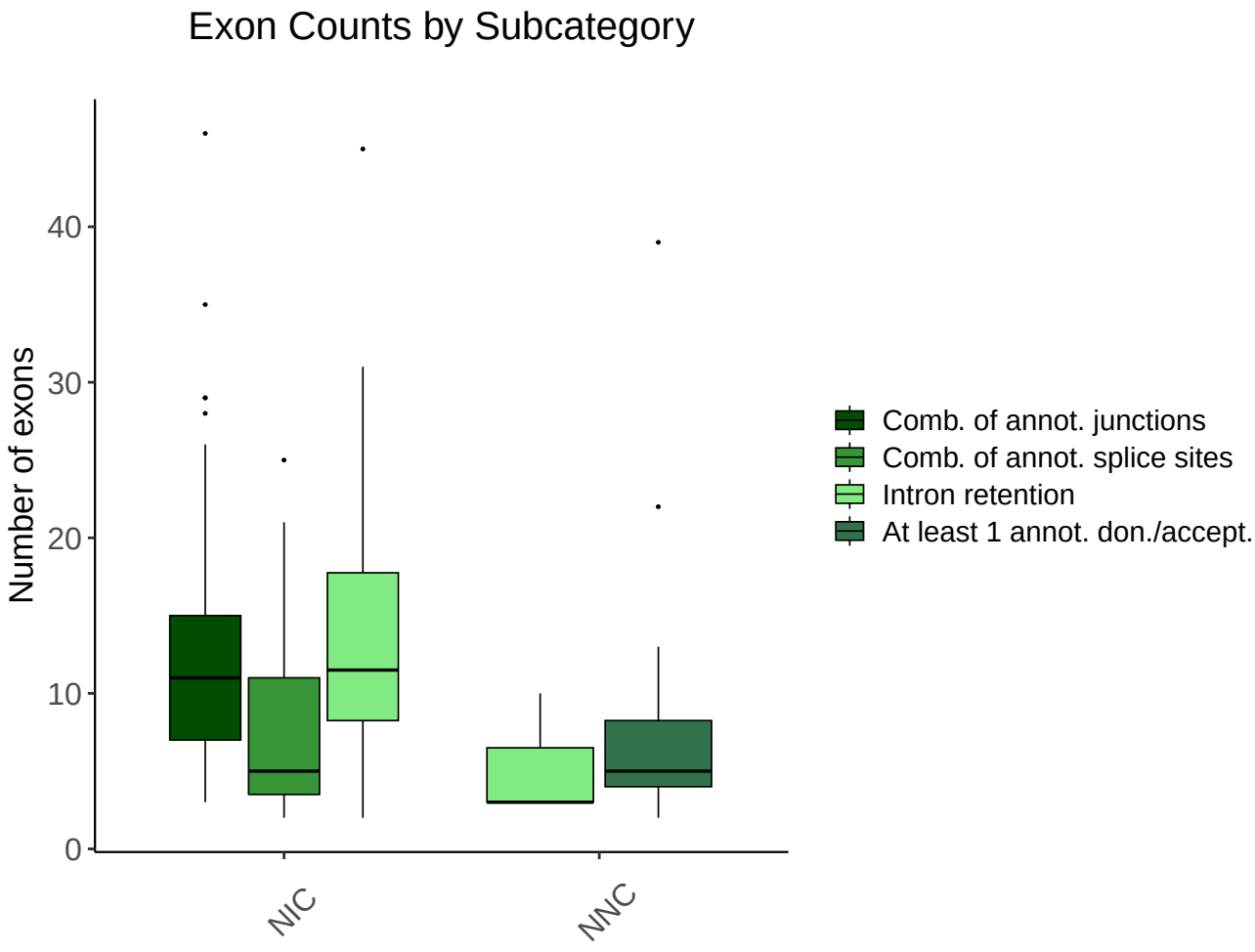

Transcript Lengths by Subcategory
Transcript Length (bp)
30000
20000
10000
0
FSM
ISM
Reference match
3' fragment
Internal fragment
5' fragment
Intron retention
Mono−exon

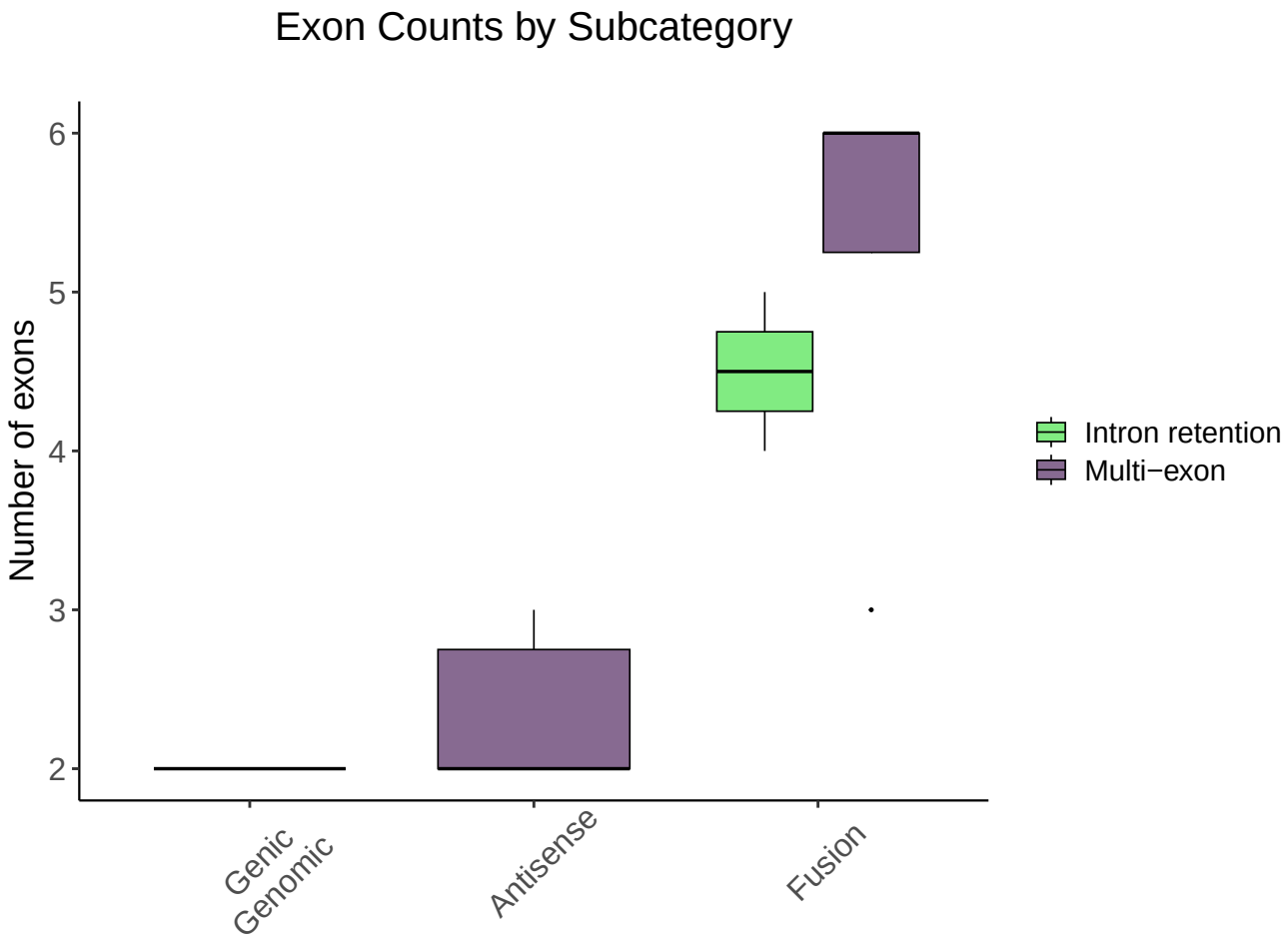

Transcript Lengths by Subcategory
Transcript Length (bp)
7500
5000
2500
NIC
NNC
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
At least 1 annot. don./accept.

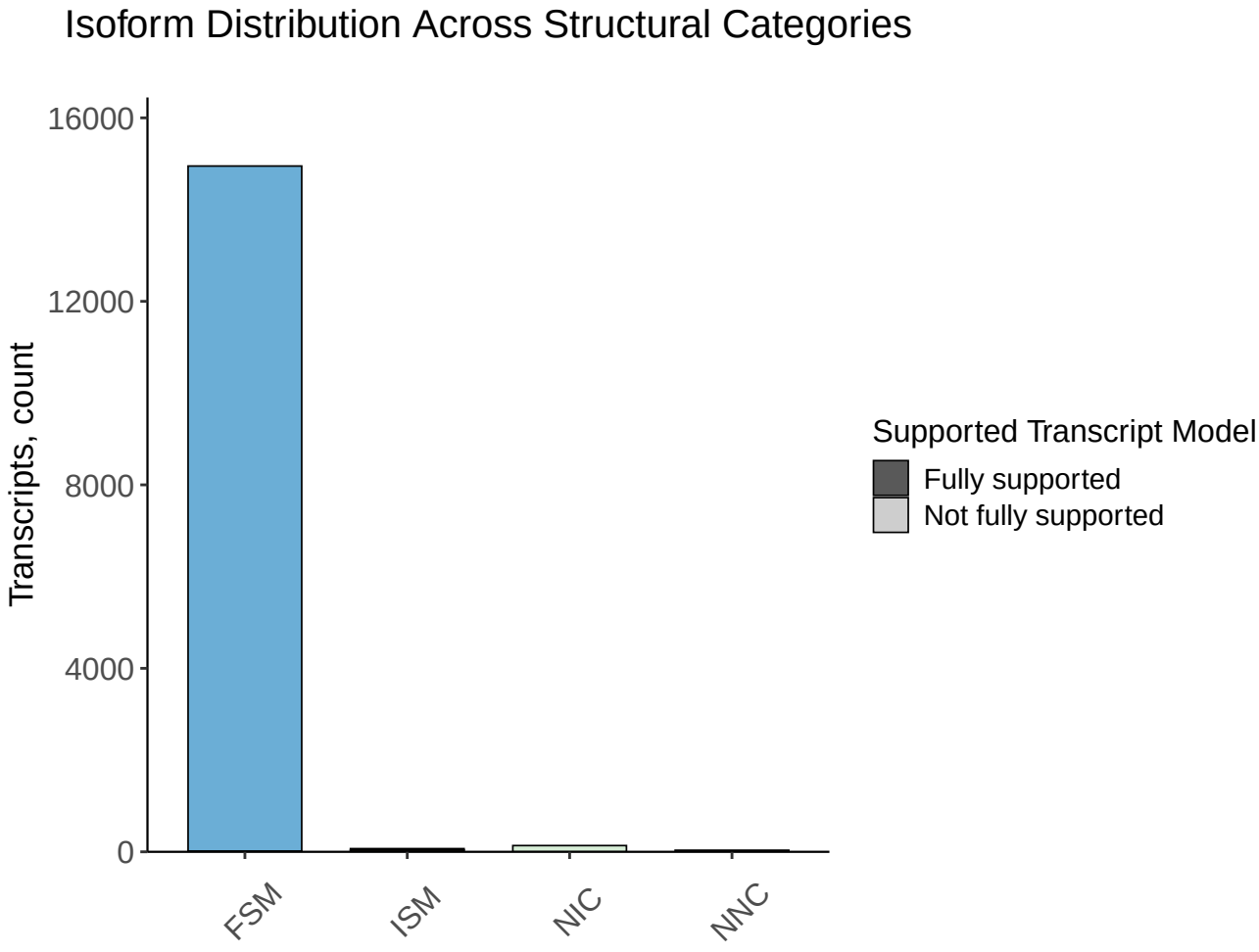

Transcript Lengths by Subcategory
Transcript Length (bp)
4000
3000
2000
1000
Genic Genomic
Antisense
Fusion
Intron retention
Multi−exon

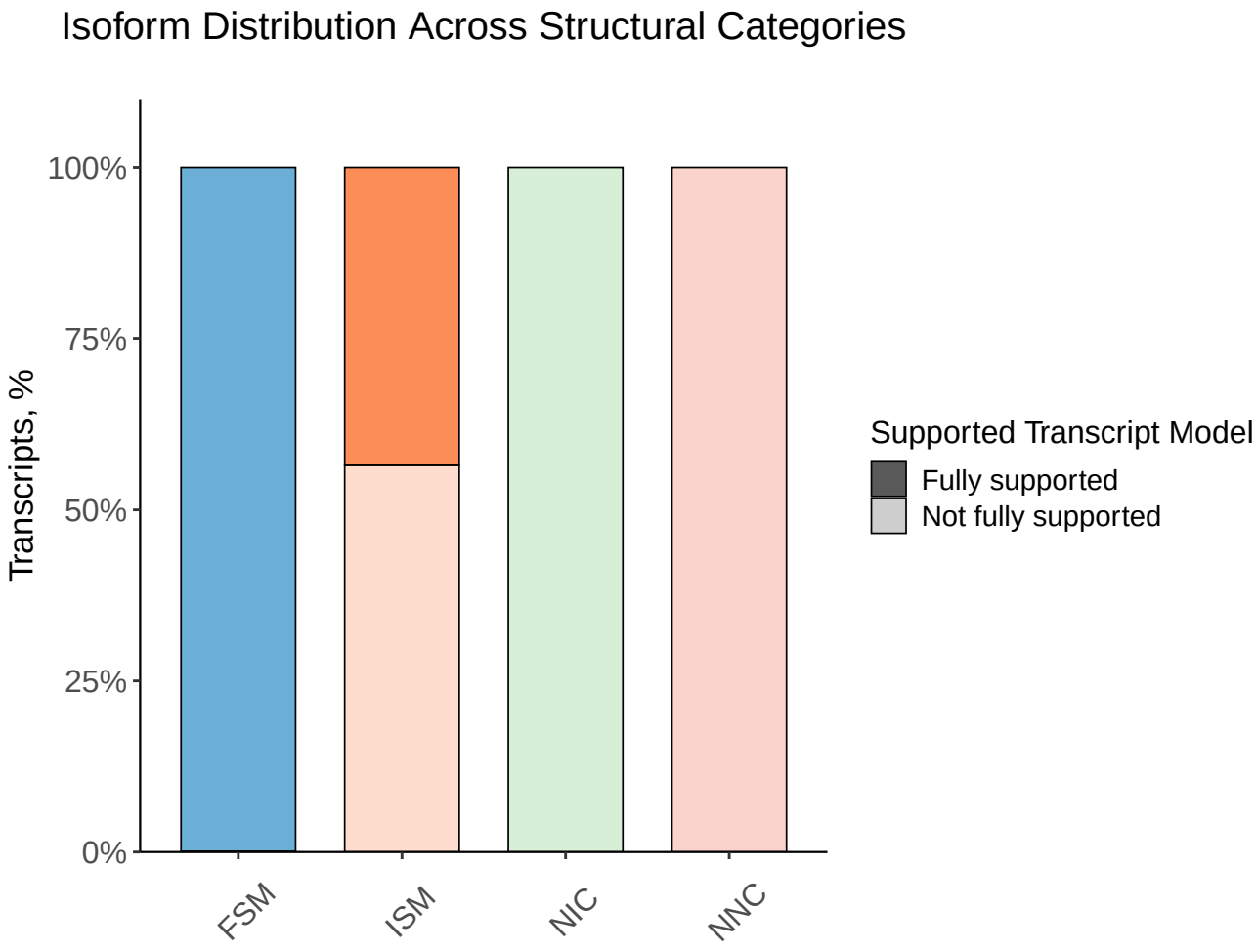

Exon Counts by Structural Classification
Number of exons
150
100
50
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Intergenic
Genic Intron

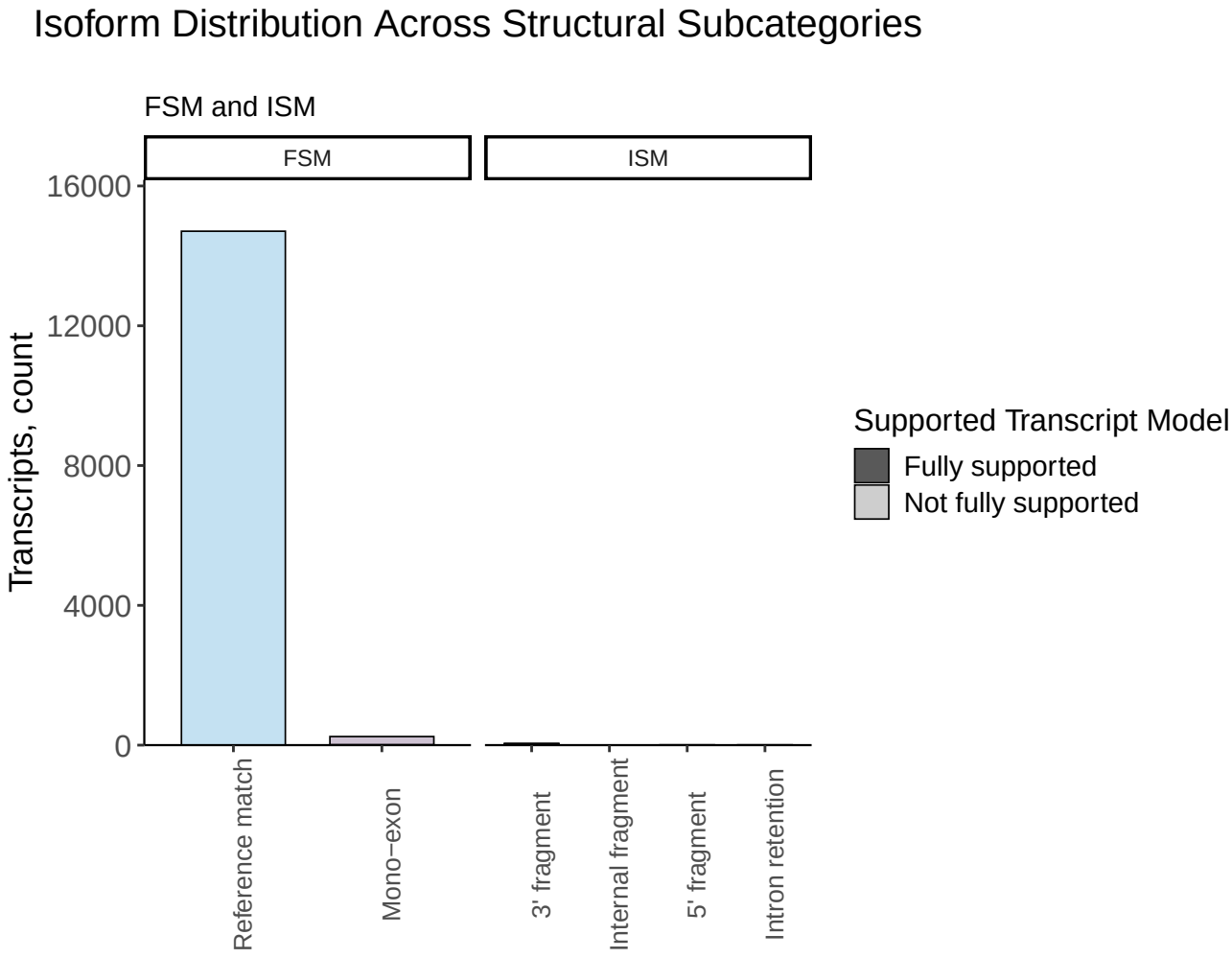

Exon Counts by Subcategory
Number of exons
150
100
50
0
FSM
ISM
Reference match
3' fragment
Internal fragment
5' fragment
Intron retention
Mono−exon


Exon Counts by Subcategory
Number of exons
0
10
20
30
40
NIC
NNC
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
At least 1 annot. don./accept.

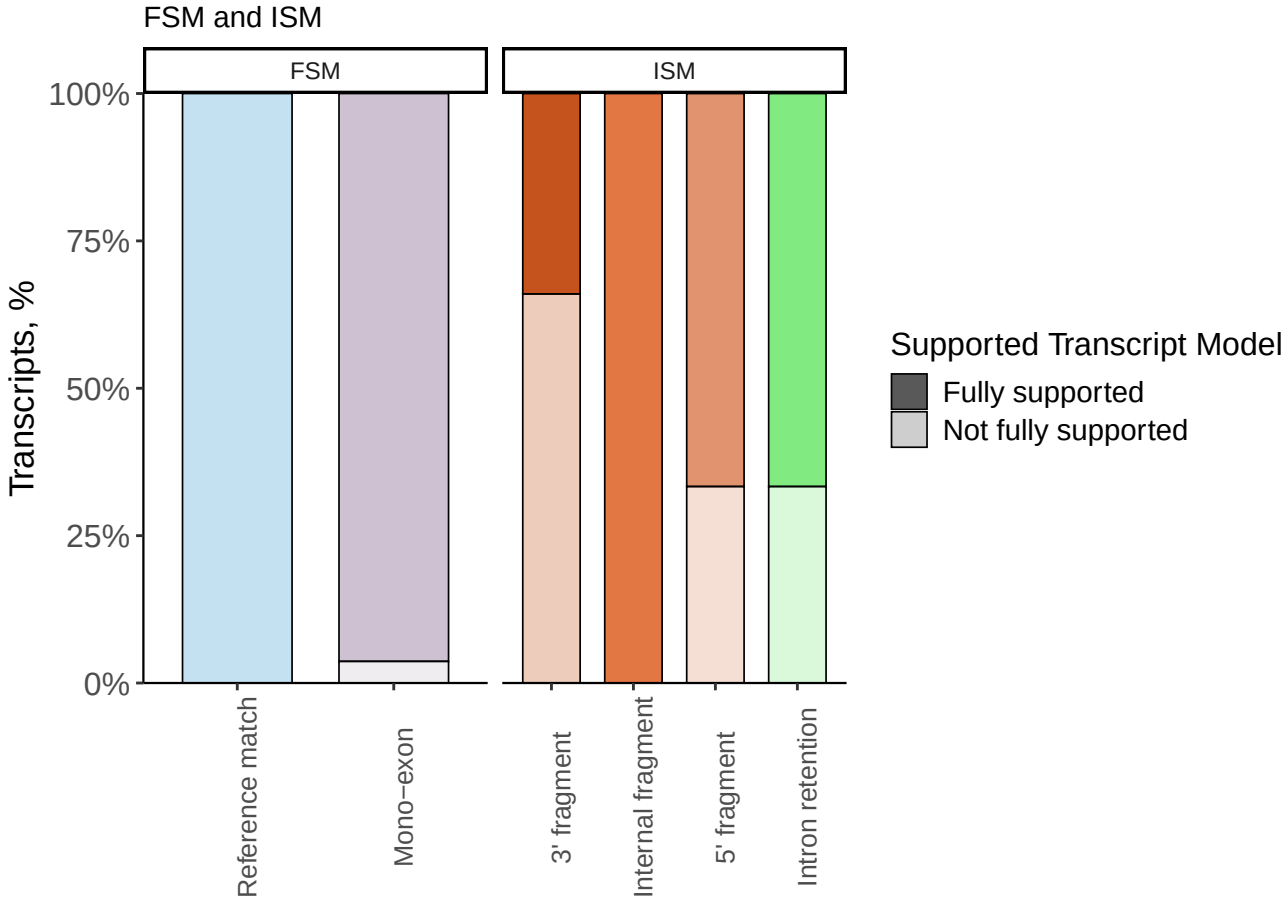

Exon Counts by Subcategory
Number of exons
6
5
4
3
2
Genic Genomic
Antisense
Fusion
Intron retention
Multi−exon

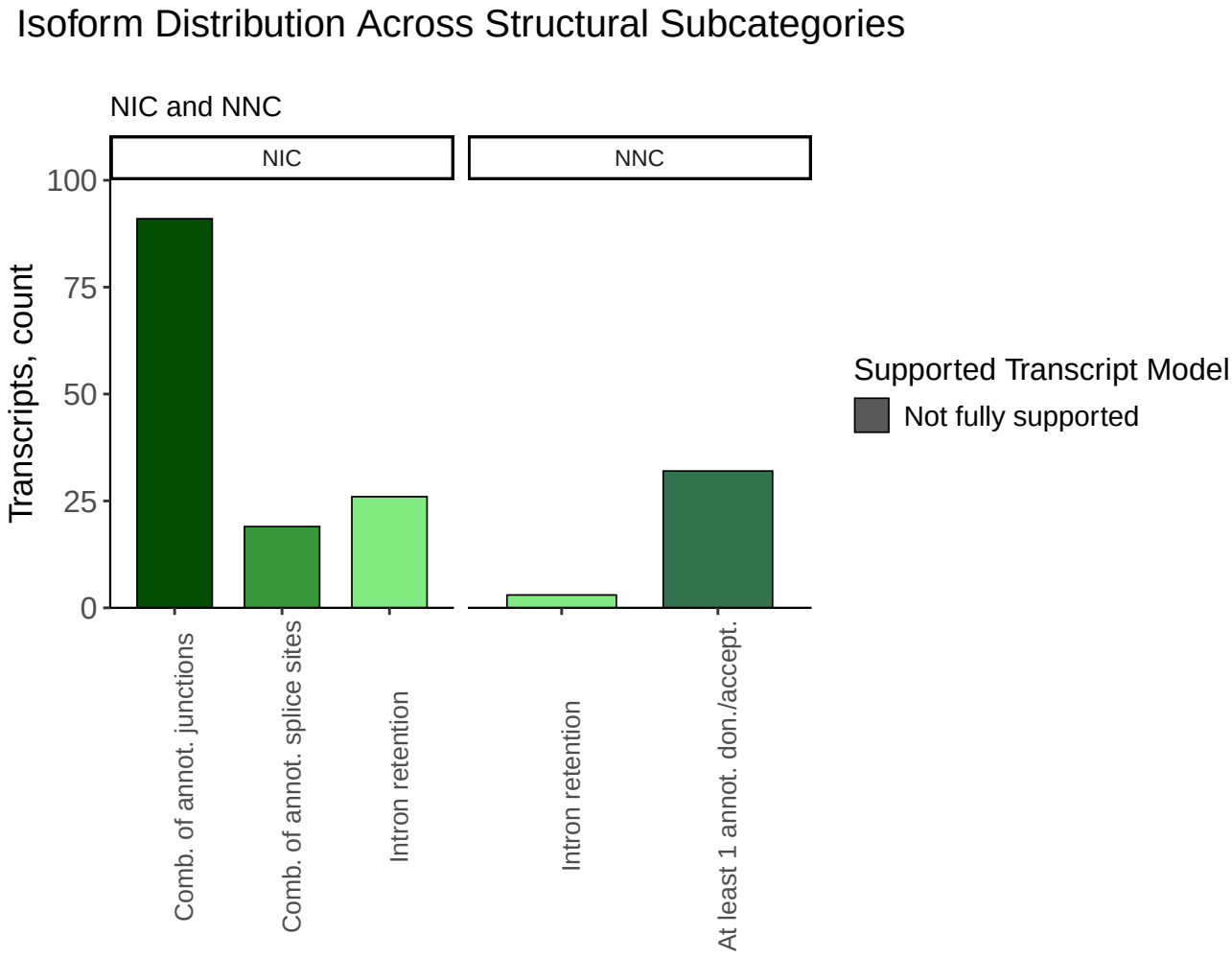

Isoform Distribution Across Structural Categories
Transcripts, count
16000
12000
8000
4000
0
FSM
ISM
NIC
NNC
Supported Transcript Model
Fully supported
Not fully supported

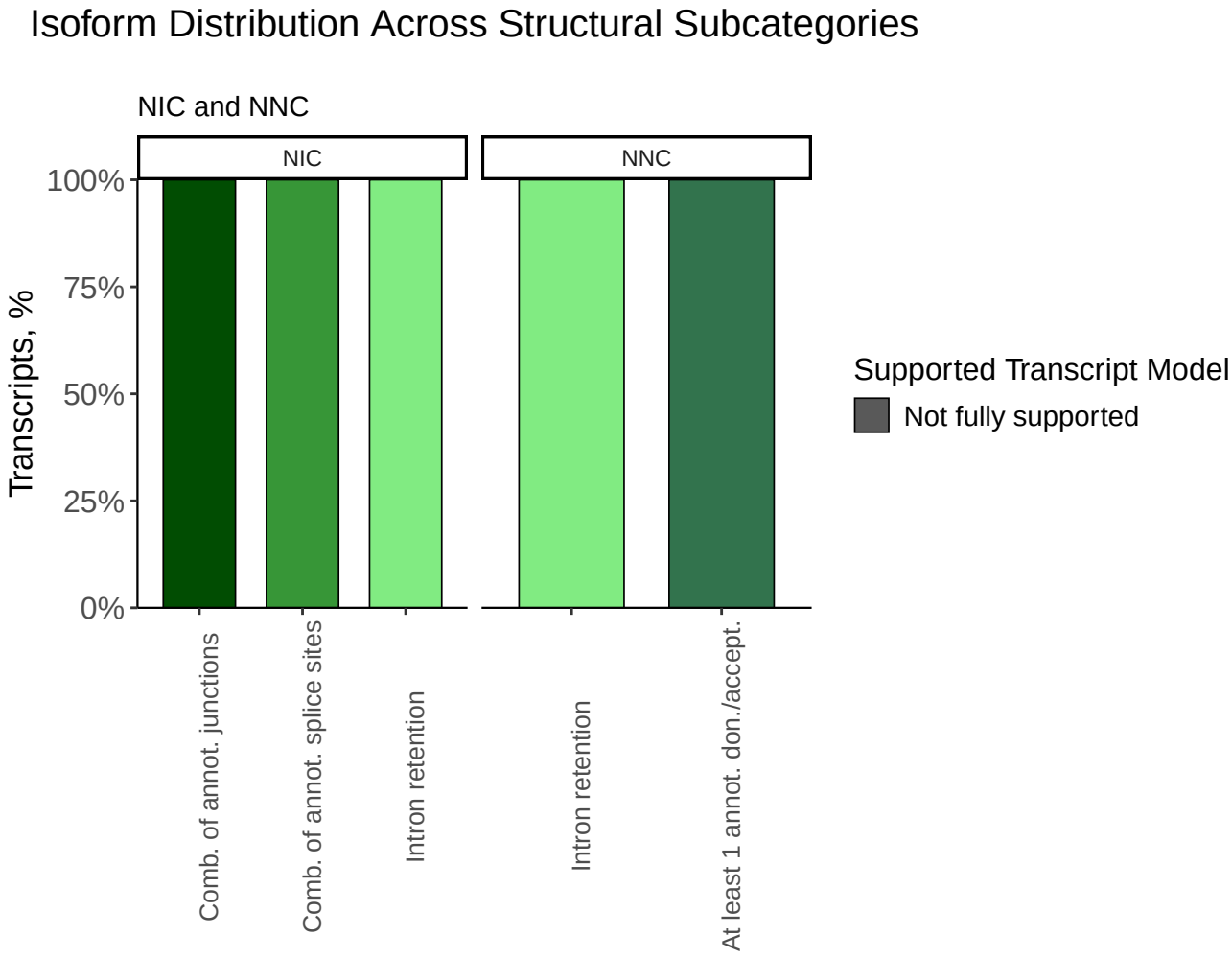

Isoform Distribution Across Structural Categories
Transcripts, %
100%
75%
50%
25%
0%
FSM
ISM
NIC
NNC
Supported Transcript Model
Fully supported
Not fully supported

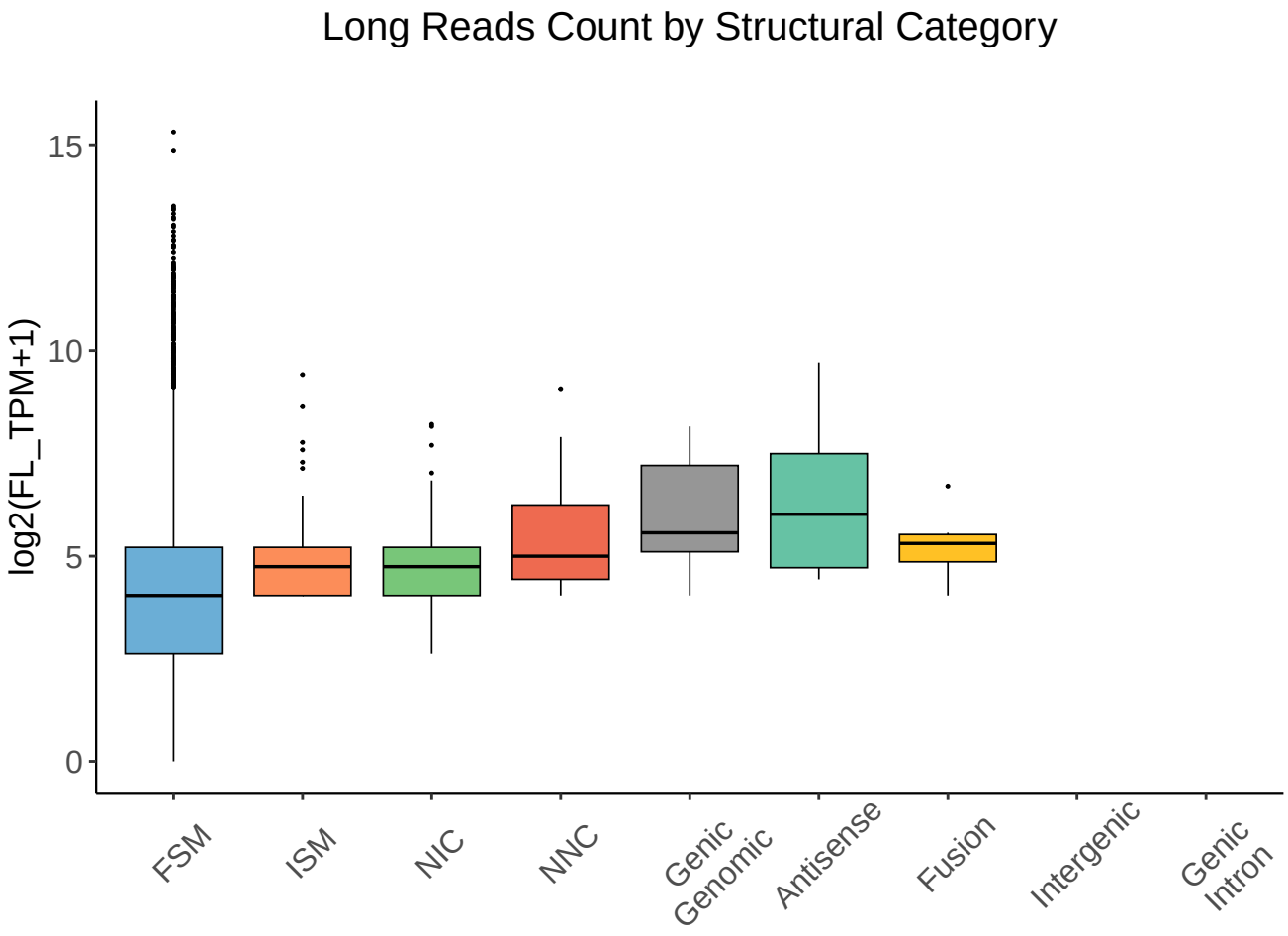

Isoform Distribution Across Structural Subcategories
FSM and ISM
FSM
ISM
Transcripts, count
16000
12000
8000
4000
0
Reference match
Mono−exon
3' fragment
Internal fragment
5' fragment
Intron retention
Supported Transcript Model
Fully supported
Not fully supported

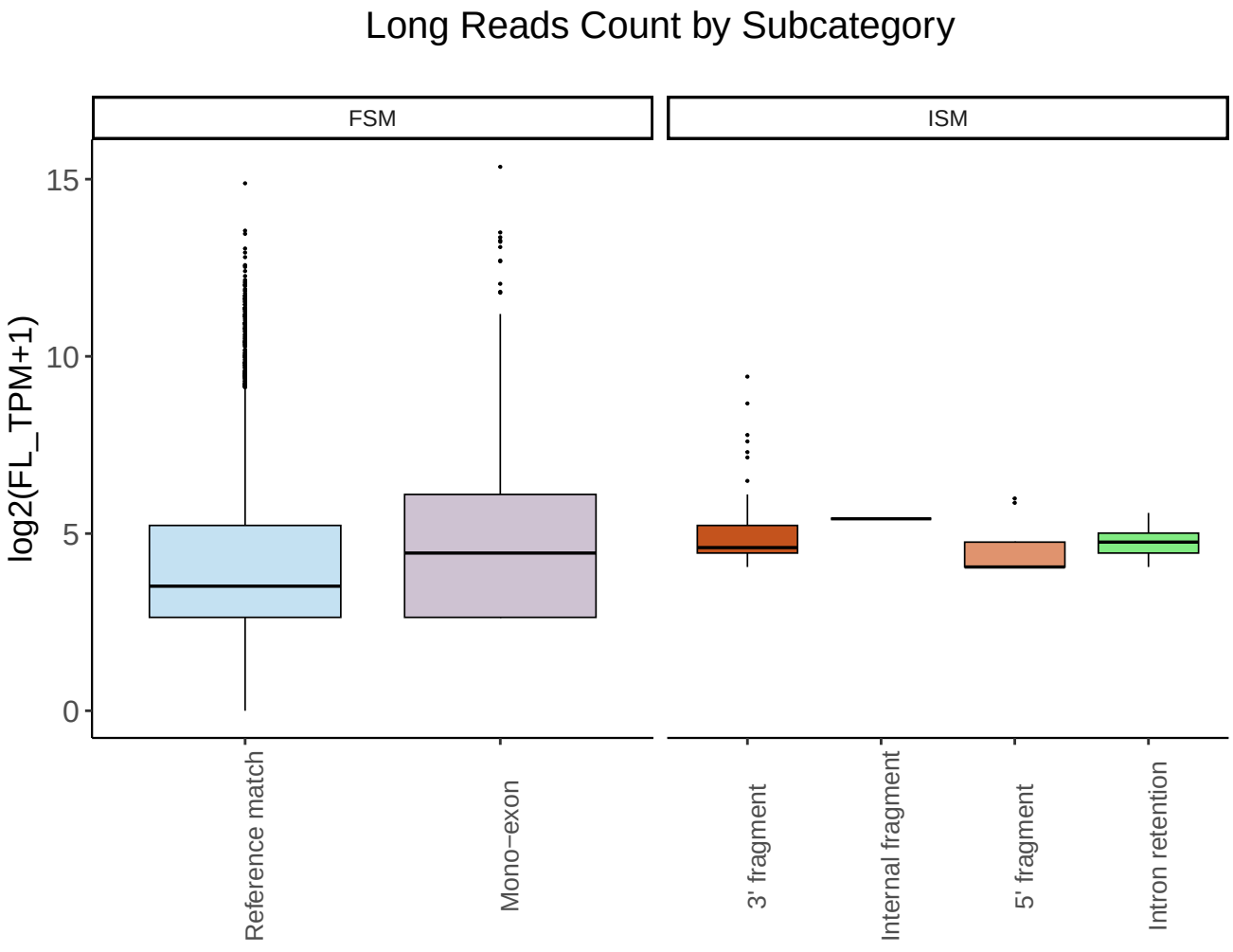
Isoform Distribution Across Structural Subcategories

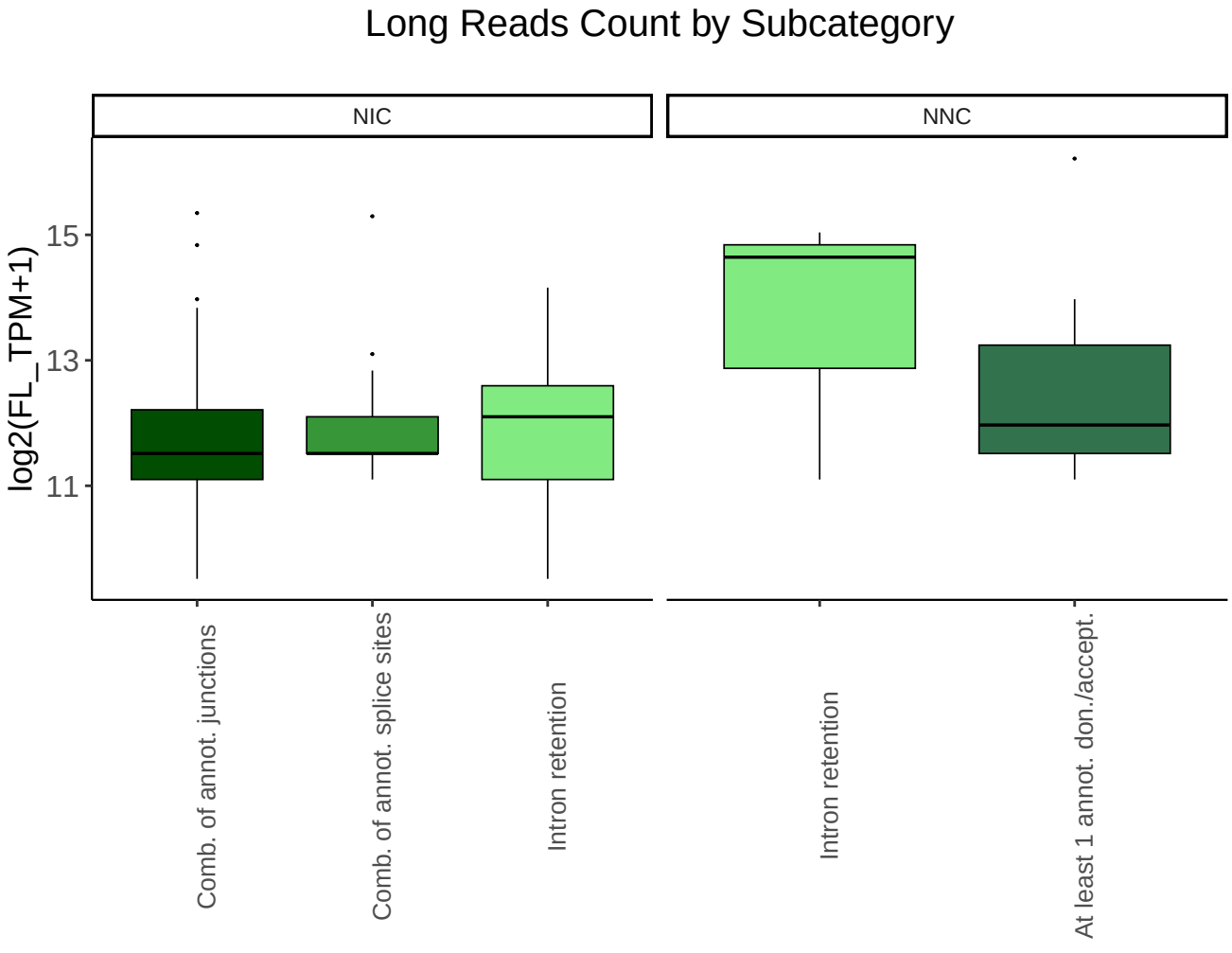


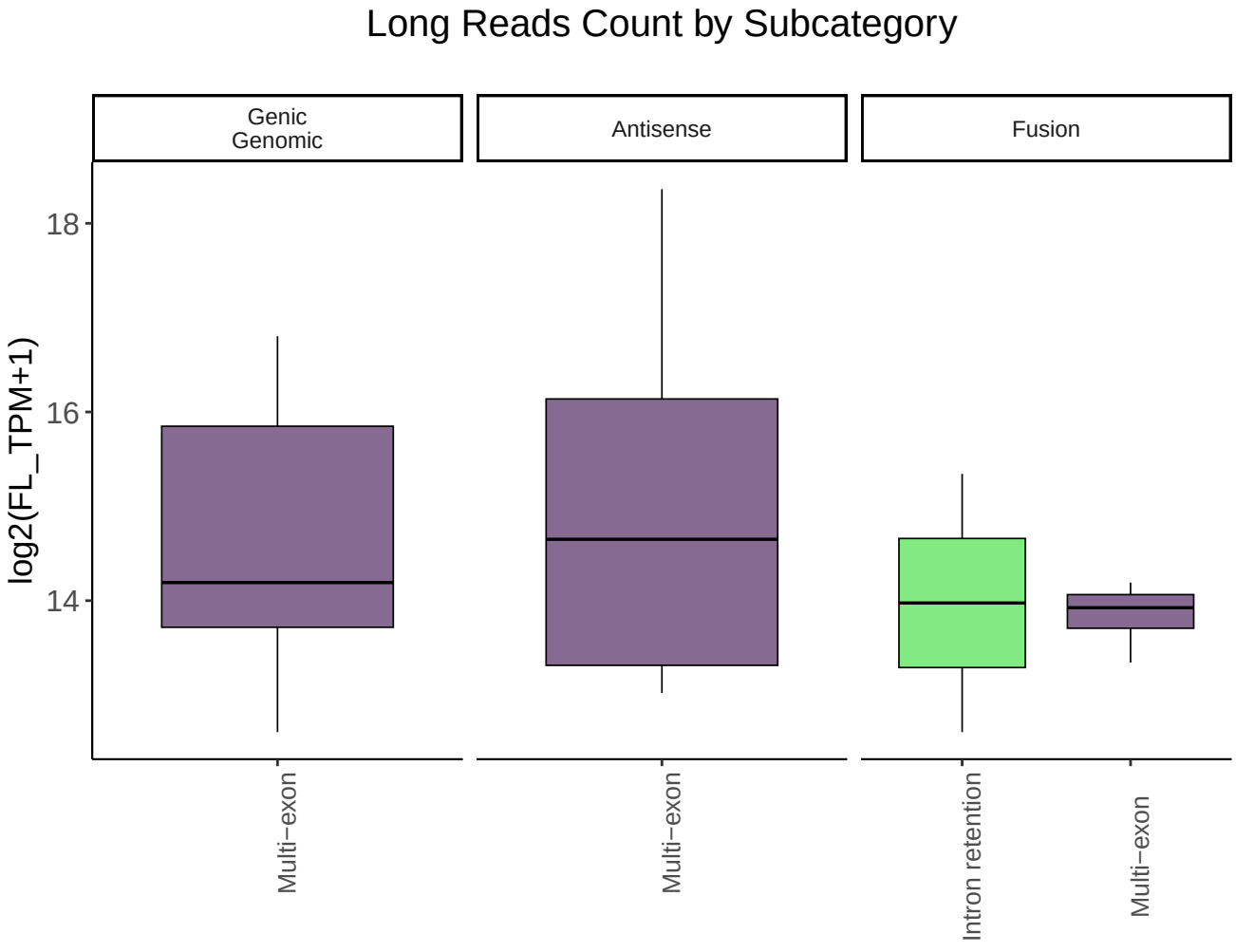

Isoform Distribution Across Structural Subcategories
NIC and NNC
NIC
NNC
Transcripts, count
100
75
50
25
0
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
Intron retention
At least 1 annot. don./accept.
Supported Transcript Model
Not fully supported

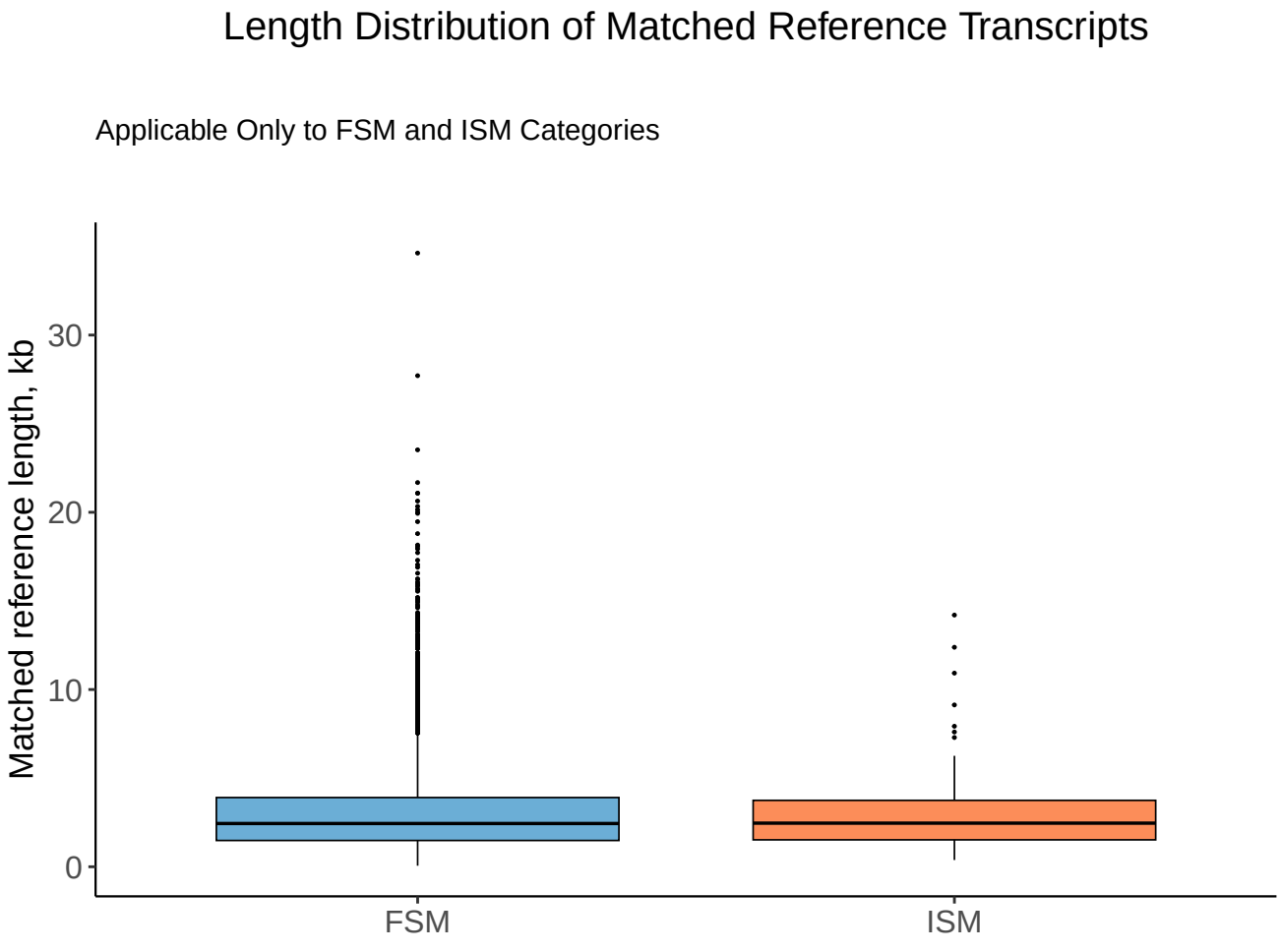

Isoform Distribution Across Structural Subcategories
NIC and NNC
NIC
NNC
Transcripts, %
100%
75%
50%
25%
0%
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
Intron retention
At least 1 annot. don./accept.
Supported Transcript Model
Not fully supported

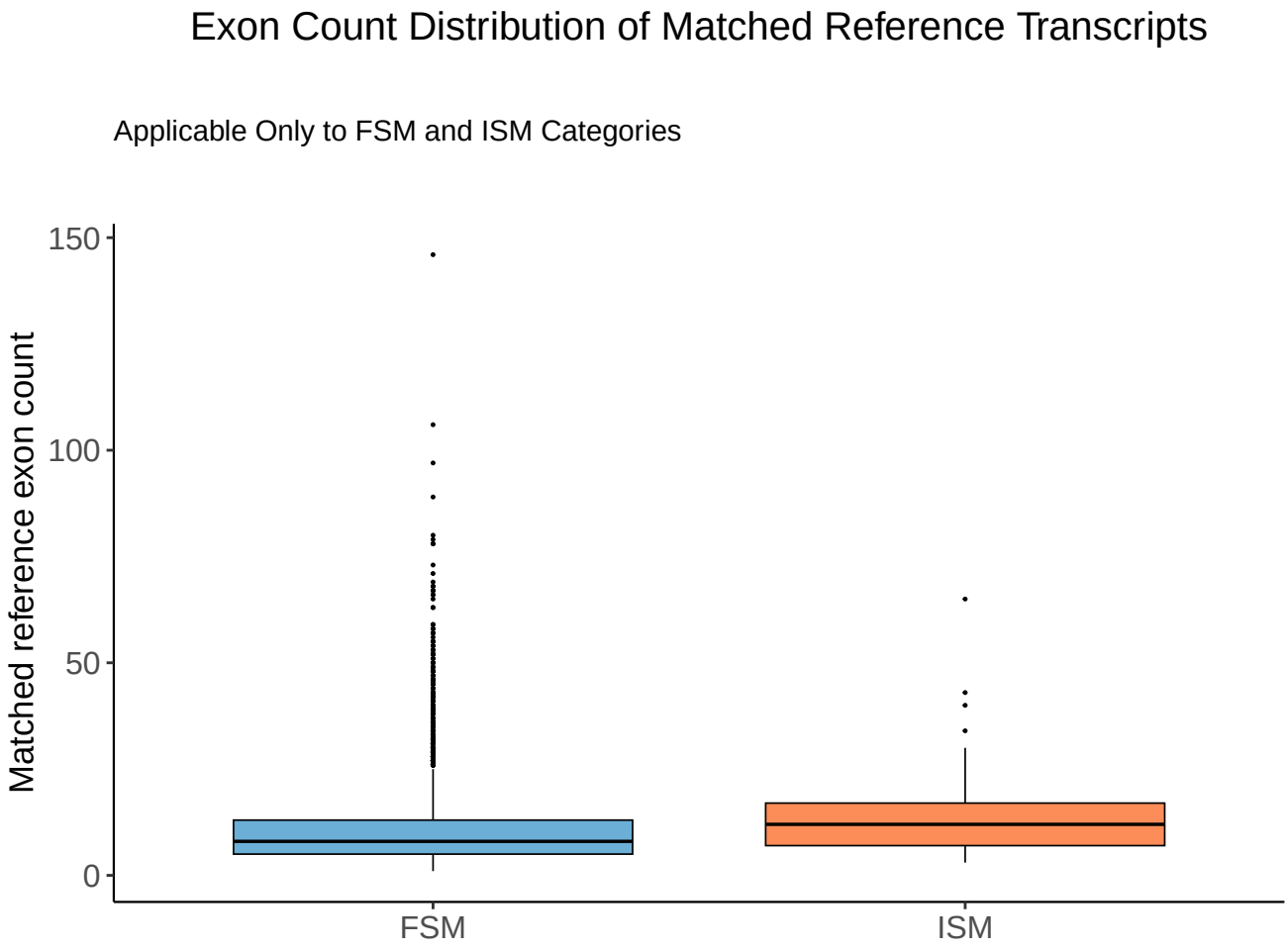

Long Reads Count by Structural Category
log2(FL_TPM+1)
15
10
5
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Intergenic
Genic Intron


Long Reads Count by Subcategory
FSM
ISM
log2(FL_TPM+1)
15
10
5
0
Reference match
Mono−exon
3' fragment
Internal fragment
5' fragment
Intron retention

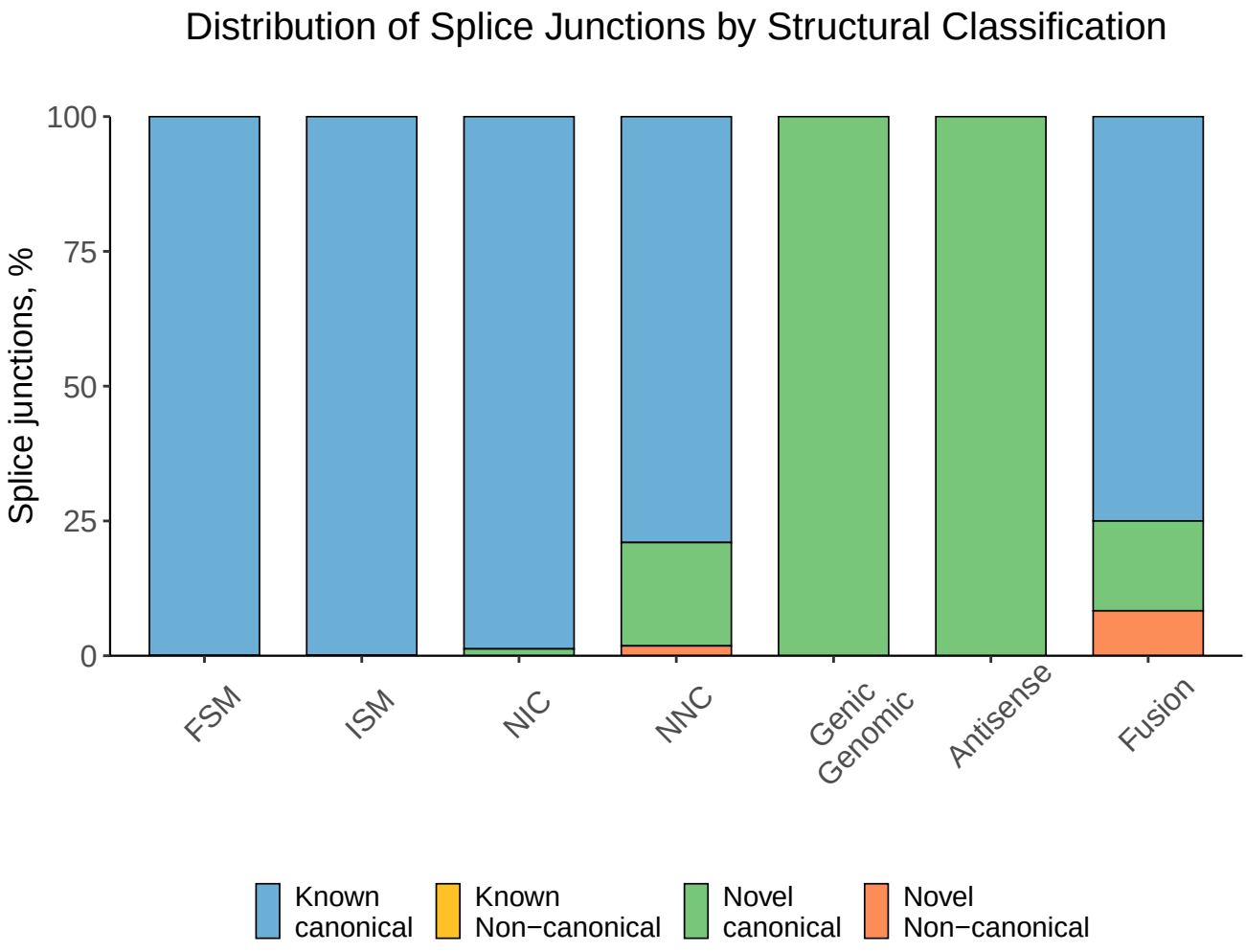

Long Reads Count by Subcategory
NIC
NNC
log2(FL_TPM+1)
15
13
11
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
Intron retention
At least 1 annot. don./accept.

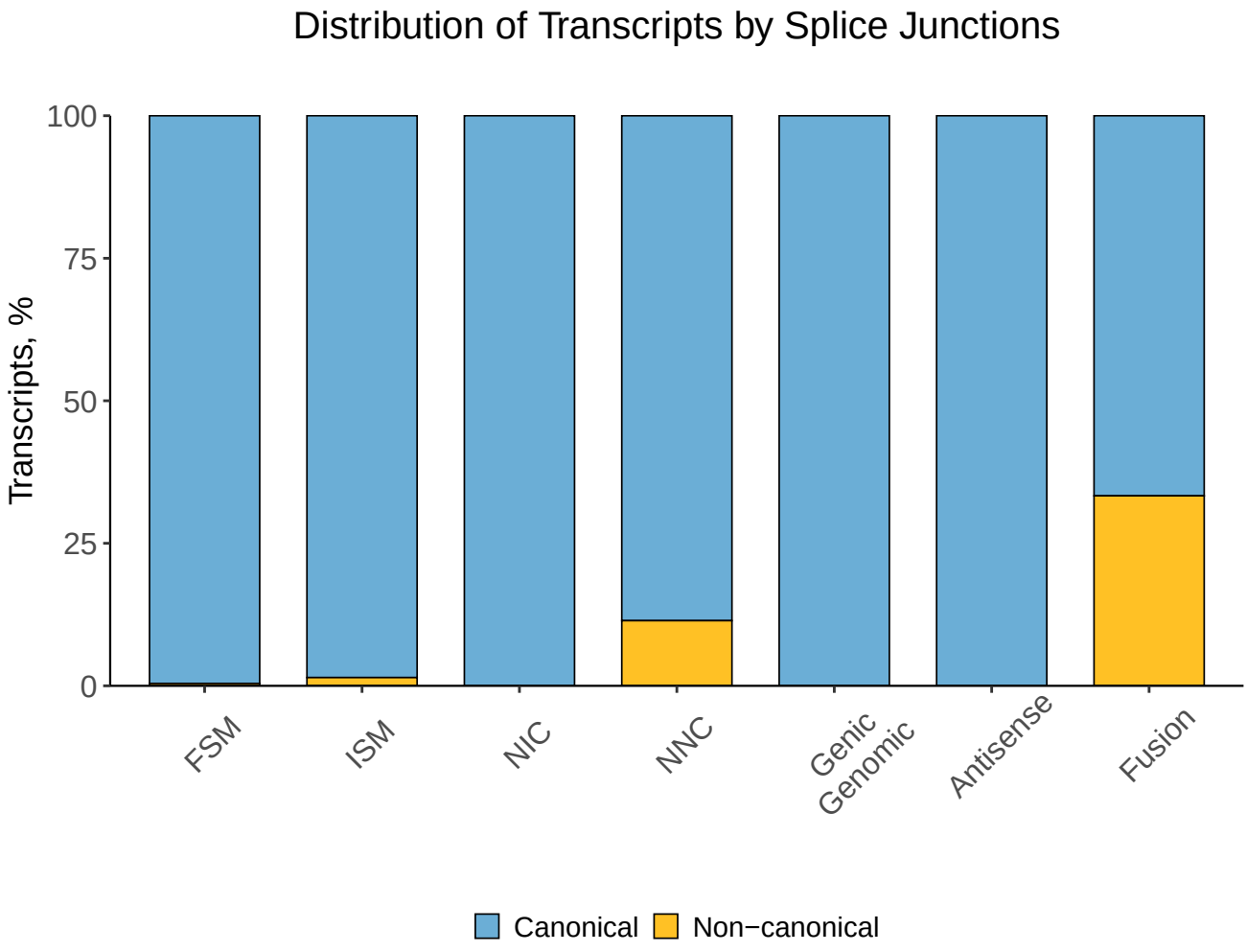

Long Reads Count by Subcategory
Genic Genomic
Antisense
Fusion
log2(FL_TPM+1)
18
16
14
Multi−exon
Multi−exon
Intron retention
Multi−exon

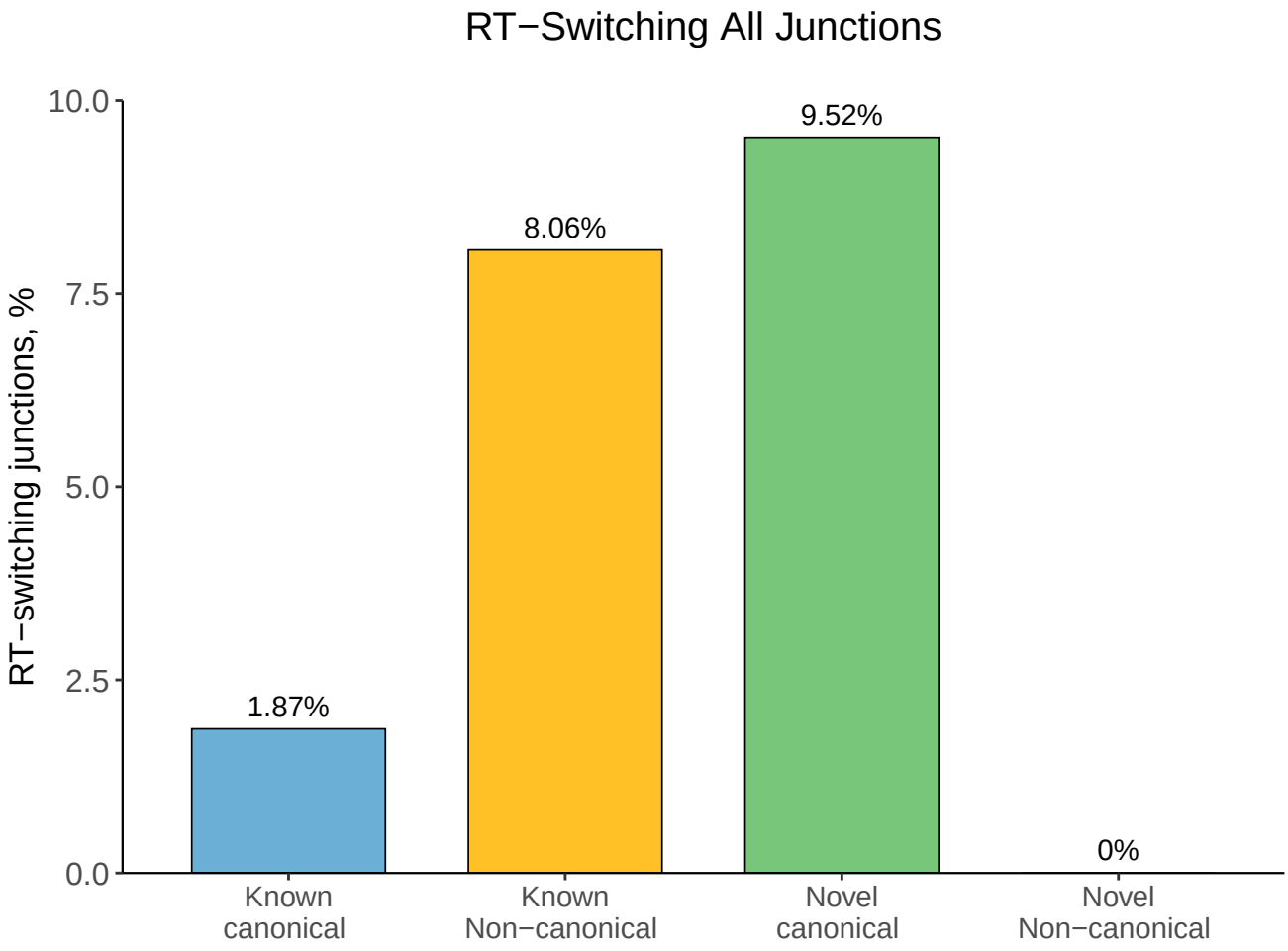

Length Distribution of Matched Reference Transcripts
Applicable Only to FSM and ISM Categories
Matched reference length, kb
30
20
10
0
FSM
ISM

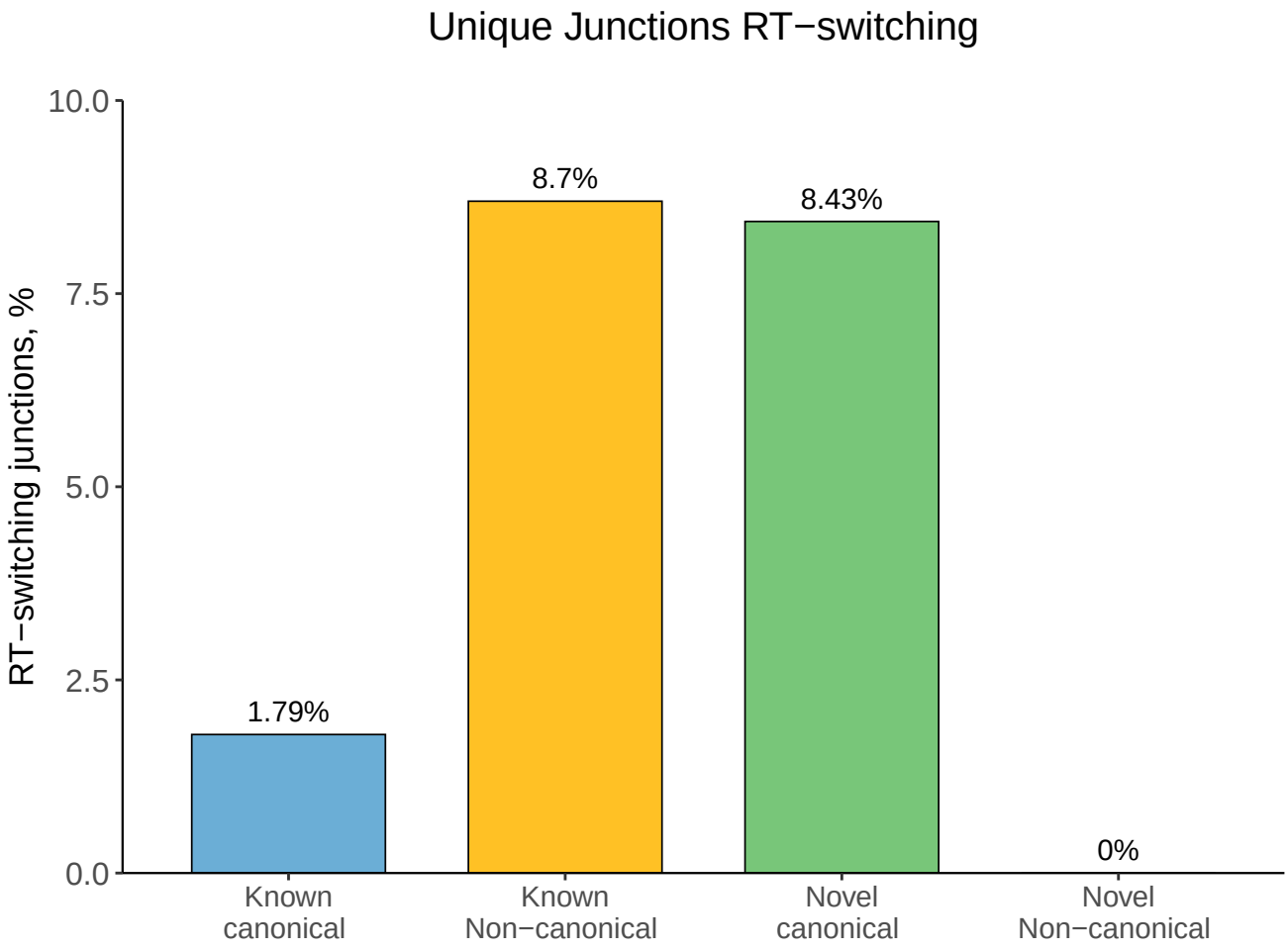

Exon Count Distribution of Matched Reference Transcripts
Applicable Only to FSM and ISM Categories
Matched reference exon count
150
100
50
0
FSM
ISM

Splice Junction Characterization

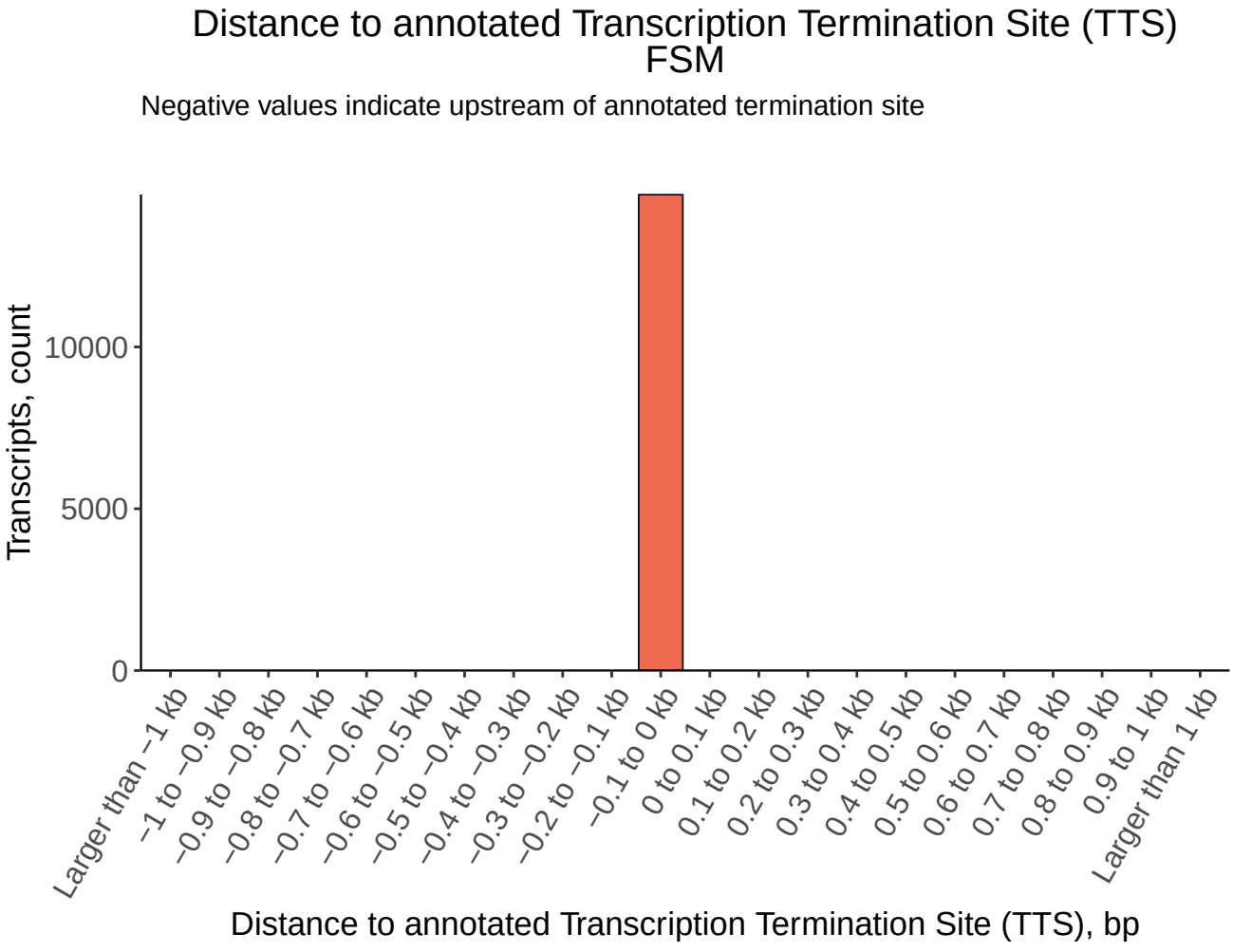

Distribution of Splice Junctions by Structural Classification
Splice junctions, %
100
75
50
25
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Known canonical
Known Non−canonical
Novel canonical
Novel Non−canonical

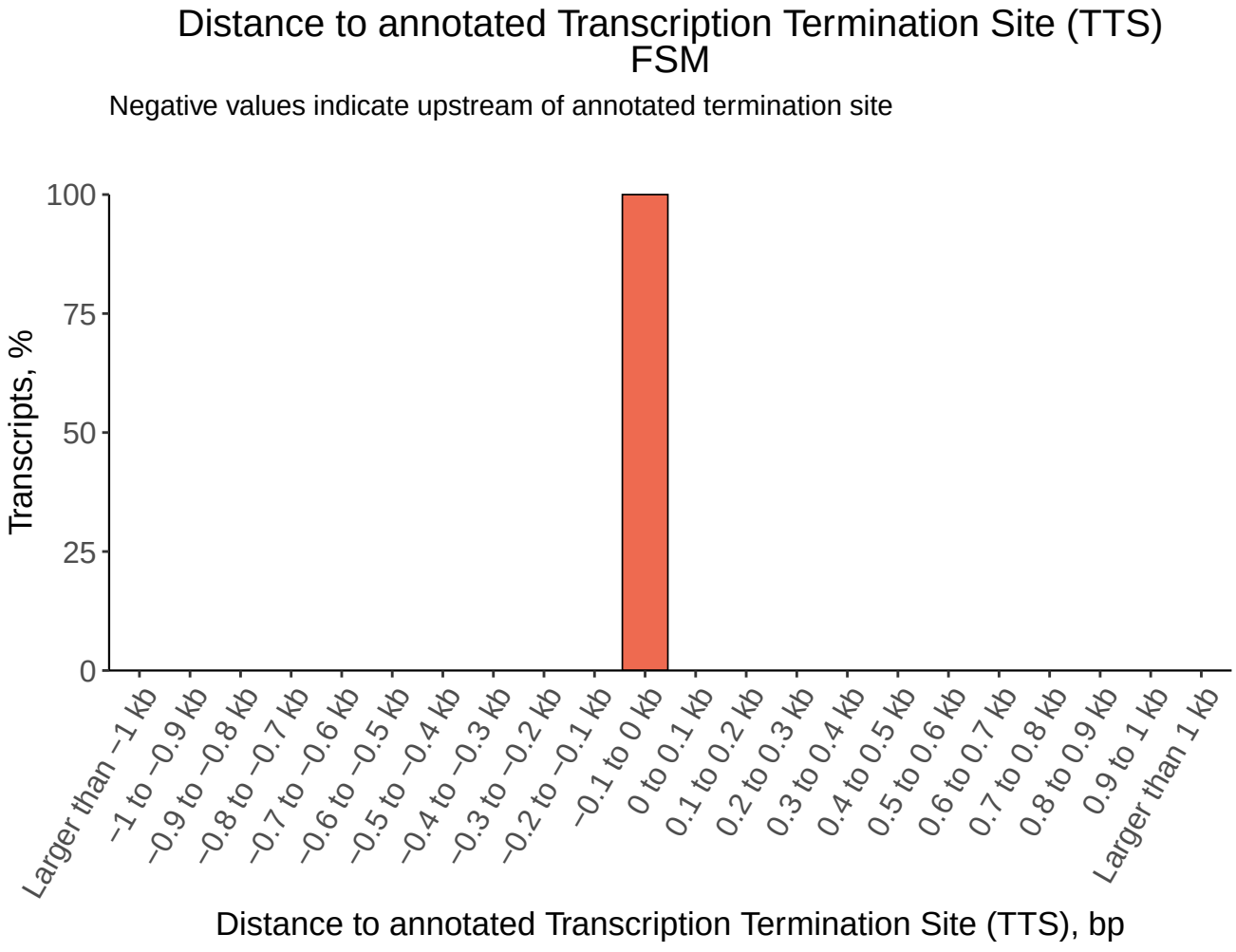

Distribution of Transcripts by Splice Junctions
Transcripts, %
100
75
50
25
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Canonical
Non−canonical

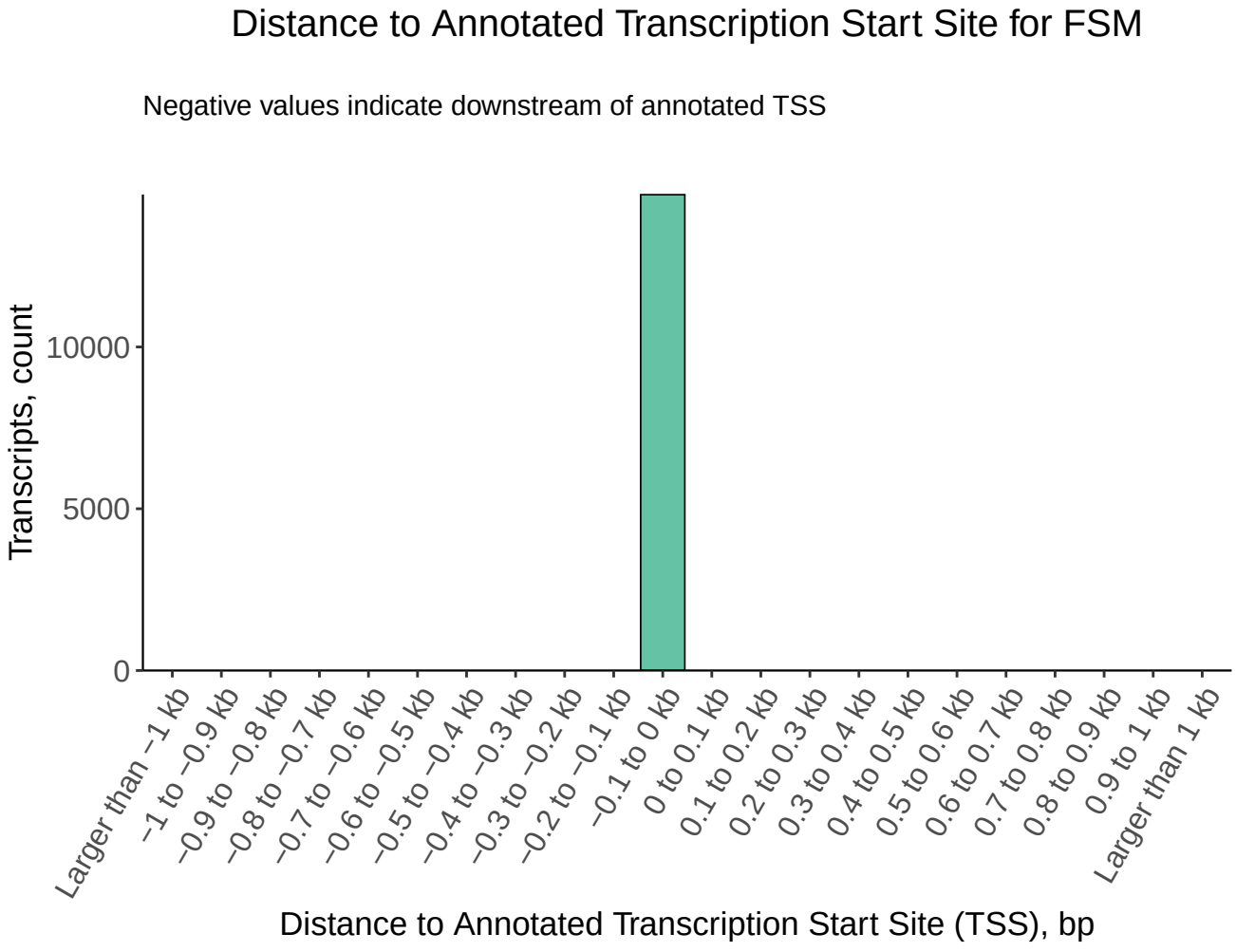

RT−Switching All Junctions
RT−switching junctions, %
10.0
7.5
5.0
2.5
0.0
1.87%
8.06%
9.52%
0%
Known canonical
Known Non−canonical
Novel canonical
Novel Non−canonical

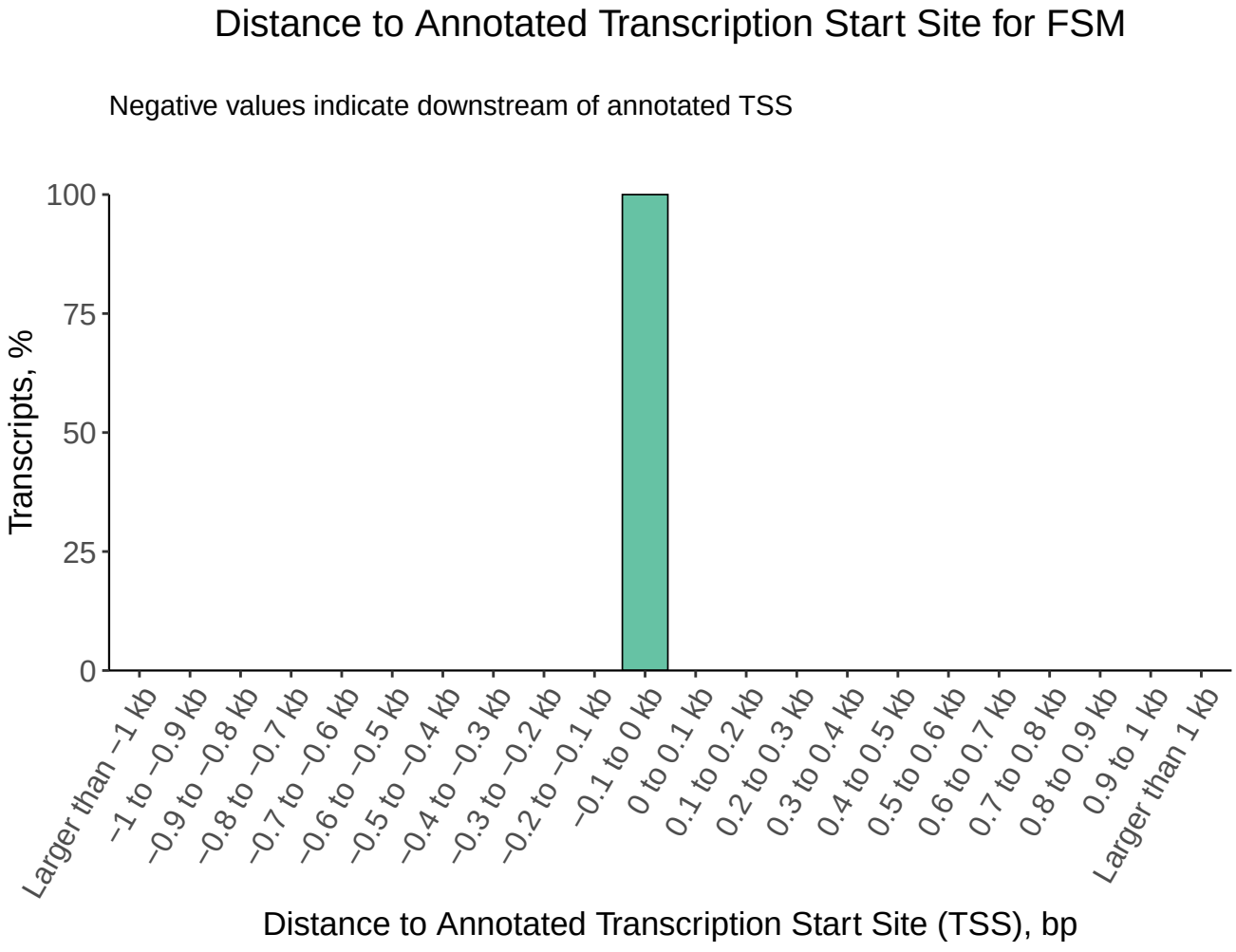

Unique Junctions RT−switching
RT−switching junctions, %
10.0
7.5
5.0
2.5
0.0
1.79%
8.7%
8.43%
0%
Known canonical
Known Non−canonical
Novel canonical
Novel Non−canonical

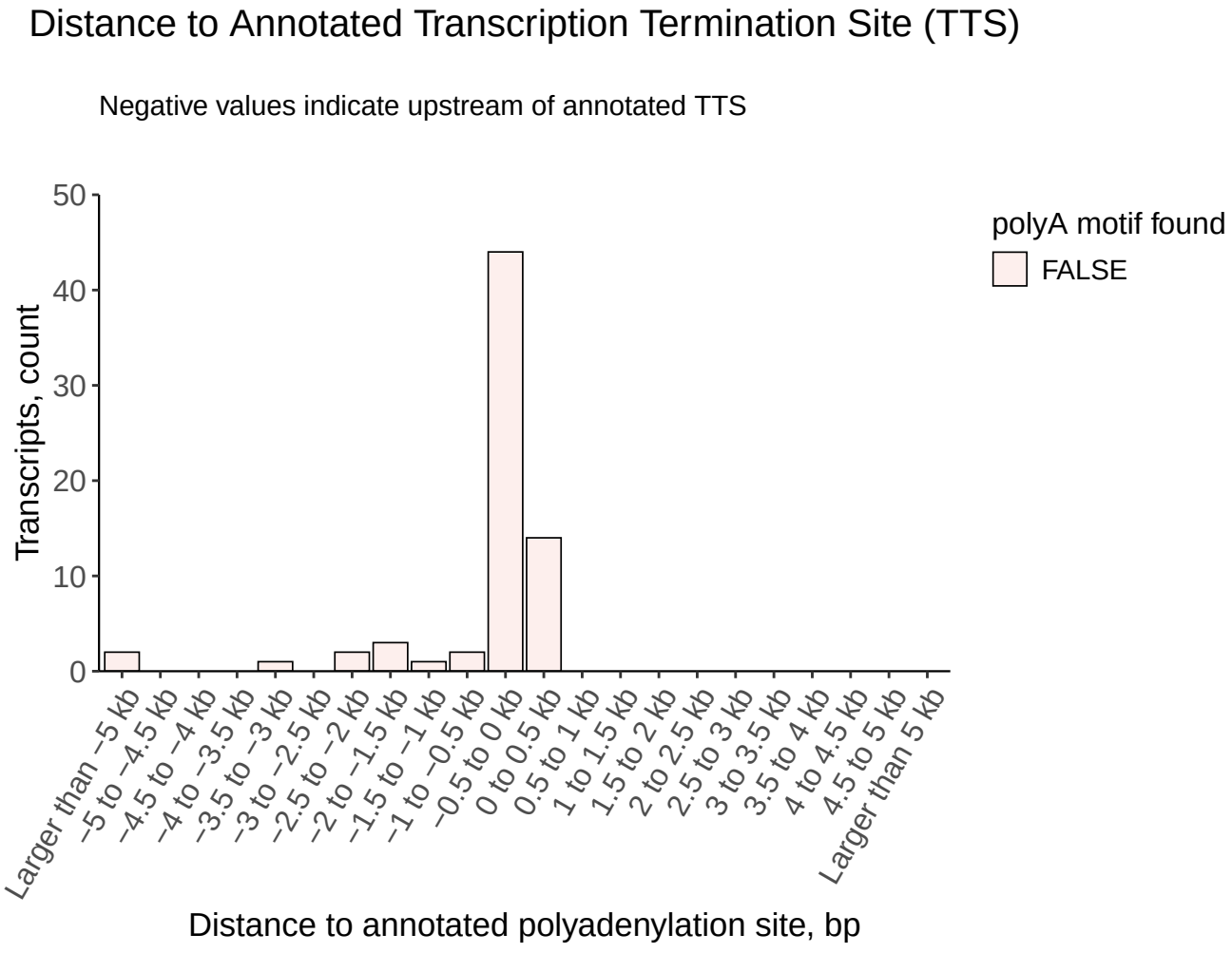
Comparison With Annotated TSS and TTS

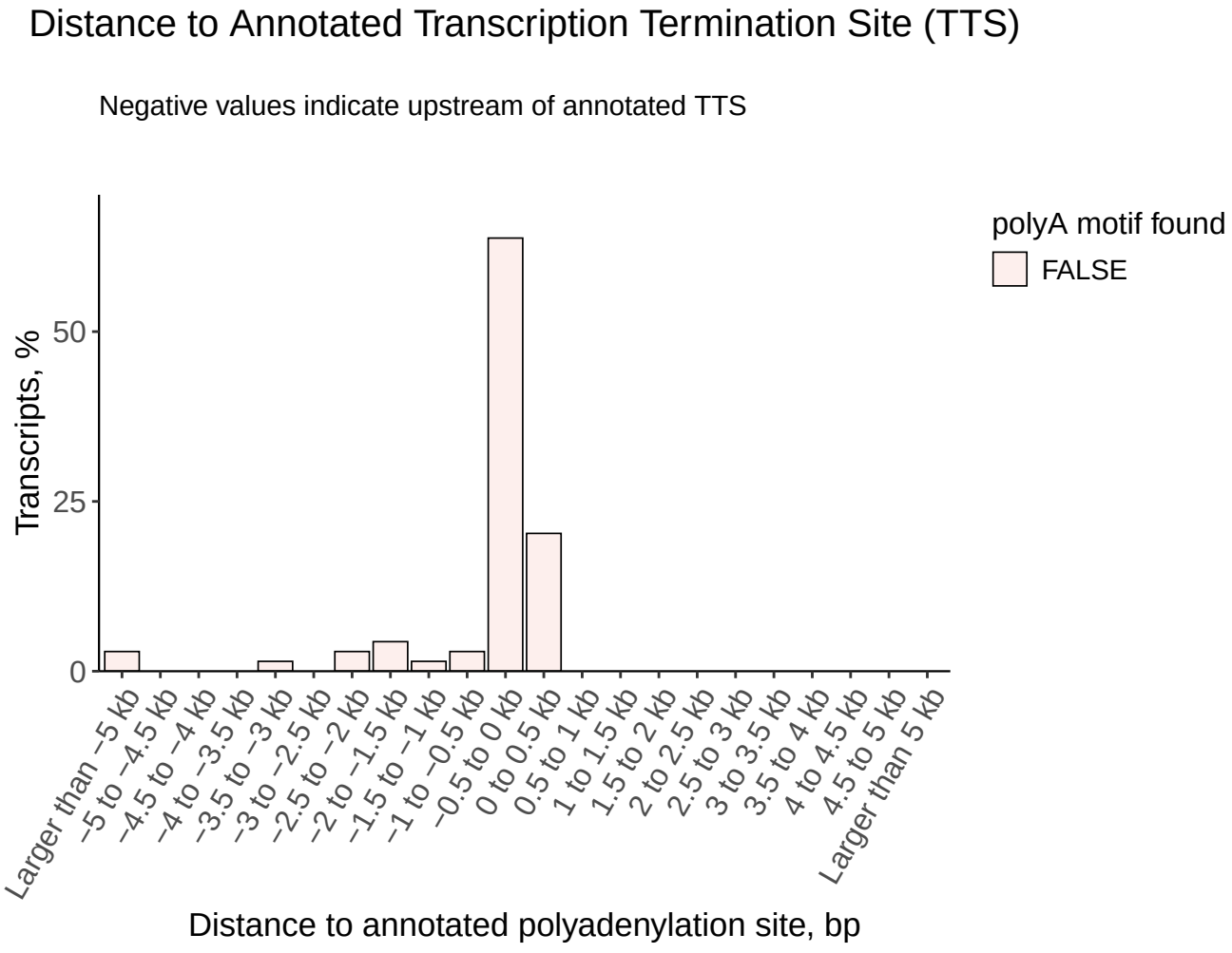

Distance to annotated Transcription Termination Site (TTS)
FSM
Negative values indicate upstream of annotated termination site
Transcripts, count
10000
5000
0
Larger than −1 kb
−1 to −0.9 kb
−0.9 to −0.8 kb
−0.8 to −0.7 kb
−0.7 to −0.6 kb
−0.6 to −0.5 kb
−0.5 to −0.4 kb
−0.4 to −0.3 kb
−0.3 to −0.2 kb
−0.2 to −0.1 kb
−0.1 to 0 kb
0 to 0.1 kb
0.1 to 0.2 kb
0.2 to 0.3 kb
0.3 to 0.4 kb
0.4 to 0.5 kb
0.5 to 0.6 kb
0.6 to 0.7 kb
0.7 to 0.8 kb
0.8 to 0.9 kb
0.9 to 1 kb
Larger than 1 kb
Distance to annotated Transcription Termination Site (TTS), bp


Distance to annotated Transcription Termination Site (TTS)
FSM
Negative values indicate upstream of annotated termination site
Transcripts, %
100
75
50
25
0
Larger than −1 kb
−1 to −0.9 kb
−0.9 to −0.8 kb
−0.8 to −0.7 kb
−0.7 to −0.6 kb
−0.6 to −0.5 kb
−0.5 to −0.4 kb
−0.4 to −0.3 kb
−0.3 to −0.2 kb
−0.2 to −0.1 kb
−0.1 to 0 kb
0 to 0.1 kb
0.1 to 0.2 kb
0.2 to 0.3 kb
0.3 to 0.4 kb
0.4 to 0.5 kb
0.5 to 0.6 kb
0.6 to 0.7 kb
0.7 to 0.8 kb
0.8 to 0.9 kb
0.9 to 1 kb
Larger than 1 kb
Distance to annotated Transcription Termination Site (TTS), bp


Distance to Annotated Transcription Start Site for FSM
Negative values indicate downstream of annotated TSS
Transcripts, count
10000
5000
0
Larger than −1 kb
−1 to −0.9 kb
−0.9 to −0.8 kb
−0.8 to −0.7 kb
−0.7 to −0.6 kb
−0.6 to −0.5 kb
−0.5 to −0.4 kb
−0.4 to −0.3 kb
−0.3 to −0.2 kb
−0.2 to −0.1 kb
−0.1 to 0 kb
0 to 0.1 kb
0.1 to 0.2 kb
0.2 to 0.3 kb
0.3 to 0.4 kb
0.4 to 0.5 kb
0.5 to 0.6 kb
0.6 to 0.7 kb
0.7 to 0.8 kb
0.8 to 0.9 kb
0.9 to 1 kb
Larger than 1 kb
Distance to Annotated Transcription Start Site (TSS), bp

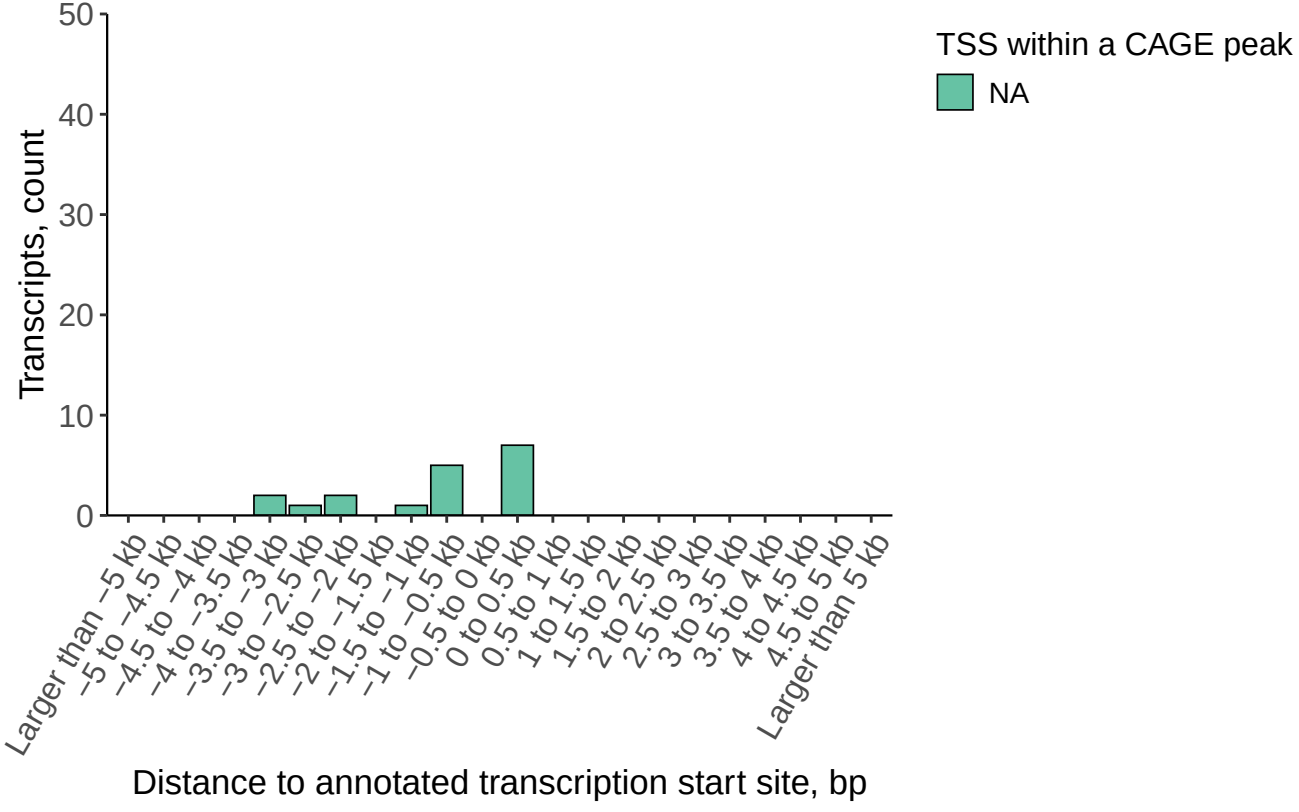

Distance to Annotated Transcription Start Site for FSM
Negative values indicate downstream of annotated TSS
Transcripts, %
100
75
50
25
0
Larger than −1 kb
−1 to −0.9 kb
−0.9 to −0.8 kb
−0.8 to −0.7 kb
−0.7 to −0.6 kb
−0.6 to −0.5 kb
−0.5 to −0.4 kb
−0.4 to −0.3 kb
−0.3 to −0.2 kb
−0.2 to −0.1 kb
−0.1 to 0 kb
0 to 0.1 kb
0.1 to 0.2 kb
0.2 to 0.3 kb
0.3 to 0.4 kb
0.4 to 0.5 kb
0.5 to 0.6 kb
0.6 to 0.7 kb
0.7 to 0.8 kb
0.8 to 0.9 kb
0.9 to 1 kb
Larger than 1 kb
Distance to Annotated Transcription Start Site (TSS), bp

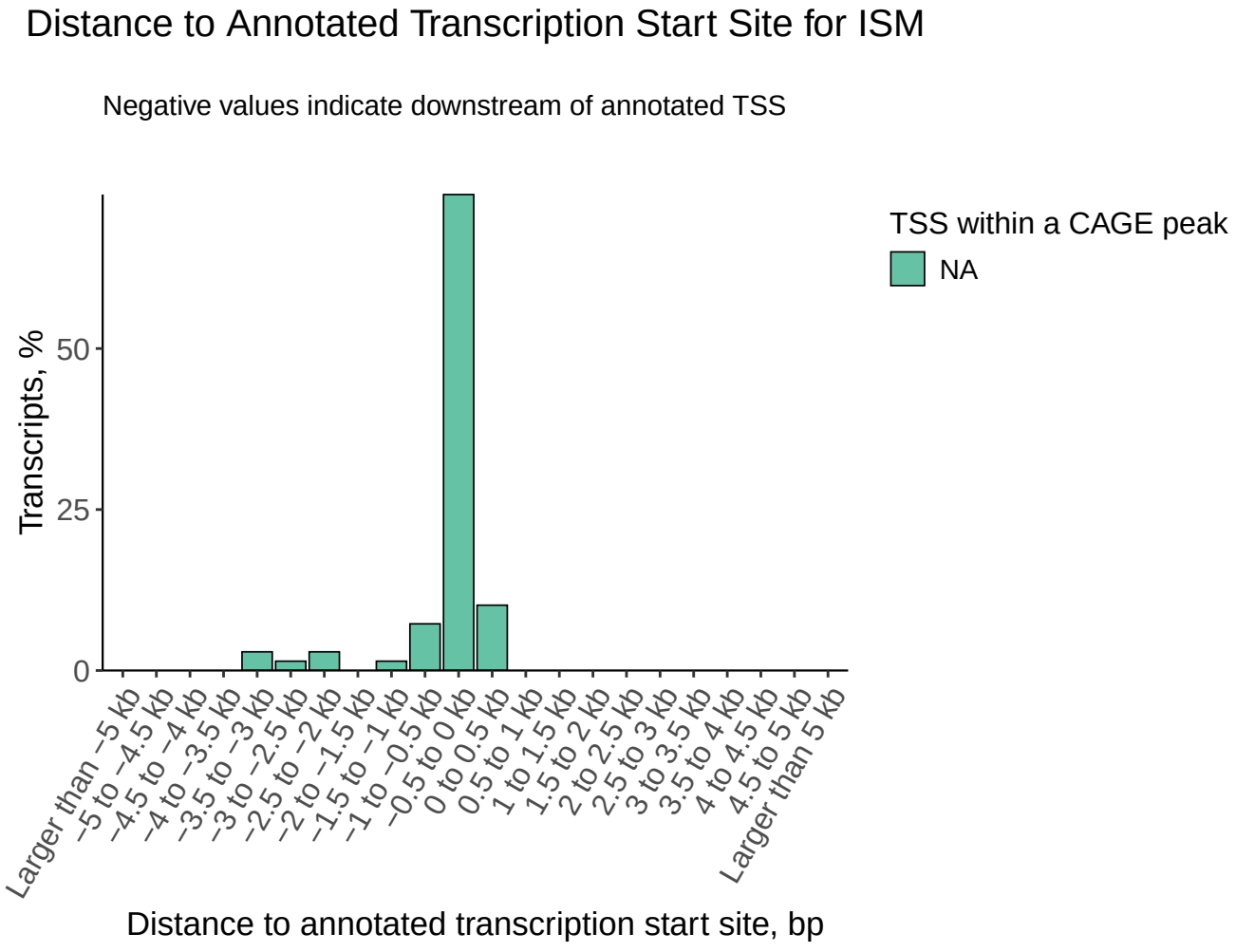

Distance to Annotated Transcription Termination Site (TTS)
Negative values indicate upstream of annotated TTS
Transcripts, count
50
40
30
20
10
0
Larger than −5 kb
−5 to −4.5 kb
−4.5 to −4 kb
−4 to −3.5 kb
−3.5 to −3 kb
−3 to −2.5 kb
−2.5 to −2 kb
−2 to −1.5 kb
−1.5 to −1 kb
−1 to −0.5 kb
−0.5 to 0 kb
0 to 0.5 kb
0.5 to 1 kb
1 to 1.5 kb
1.5 to 2 kb
2 to 2.5 kb
2.5 to 3 kb
3 to 3.5 kb
3.5 to 4 kb
4 to 4.5 kb
4.5 to 5 kb
Larger than 5 kb
Distance to annotated polyadenylation site, bp
polyA motif found
FALSE


Distance to Annotated Transcription Termination Site (TTS)
Negative values indicate upstream of annotated TTS
polyA motif found
FALSE
Transcripts, %
0
25
50
Larger than −5 kb
−5 to −4.5 kb
−4.5 to −4 kb
−4 to −3.5 kb
−3.5 to −3 kb
−3 to −2.5 kb
−2.5 to −2 kb
−2 to −1.5 kb
−1.5 to −1 kb
−1 to −0.5 kb
−0.5 to 0 kb
0 to 0.5 kb
0.5 to 1 kb
1 to 1.5 kb
1.5 to 2 kb
2 to 2.5 kb
2.5 to 3 kb
3 to 3.5 kb
3.5 to 4 kb
4 to 4.5 kb
4.5 to 5 kb
Larger than 5 kb
Distance to annotated polyadenylation site, bp

Distance to Annotated Transcription Start Site for ISM

Negative values indicate downstream of annotated TSS

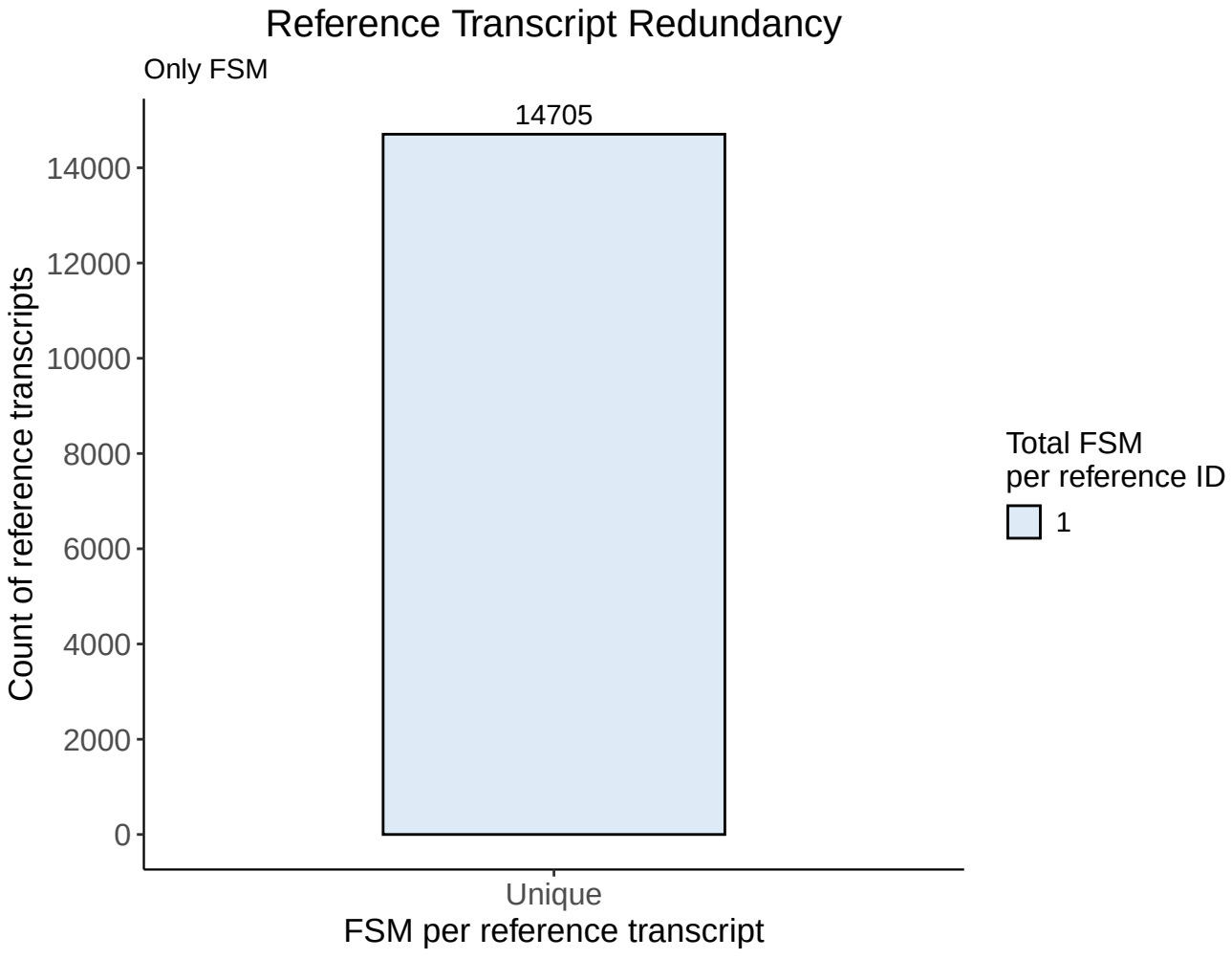


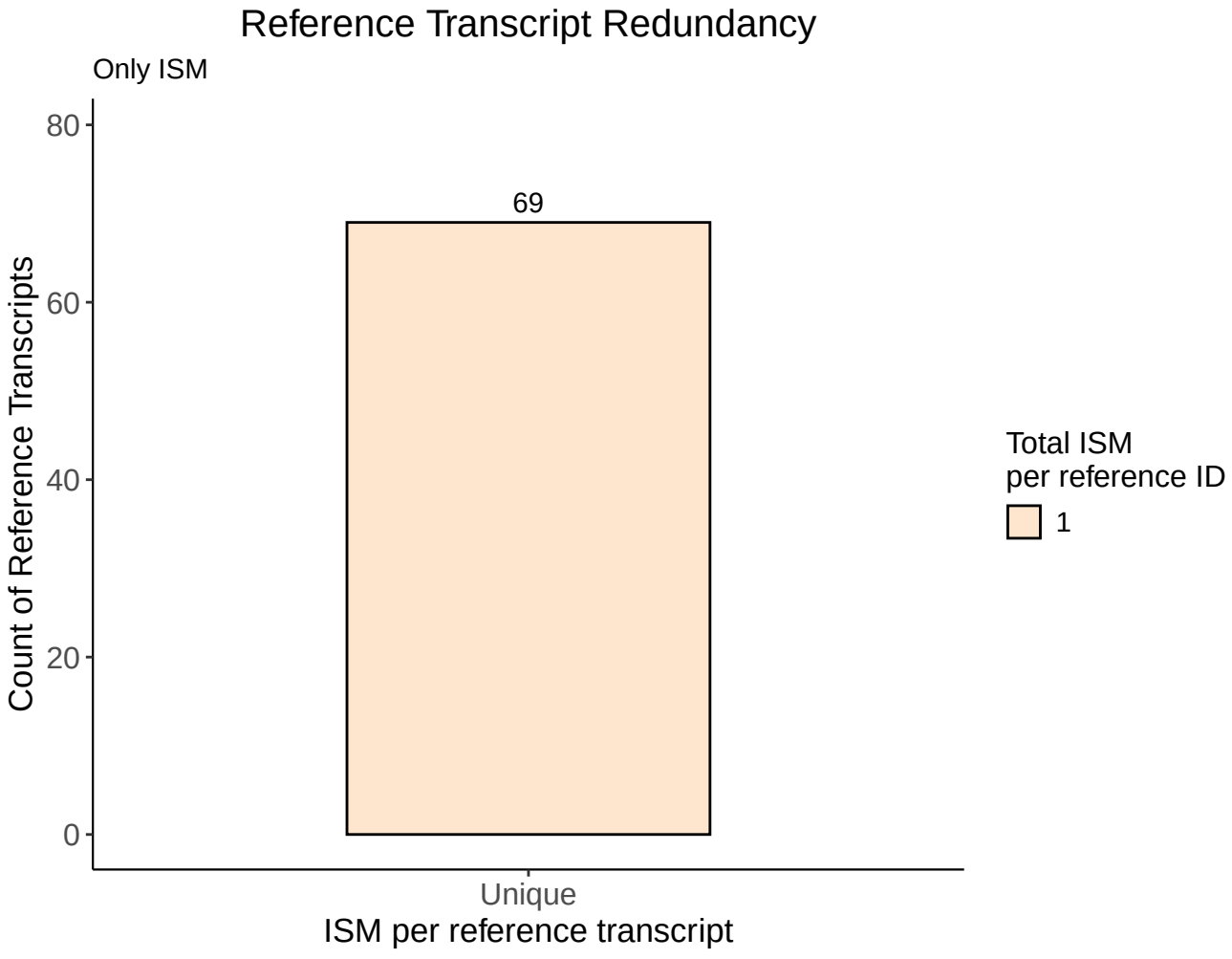

Distance to Annotated Transcription Start Site for ISM
Negative values indicate downstream of annotated TSS
Transcripts, %
0
25
50
Larger than −5 kb
−5 to −4.5 kb
−4.5 to −4 kb
−4 to −3.5 kb
−3.5 to −3 kb
−3 to −2.5 kb
−2.5 to −2 kb
−2 to −1.5 kb
−1.5 to −1 kb
−1 to −0.5 kb
−0.5 to 0 kb
0 to 0.5 kb
0.5 to 1 kb
1 to 1.5 kb
1.5 to 2 kb
2 to 2.5 kb
2.5 to 3 kb
3 to 3.5 kb
3.5 to 4 kb
4 to 4.5 kb
4.5 to 5 kb
Larger than 5 kb
Distance to annotated transcription start site, bp
TSS within a CAGE peak
NA

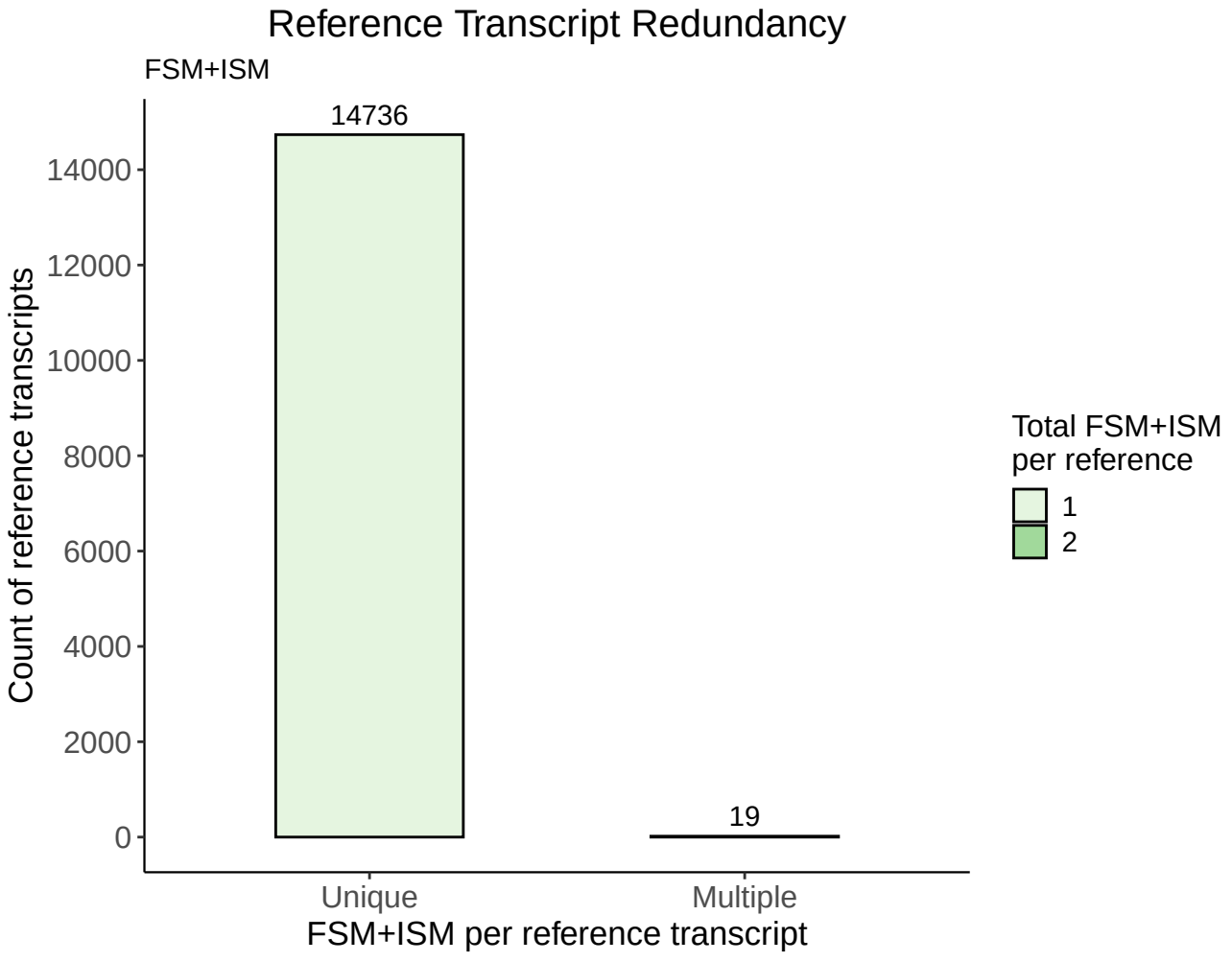
Comparison With Annotated TSS and TTS by Subcategories

PolyA Distance Analysis

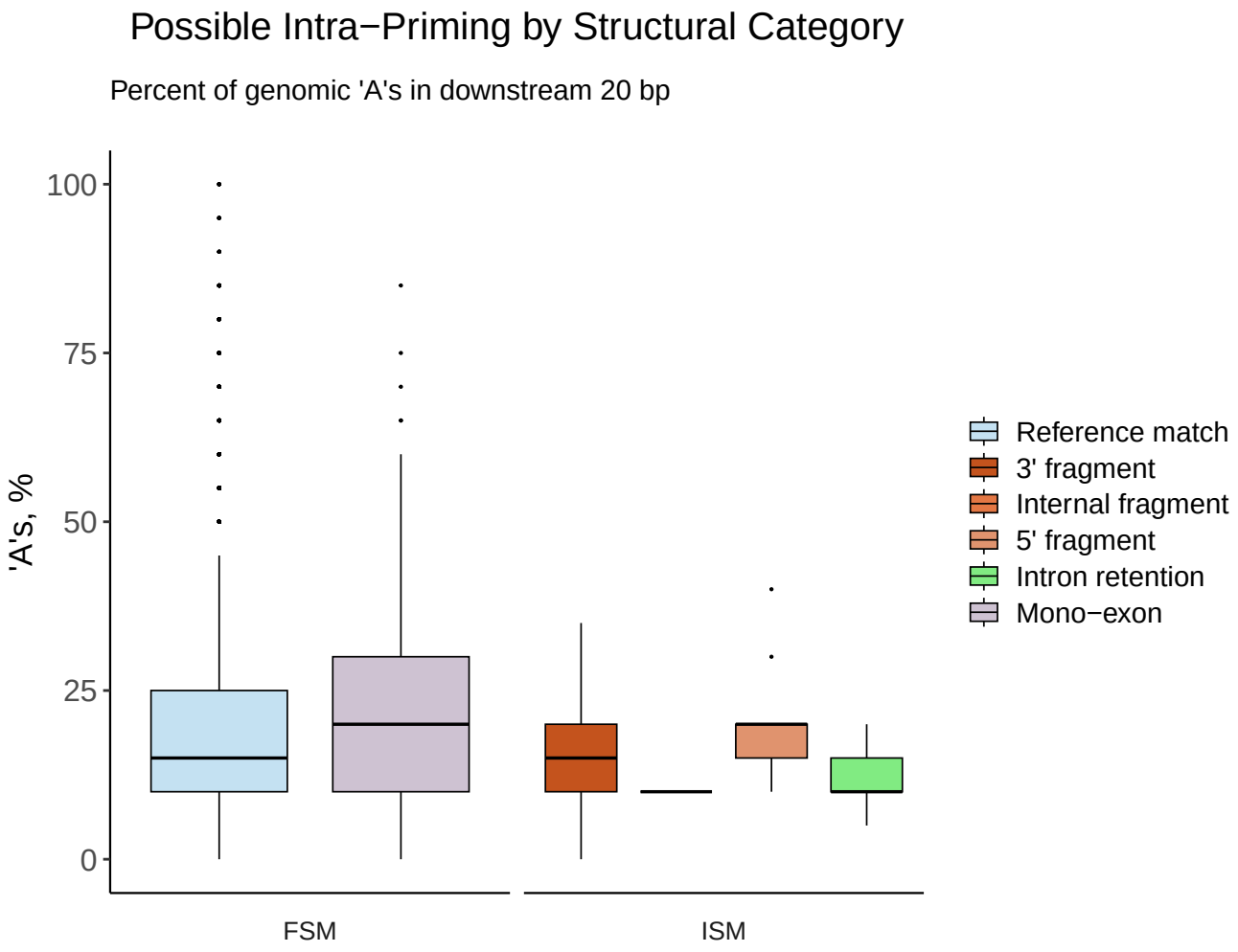
Redundancy Analysis

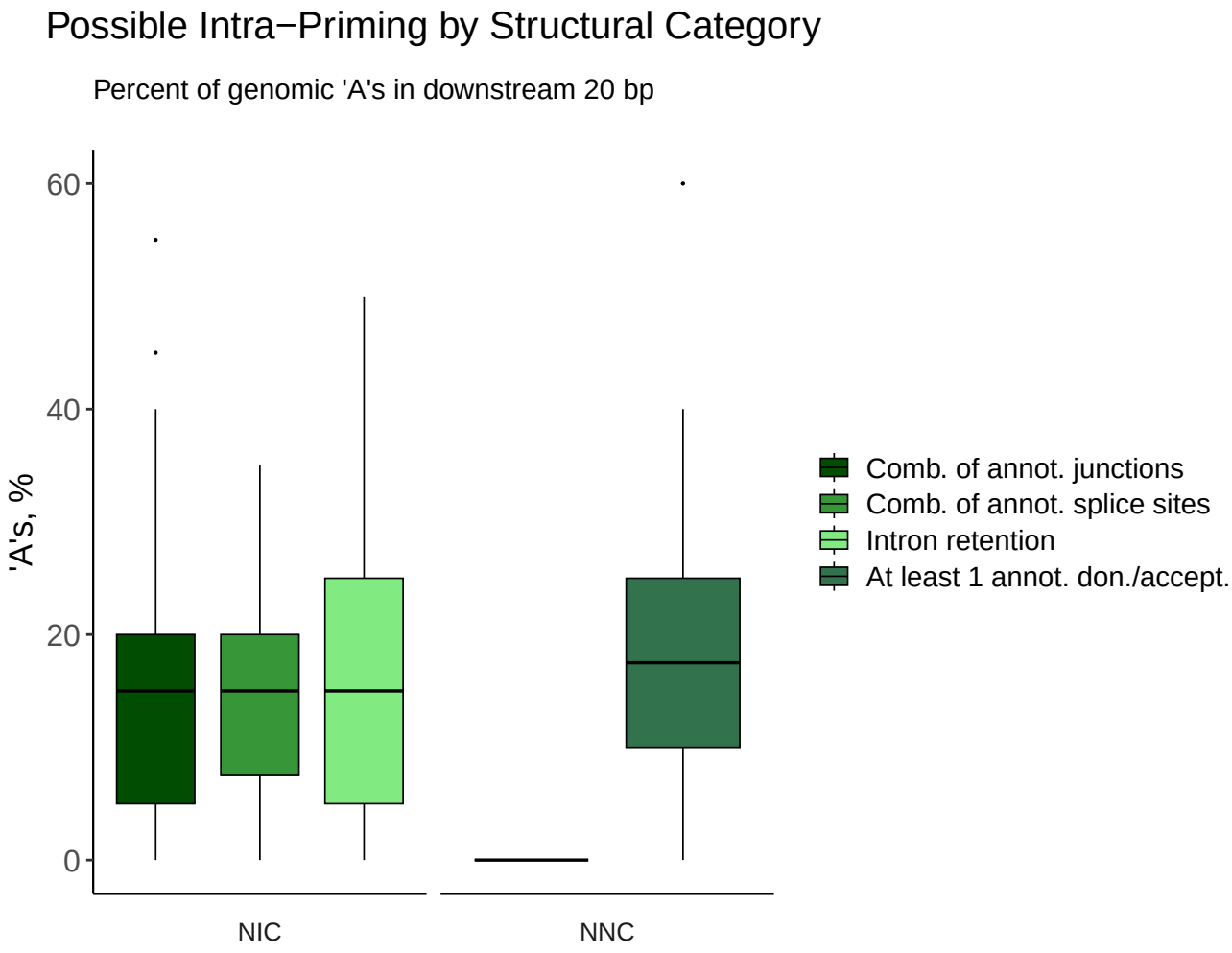

Reference Transcript Redundancy
Only FSM
14705
14000
12000
10000
8000
6000
4000
2000
0
Count of reference transcripts
Unique
FSM per reference transcript
Total FSM
per reference ID
1

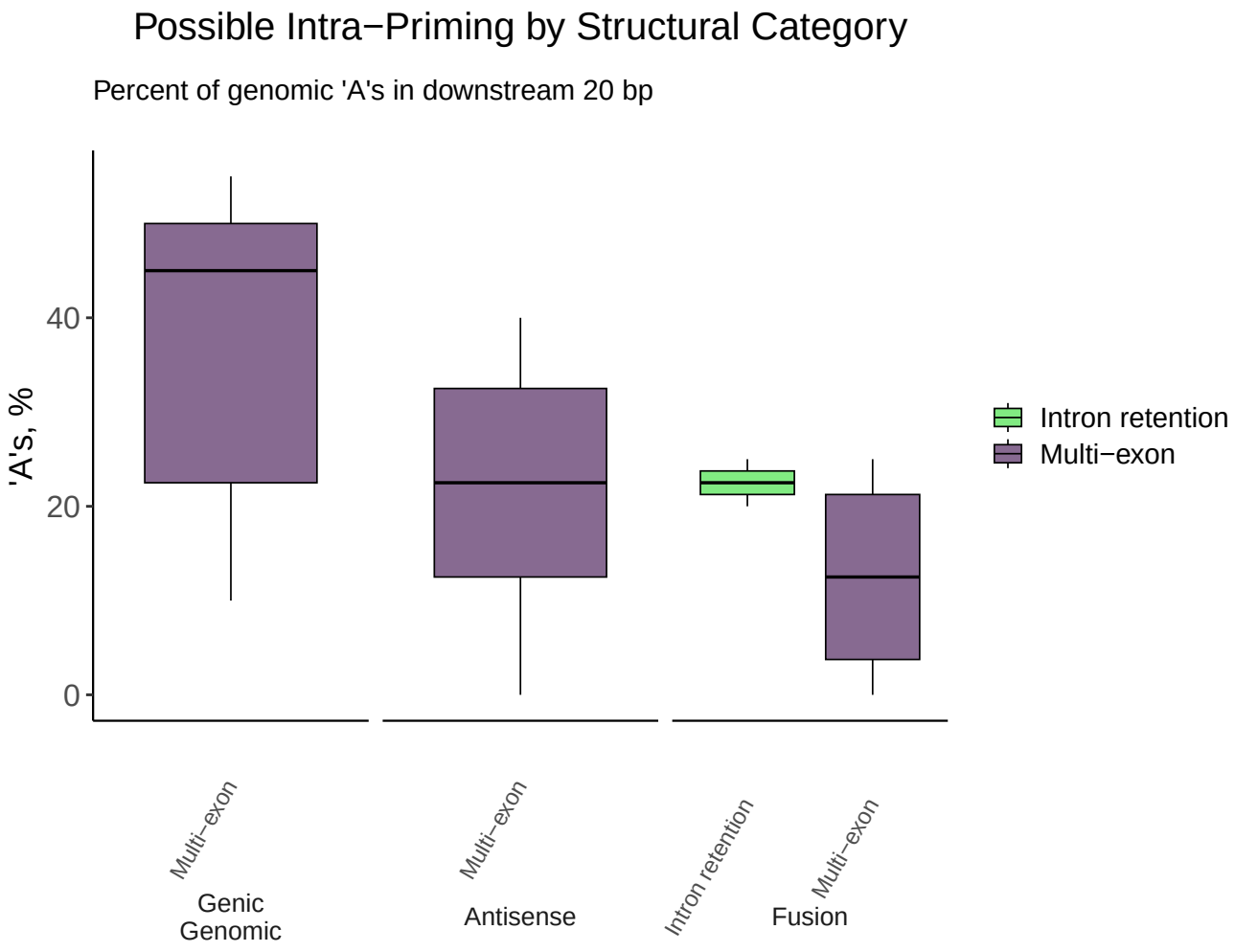

Reference Transcript Redundancy
Only ISM
80
60
40
20
0
69
Count of Reference Transcripts
Unique
ISM per reference transcript
Total ISM
per reference ID
1

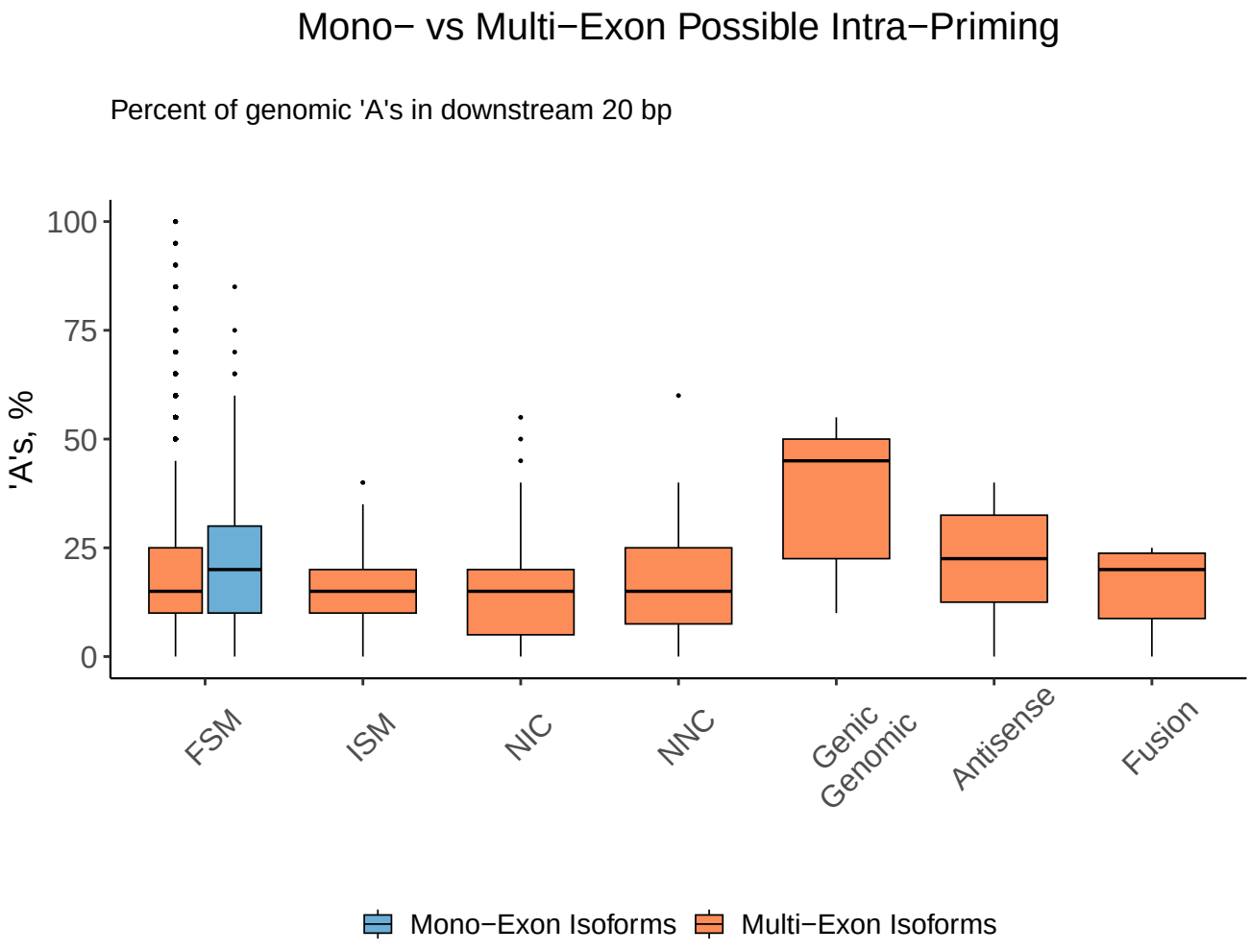

Reference Transcript Redundancy
FSM+ISM
14736
19
14000
12000
10000
8000
6000
4000
2000
0
Count of reference transcripts
Unique
Multiple
FSM+ISM per reference transcript
Total FSM+ISM per reference
1
2

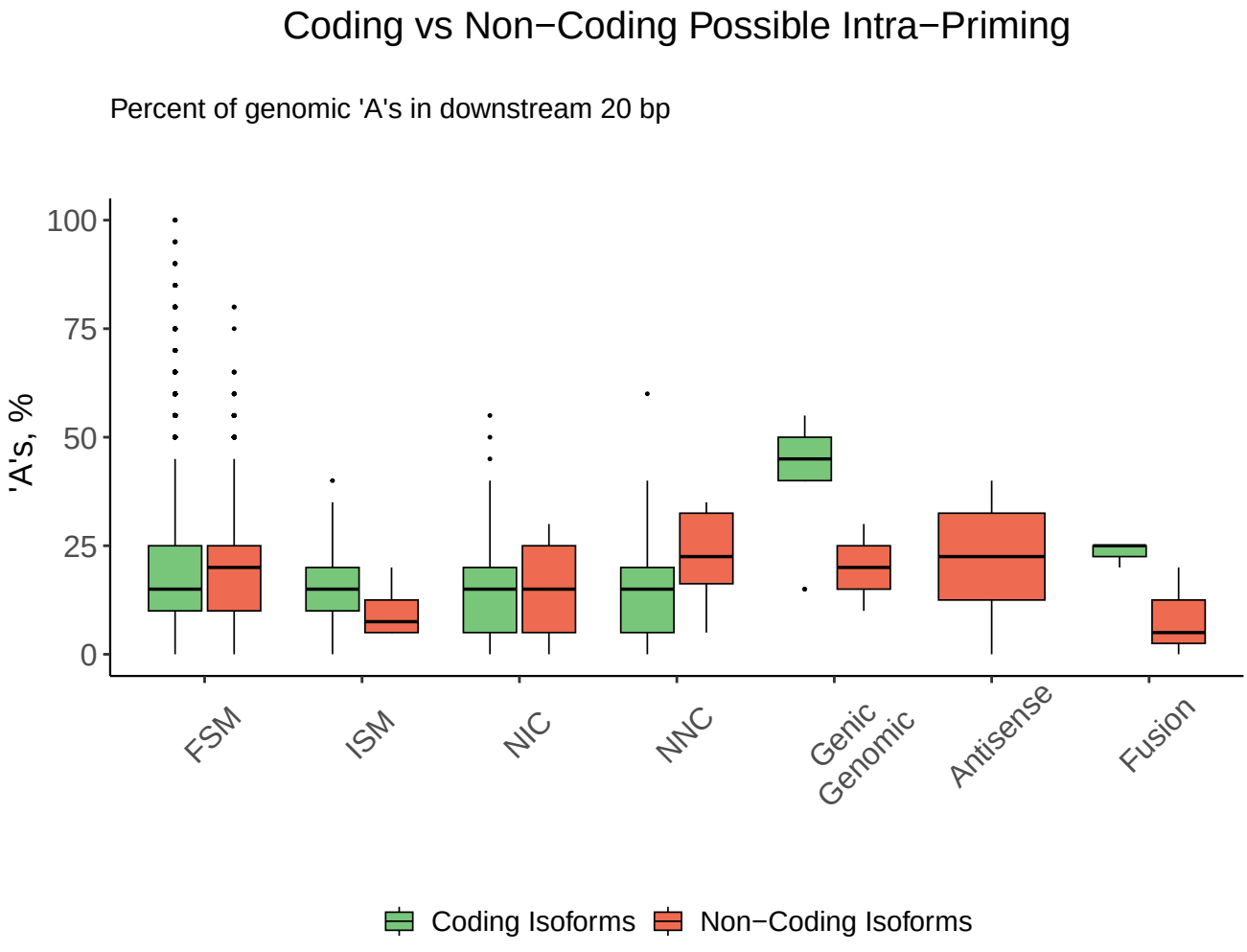
Intra−Priming Quality Check


Possible Intra−Priming by Structural Category
Percent of genomic 'A's in downstream 20 bp
'A's, %
100
75
50
25
0
FSM
ISM
Reference match
3' fragment
Internal fragment
5' fragment
Intron retention
Mono−exon

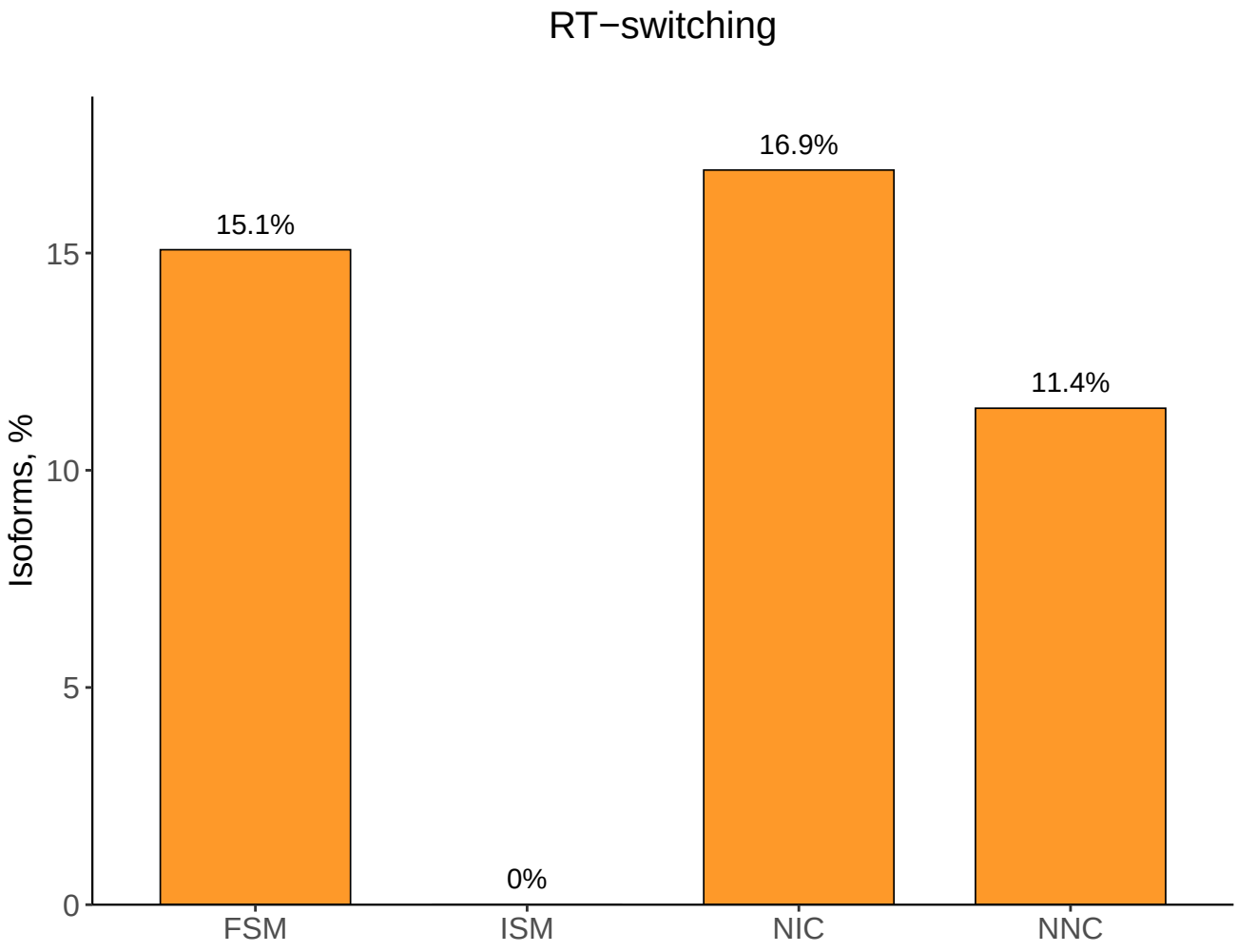

Possible Intra−Priming by Structural Category
Percent of genomic 'A's in downstream 20 bp
'A's, %
60
40
20
0
NIC
NNC
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
At least 1 annot. don./accept.

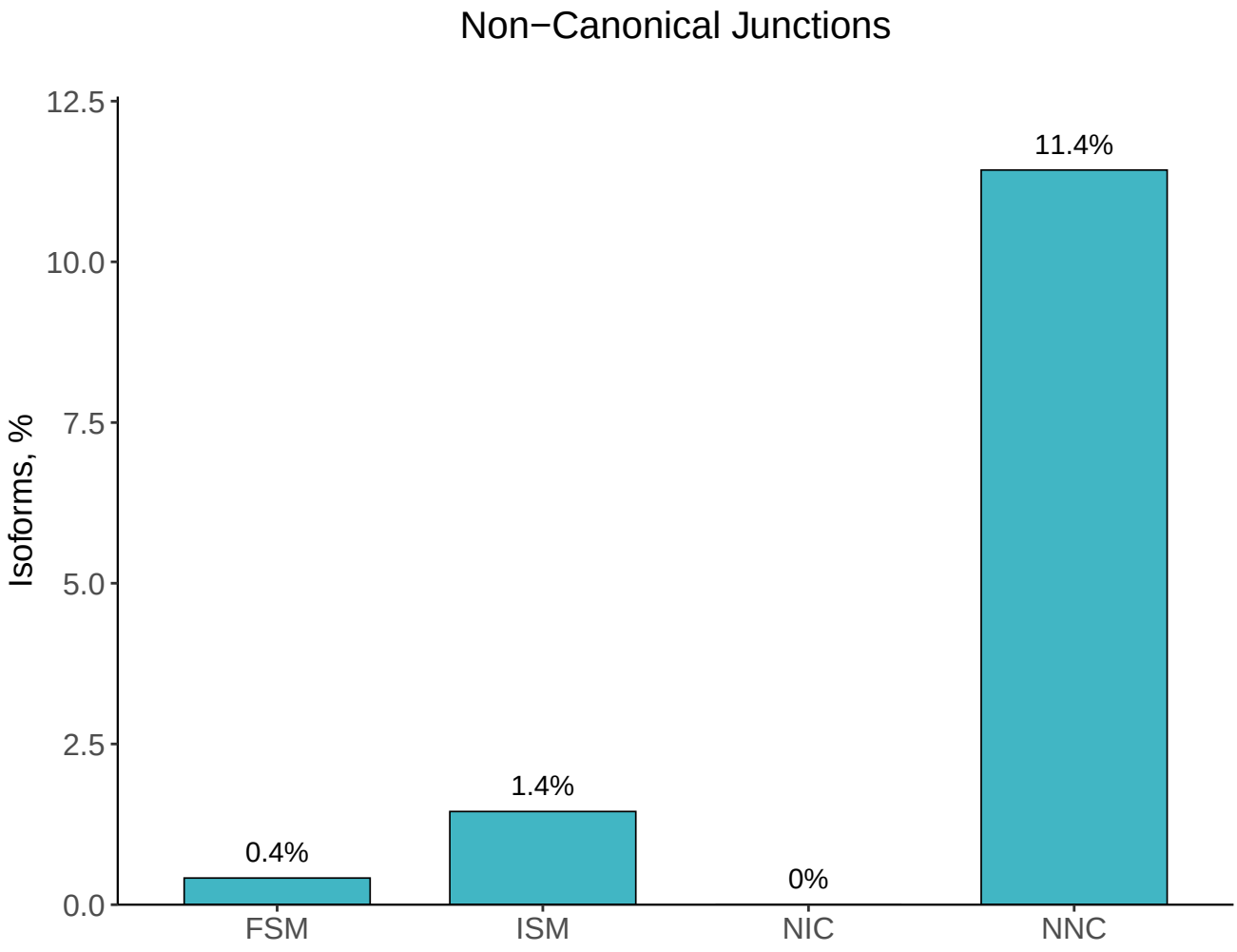

Possible Intra−Priming by Structural Category
Percent of genomic 'A's in downstream 20 bp
'A's, %
40
20
0
Multi−exon
Genic Genomic
Multi−exon
Antisense
Intron retention
Multi−exon
Fusion
Intron retention
Multi−exon

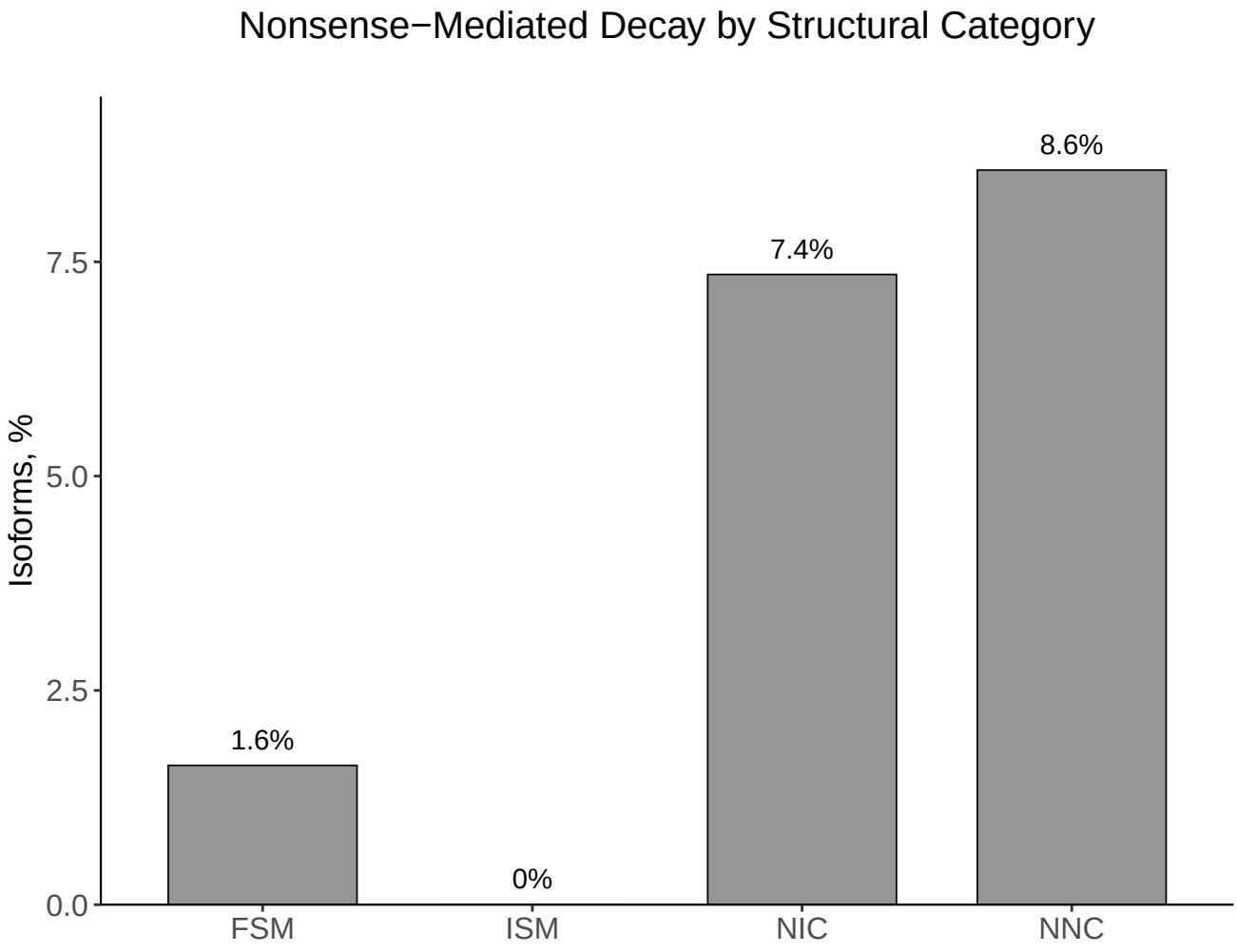

Mono− vs Multi−Exon Possible Intra−Priming
Percent of genomic 'A's in downstream 20 bp
'A's, %
100
75
50
25
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Mono−Exon Isoforms
Multi−Exon Isoforms

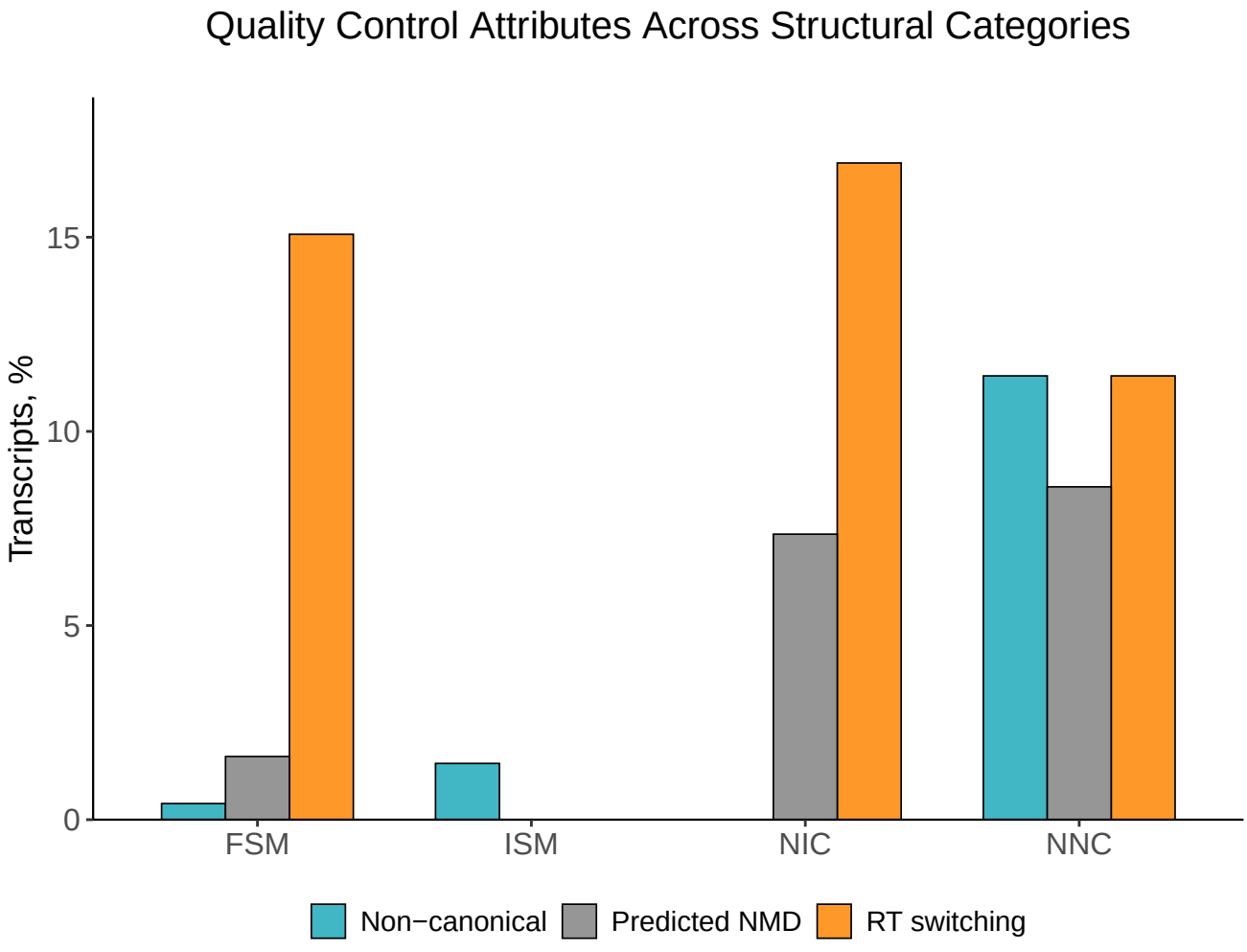

Coding vs Non−Coding Possible Intra−Priming
Percent of genomic 'A's in downstream 20 bp
'A's, %
100
75
50
25
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Coding Isoforms
Non−Coding Isoforms

Features of Bad Quality

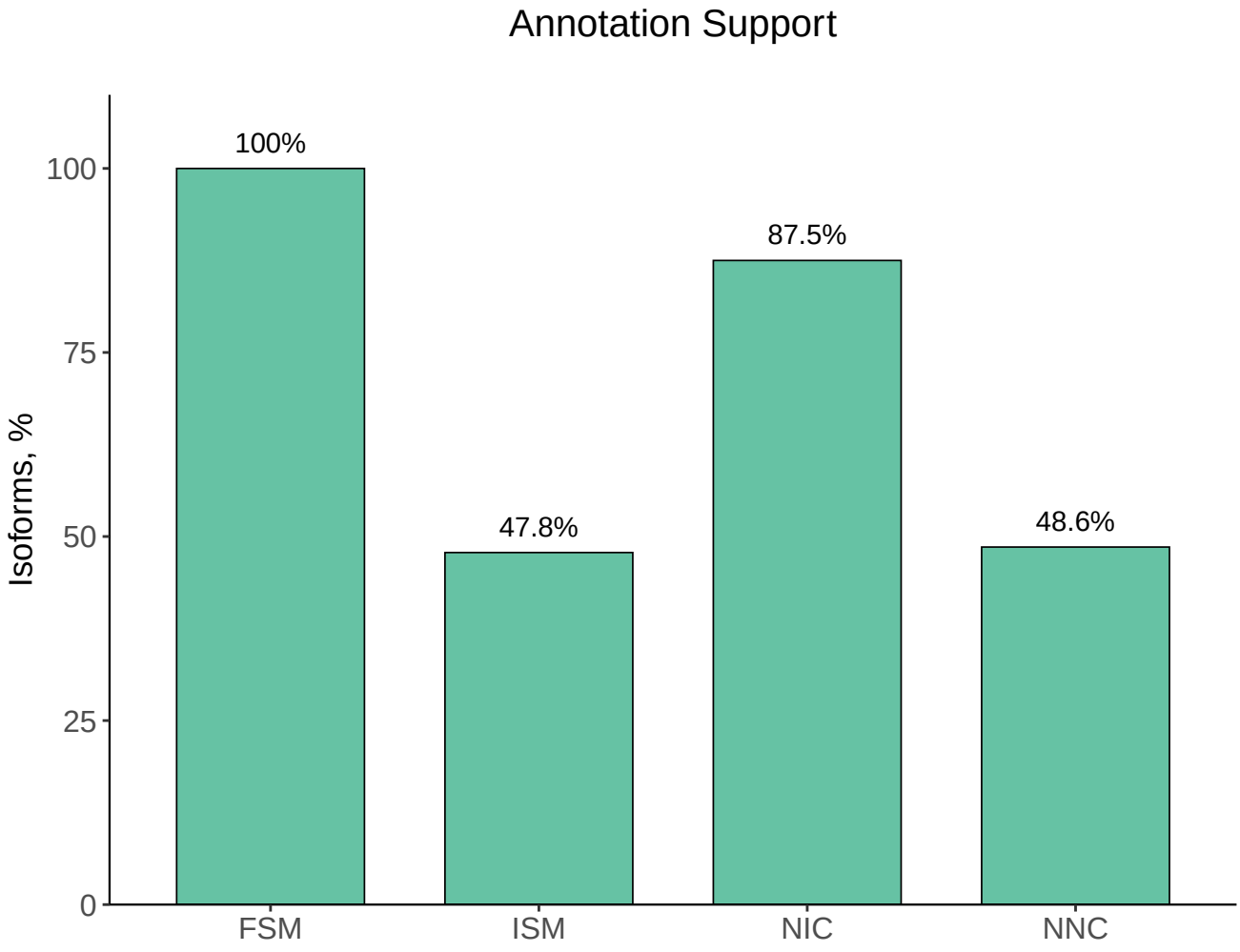

RT−switching
Isoforms, %
0
5
10
15
15.1%
0%
16.9%
11.4%
FSM
ISM
NIC
NNC

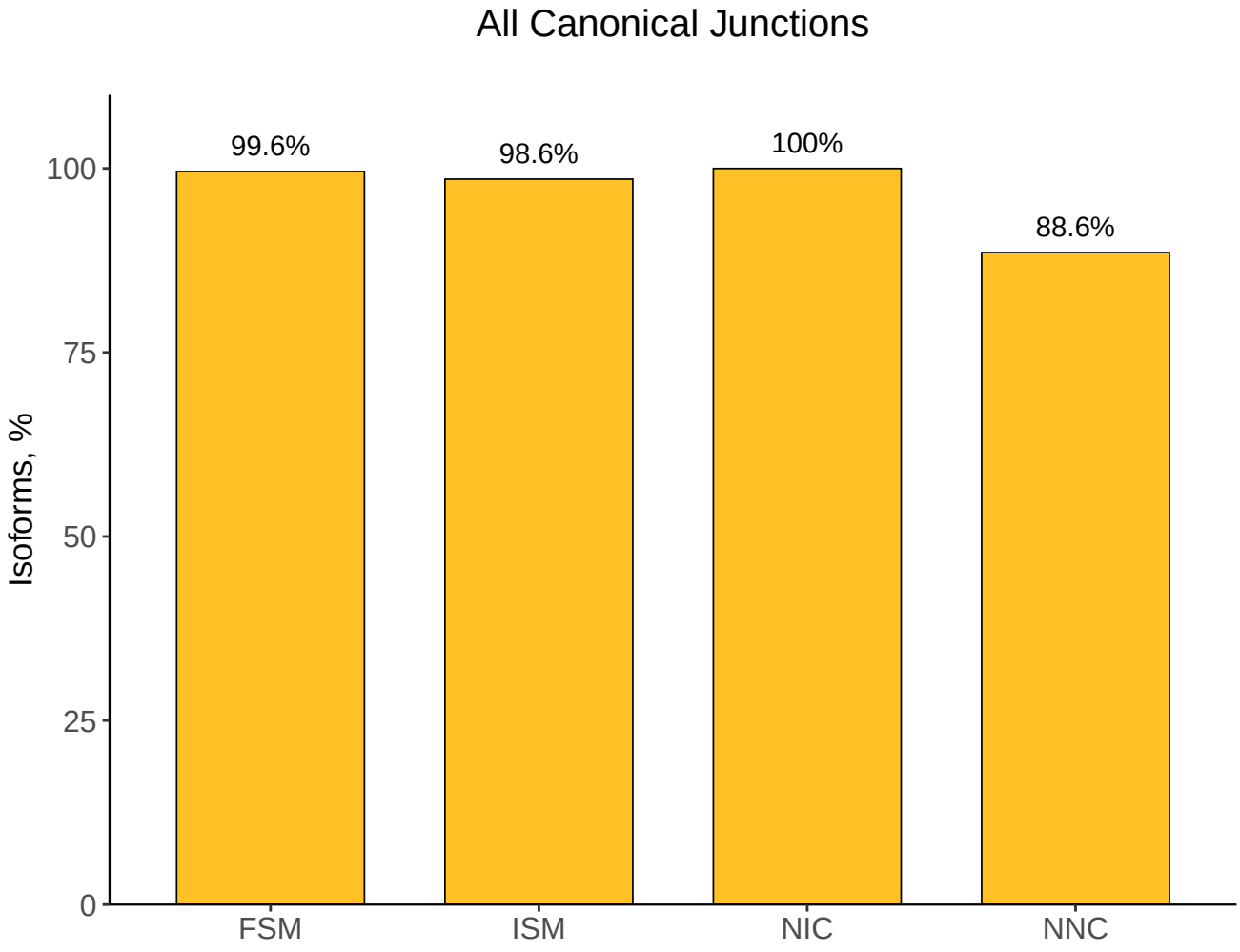

Non−Canonical Junctions
Isoforms, %
12.5
10.0
7.5
5.0
2.5
0.0
0.4%
1.4%
0%
11.4%
FSM
ISM
NIC
NNC

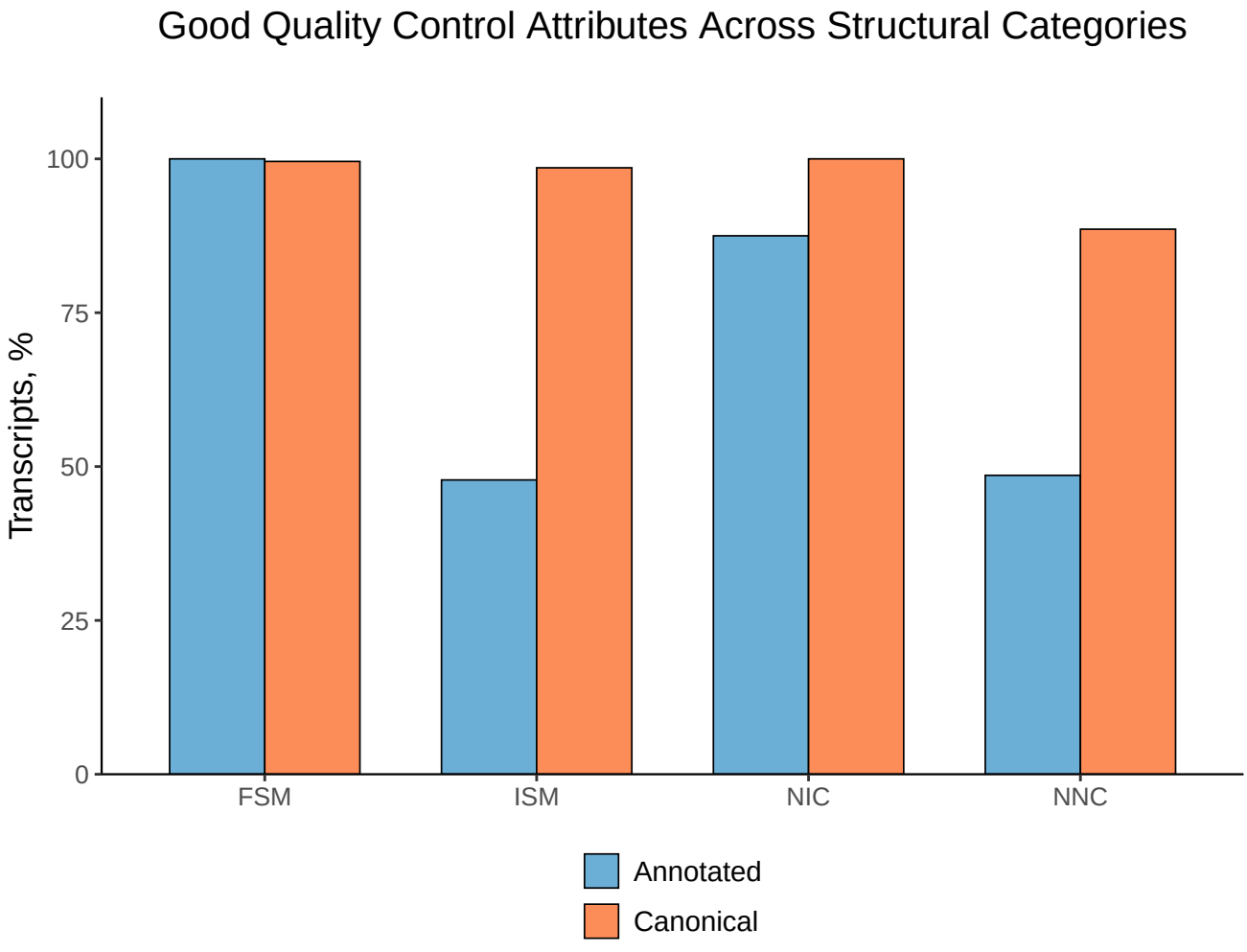

Nonsense−Mediated Decay by Structural Category
Isoforms, %
0.0
2.5
5.0
7.5
1.6%
0%
7.4%
8.6%
FSM
ISM
NIC
NNC


Quality Control Attributes Across Structural Categories
Transcripts, %
0
5
10
15
FSM
ISM
NIC
NNC
Non−canonical
Predicted NMD
RT switching

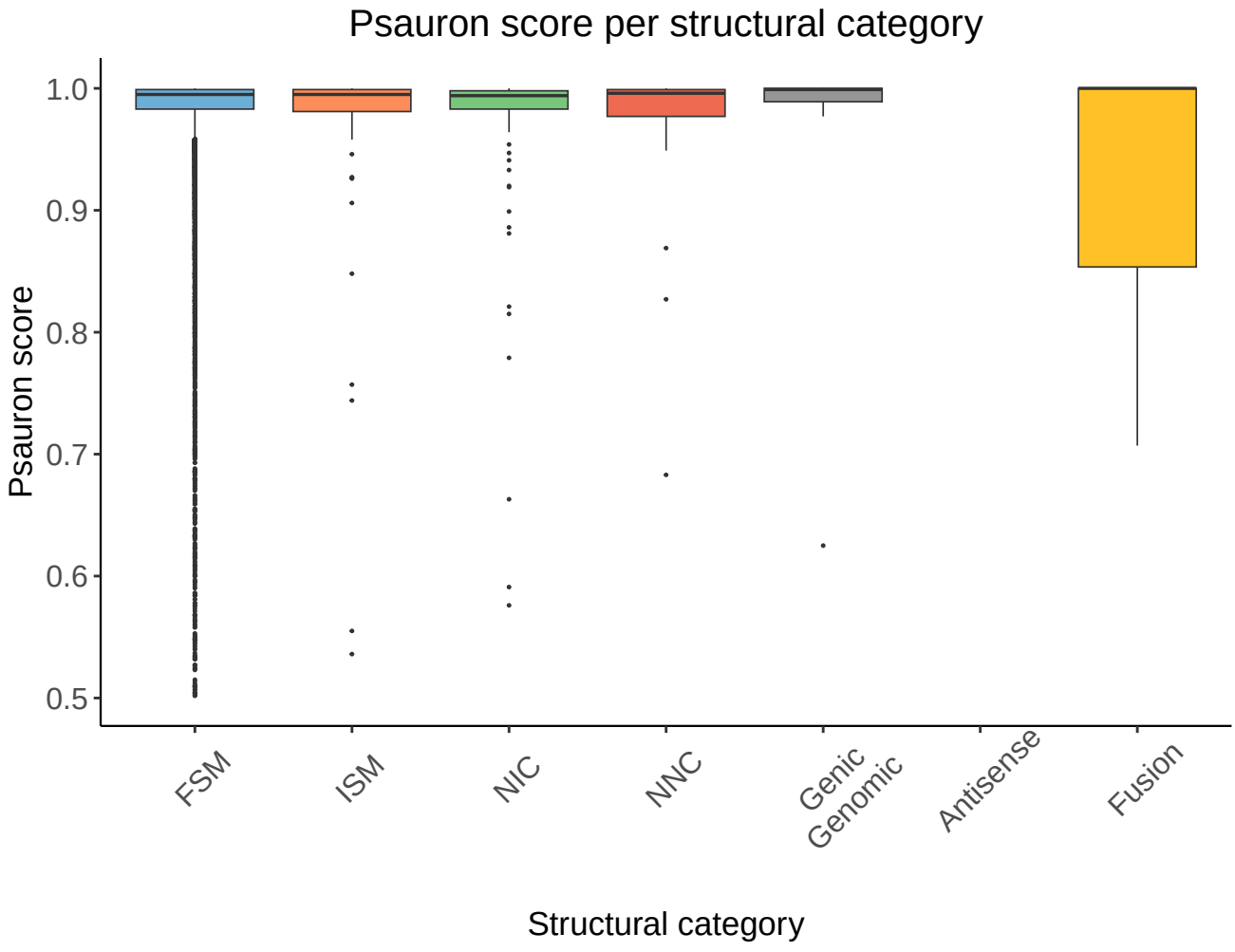
Features of Good Quality

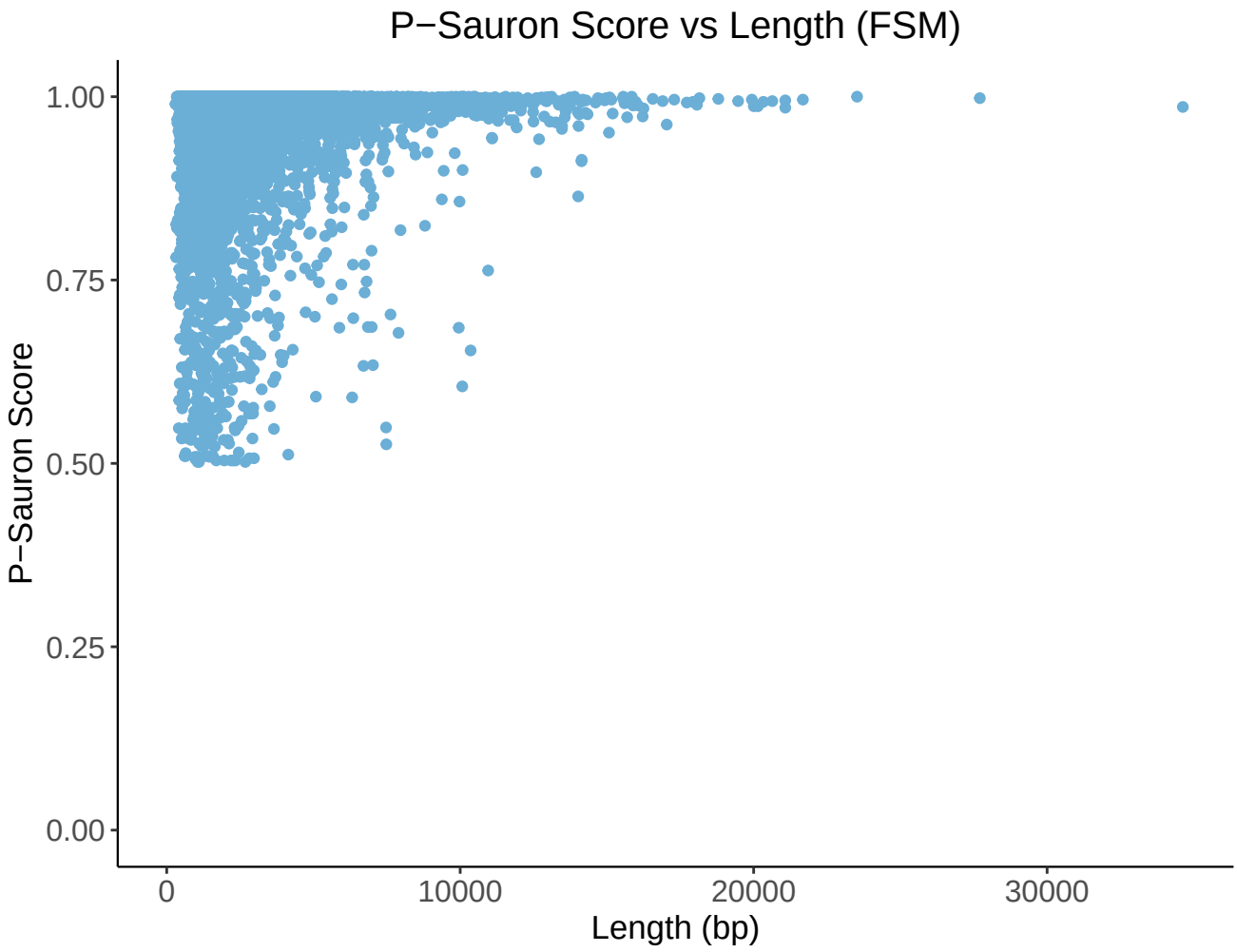

Annotation Support
Isoforms, %
100
75
50
25
0
100%
47.8%
87.5%
48.6%
FSM
ISM
NIC
NNC

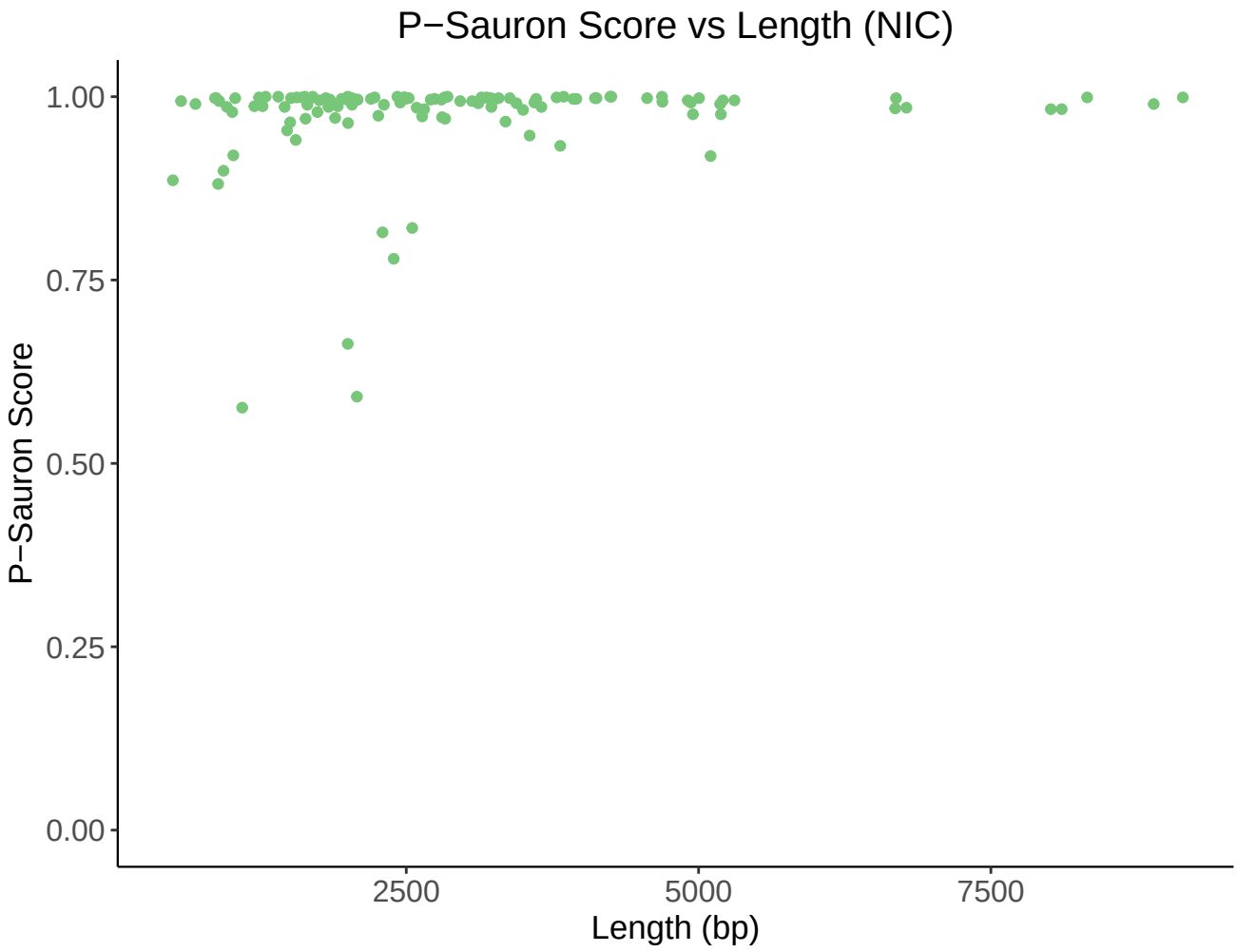

All Canonical Junctions
Isoforms, %
100
75
50
25
0
99.6%
98.6%
100%
88.6%
FSM
ISM
NIC
NNC

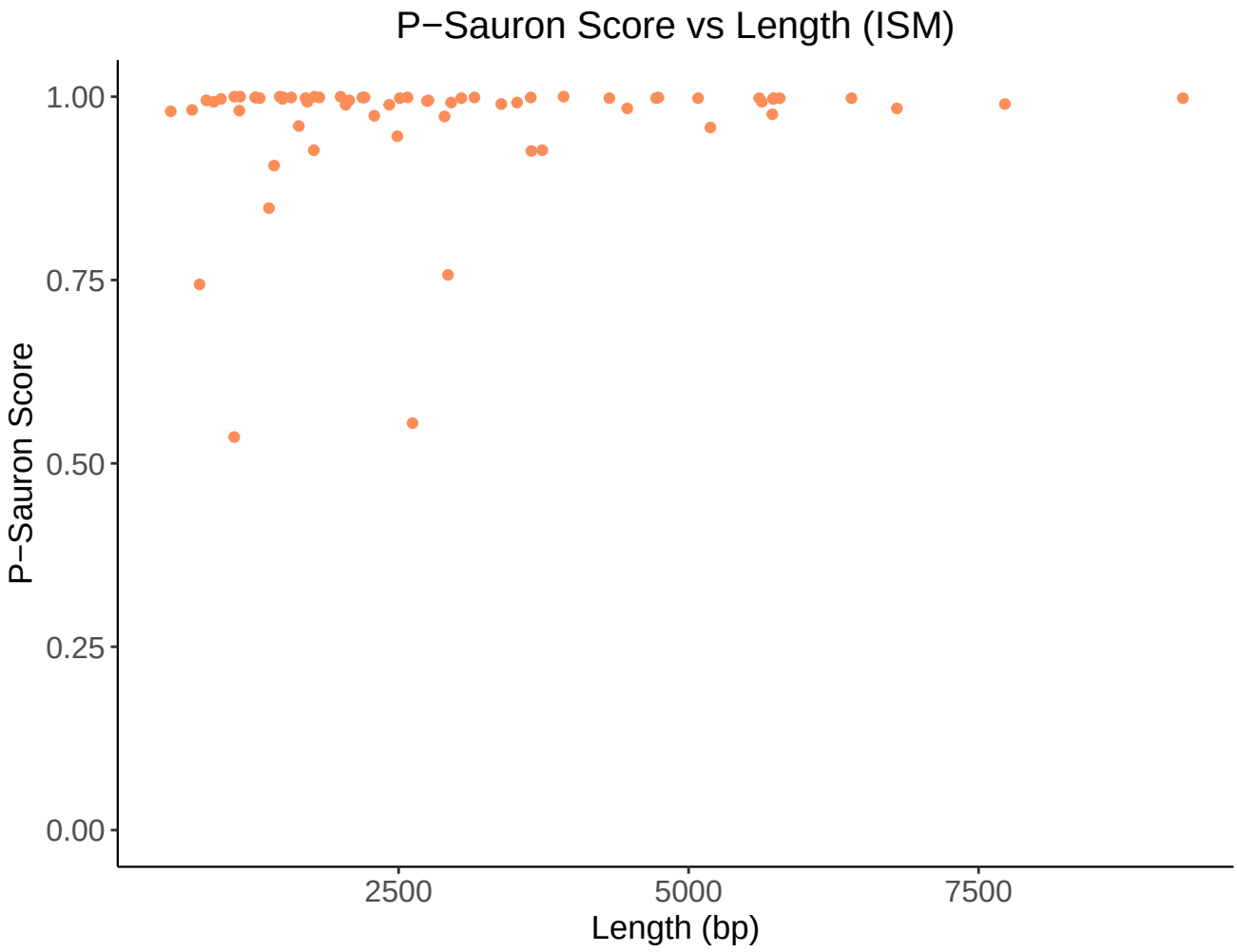

Good Quality Control Attributes Across Structural Categories
Transcripts, %
100
75
50
25
0
FSM
ISM
NIC
NNC
Annotated
Canonical

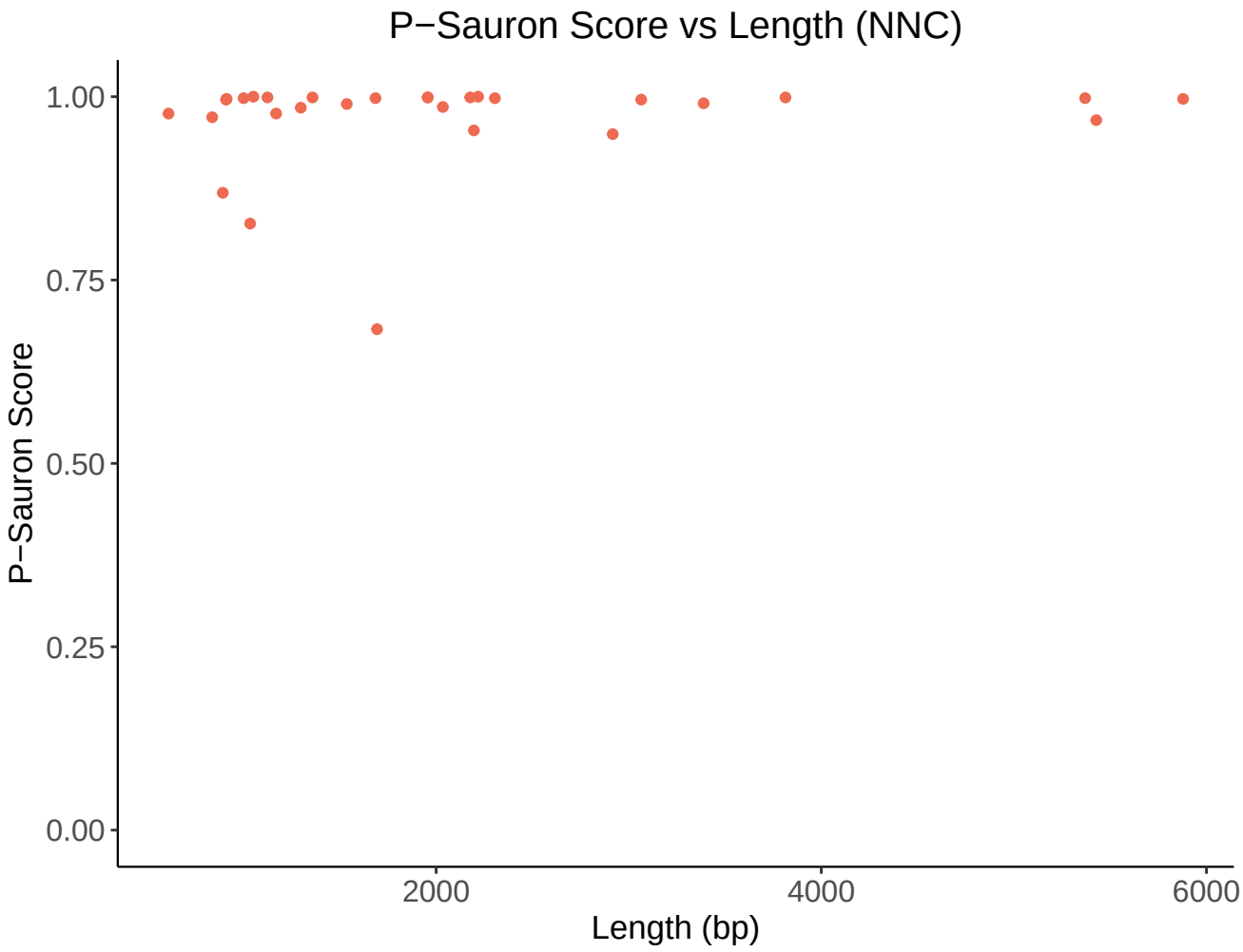
TransDecoder2 CDS prediction statistics

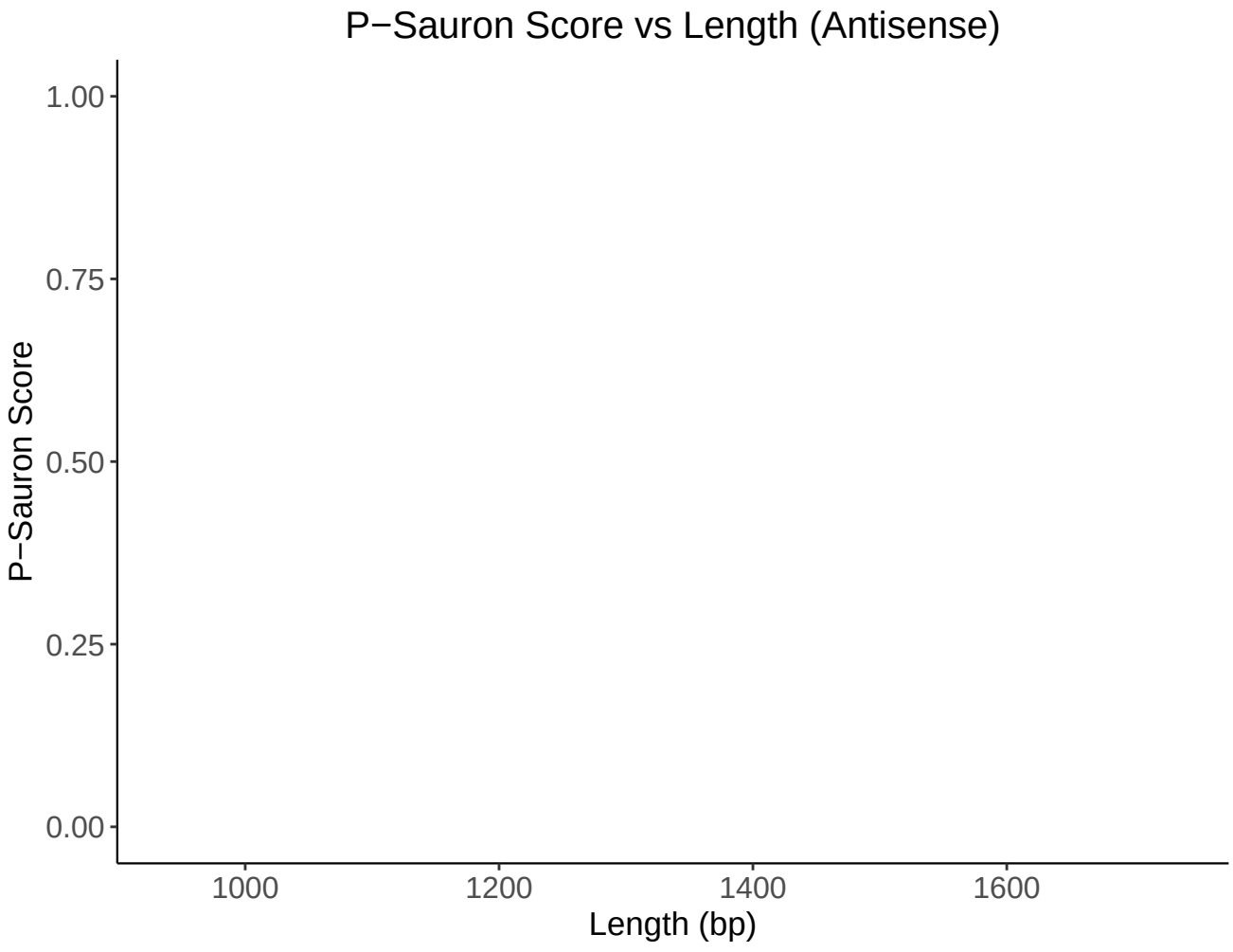

Psauron score per structural category
Psauron score
1.0
0.9
0.8
0.7
0.6
0.5
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Structural category

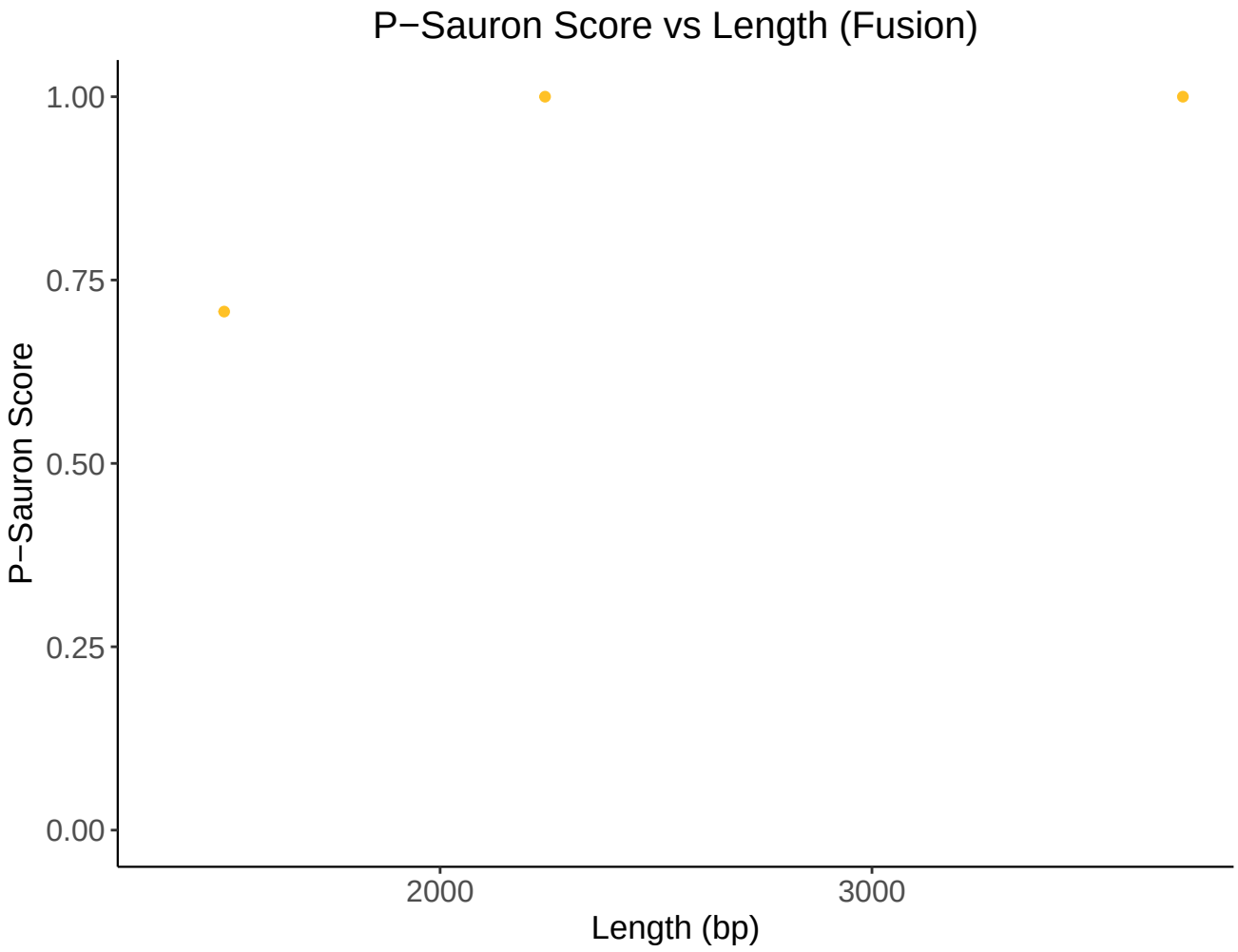

P−Sauron Score vs Length (FSM)
P−Sauron Score
1.00
0.75
0.50
0.25
0.00
0
10000
20000
30000
Length (bp)

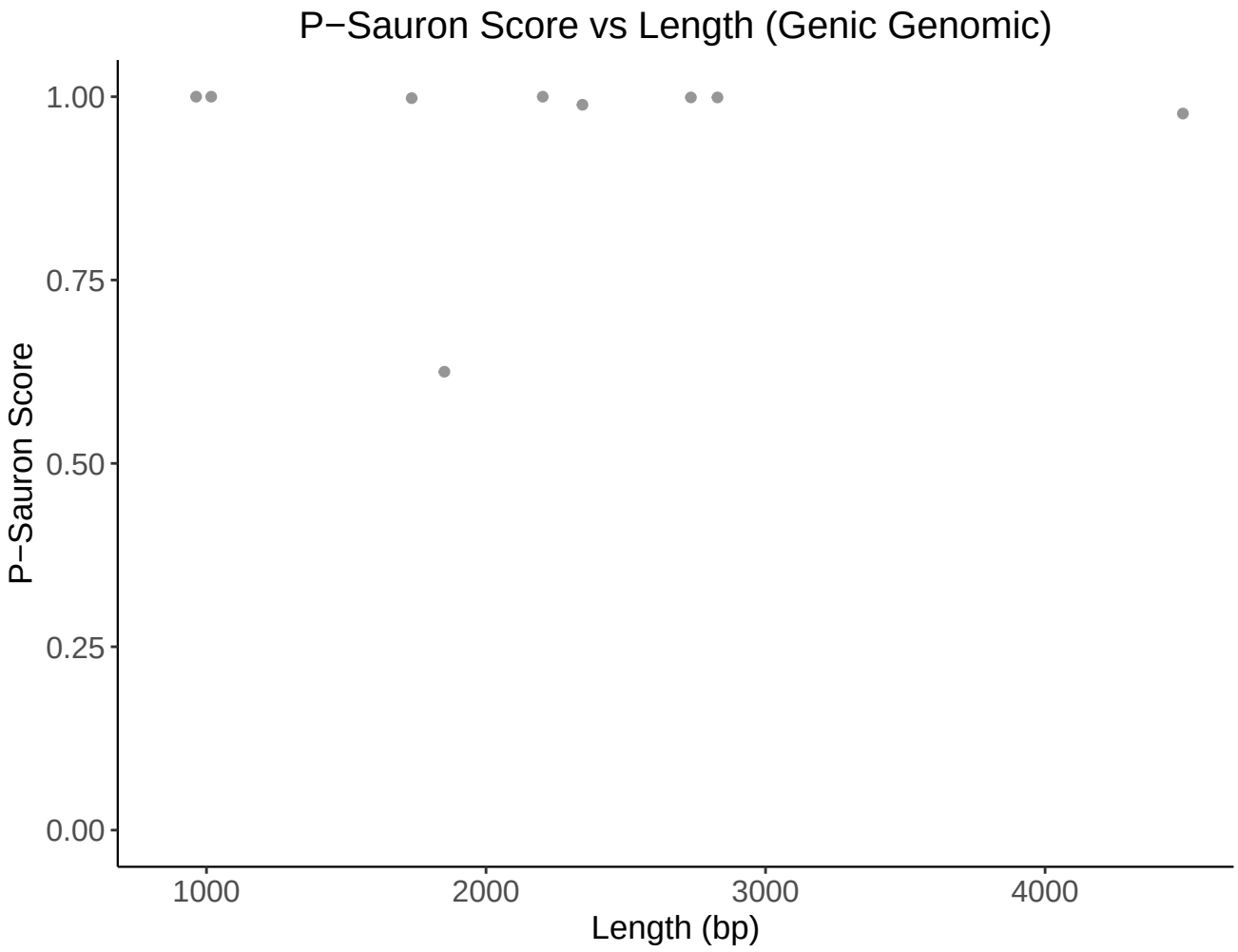

P−Sauron Score vs Length (NIC)
P−Sauron Score
1.00
0.75
0.50
0.25
0.00
2500
5000
7500
Length (bp)

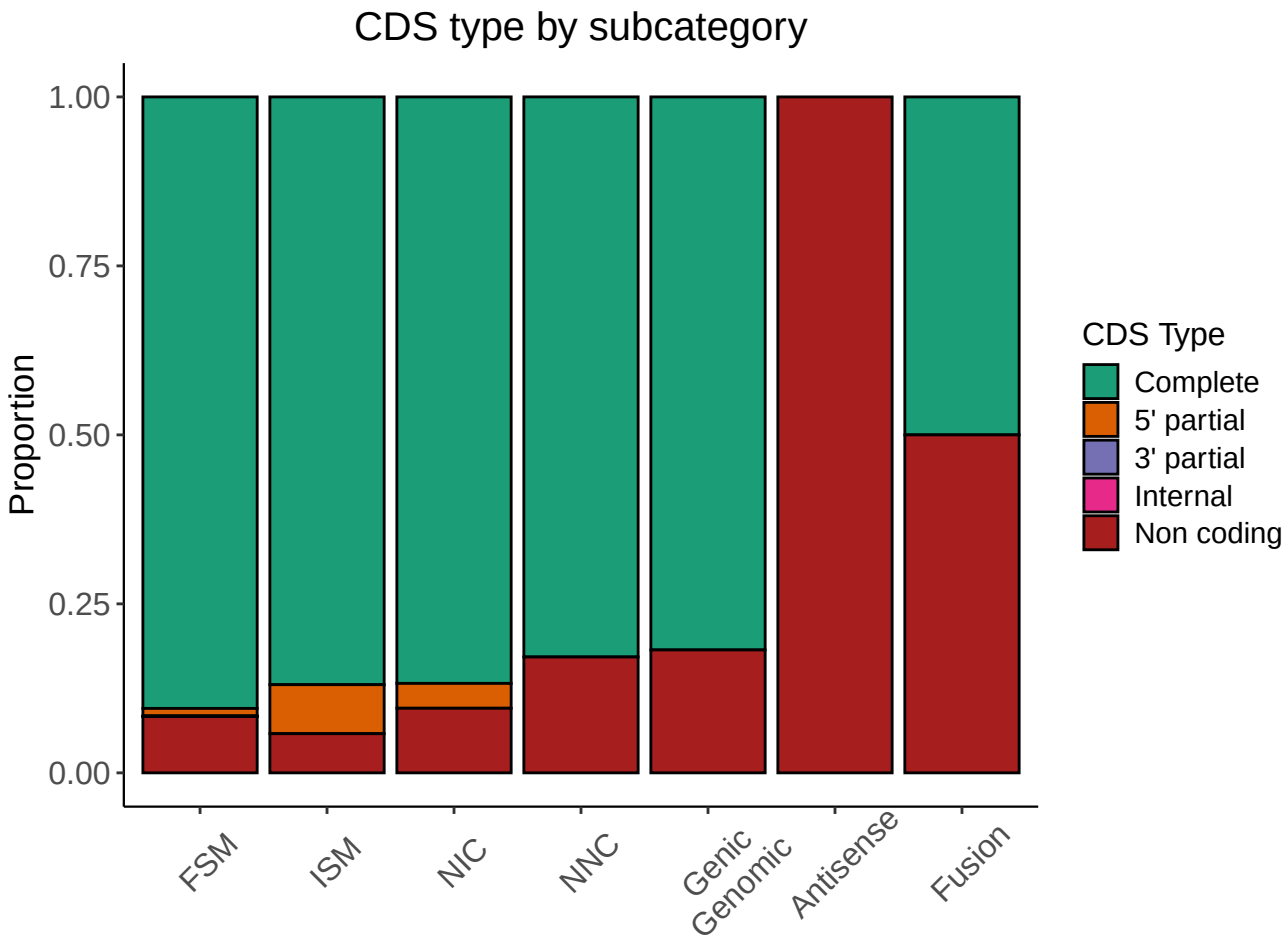

P−Sauron Score vs Length (ISM)
P−Sauron Score
1.00
0.75
0.50
0.25
0.00
2500
5000
7500
Length (bp)

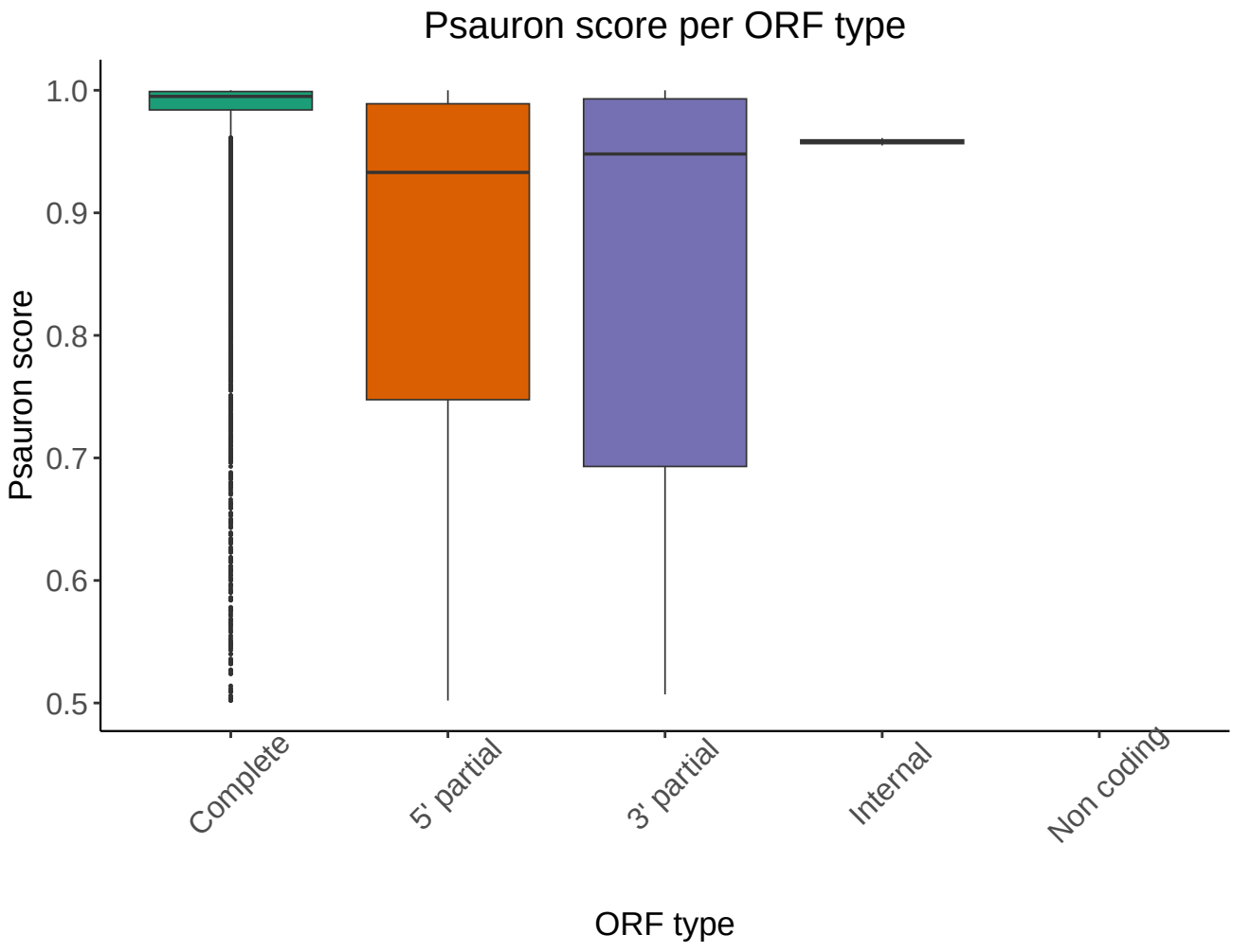

P−Sauron Score vs Length (NNC)
P−Sauron Score
1.00
0.75
0.50
0.25
0.00
2000
4000
6000
Length (bp)


P−Sauron Score vs Length (Antisense)
P−Sauron Score
1.00
0.75
0.50
0.25
0.00
1000
1200
1400
1600
Length (bp)


P−Sauron Score vs Length (Fusion)
1.00
0.75
0.50
0.25
0.00
P−Sauron Score
2000
3000
Length (bp)


P−Sauron Score vs Length (Genic Genomic)
P−Sauron Score
1.00
0.75
0.50
0.25
0.00
1000
2000
3000
4000
Length (bp)


CDS type by subcategory
Proportion
1.00
0.75
0.50
0.25
0.00
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
CDS Type
Complete
5' partial
3' partial
Internal
Non coding


Psauron score per ORF type
Psauron score
1.0
0.9
0.8
0.7
0.6
0.5
Complete
5' partial
3' partial
Internal
Non coding
ORF type